Panasonic

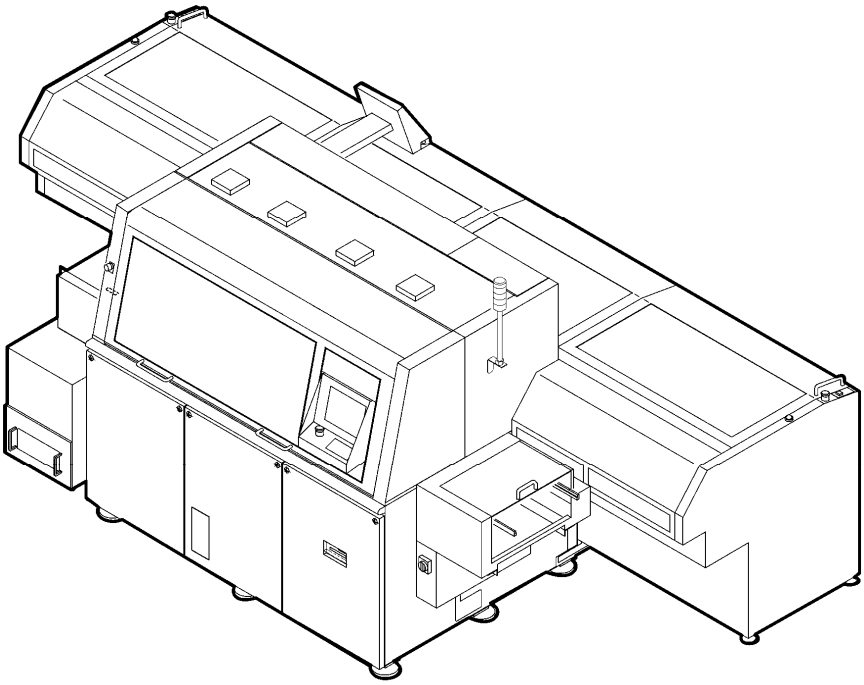
Operating Instructions

Programming Manual

High-Speed Axial Lead Component Insertion Machine

AV131

Model No. **NM-EJA2A**
NM-EJA3A



Thank you for purchasing AV131, the high-speed axial lead component insertion machine.
Before placing the machine in service, be sure to read the instruction manual for proper usage.
After that, store it carefully and read it if necessary.

N7201A233E00

Marking Samples for Product Safety

Identification for HHS (U.S.A.)
Manufactured: xxxx(xx),xxxx
Panasonic Factory Solutions Co., Ltd.
Kadoma, Osaka, 571-8502, Japan
Certification for HHS (U.S.A.)
Complies with FDA radiation performance standards, 21 CFR Sub-chapter J.
PS0004E2

HHS-Mark for U.S.A.

CE
TÜV PRODUCT SERVICE
Classification of ISM equipment: Group 1 Class A
Panasonic Factory Solutions Co., Ltd.
Kadoma, Osaka, 571-8502, Japan
PS0002E2

CE-Mark

USAGE OF WARNING SHEETS IN MAINTENANCE

Before performing maintenance, affix the included warning sheets (see below) to prominent positions, such as the operating panel or the vicinity of it, for safety.
During the maintenance by two or more persons in particular, as well as using these warning sheets to ensure safety, be sure to send signals to each other.

This warning sheet uses a magnet. Do not use it or store it near magnetic media or any other items susceptible to magnetic influence.

Use of this machine:
This machine is designed and constructed to be used for placing electric and electronic components on PCB.
This machine shall be used for this purpose only.

It shall not be used for any other purpose.

Operation of this machine:
This machine shall be operated by professional, specially trained operators.

It shall not be operated by non-professional, not specially trained operators.

Regarding the training course, please contact Panasonic Factory Solutions Co., Ltd. or PFSE.

⚠ WARNING

Do not allow any person who has not received special training to operate the machine.

Prohibition

The machine many be operated improperly, causing injury.

- Either carry out training by yourself of consult us.

Machine was manufactured by: Panasonic Factory Solutions Co., Ltd.

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Phone number: +81-942-84-2644
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Year: 2008 CE

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1. INTRODUCTION

This chapter describes how to use this manual.

1.1. How to Use This Manual

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1.1.1 Document Summary

This section describes how the AV131 Operating Instructions are organized.

Operation Manual
Describes the guidelines for basic operations.

1. INTRODUCTION
2. SAFETY PRECAUTIONS
3. DESCRIPTION
4. START/STOP
5. PROCEDURE FOR PRODUCTION (BASICS)
6. TROUBLESHOOTING

Programming Manual
Discusses the method of creating various programs.

1. INTRODUCTION
2. DATA CREATION
3. NC DATA CREATION
4. PART DATA CREATION
5. P.C.B. DATA CREATION
6. P.C.B. RECOGNITION DATA CREATION
7. DATA CHECK PROCEDURE
8. FILE MANAGING

Reference Manual
Offers helpful information clarifying unclear points you may encounter in operating the equipment.

1. INTRODUCTION
2. OVERVIEW
3. VARIOUS FUNCTIONS
4. SCREEN TRANSITION
5. PRODUCTION MANAGEMENT INFORMATION
6. SYSTEM MANAGEMENT

Maintenance Manual
Provides essential information for performing routine maintenance.

1. INTRODUCTION
2. SAFETY PRECAUTIONS
3. INSTALLATION
4. CONTROL SYSTEM
5. SENSOR LAYOUT
6. PREPARED MATERIALS
7. INSPECTION
8. MAINTENANCE GUIDE
9. SYSTEM CHECK FUNCTIONS
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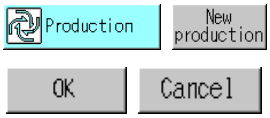
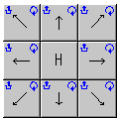
Error Code List
Contains organized error codes to be displayed when troubles occur.

1. INTRODUCTION
2. ERROR CODES

1.1.2 Conventions Used

■ Buttons, Keys and Switches

For buttons, keys and switches, this manual uses the following conventions for description of the procedure except for the functions or roles of this machine.

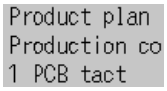
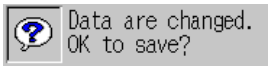
Name	Explanation	Indication example	Symbol	Example of use
Button	Pressed on the screen.		[ ]	[Production], [New production], [OK], [Cancel], etc.
Key	Used to input characters and numbers. (Keyboard)			[ESC], [BS], [ENT], etc.
	Used to move the camera etc. (Teaching keys)			[H], etc.
Switch	Used in operation other than on the screen (e.g., on control panel, sub-control panel, power-control panel).		< >	<START>, <STOP>, <ENABLING>, <PARTS EXCHANGE>, etc.

(Note) < ENABLING > + <START> indicates that you should press <START> with < ENABLING > held down.

Icons or buttons are used to show keys with no name, etc. (e.g. function keys, scroll buttons).

Function key	
Scroll button	

■ Screen Display

Item	Indication example	Symbol	Example of use
Screen name or on-screen item		“ ”	“Product plan”
Message			“Data are changed. OK to save?”

■ Cautionary Statements

Cautionary statements and cross-references are entitled or marked with an icon as follows:

Notation	Meaning
=REMARKS=	Matters requiring attention in operation
=HINT=	Helpful suggestions or available operations other than those described here
=📖=	Supplementary information for learning more about operations

These titles may be numbered as ‘=HINT=1’ in figures and footnoted.

■ Abbreviations

The names of the referential manuals are abbreviated as follows:

Ope.: Operation Manual Ref.: Reference Manual
Prg.: Programming Manual Mtn.: Maintenance Manual

The manuals are cross-referenced with the notation ‘Manual Name / Chapter Title / Section Title’.
Ex.) =📖= Refer to ‘Ope. / VARIOUS FUNCTIONS / Move Axis’ for details.

2. DATA CREATION

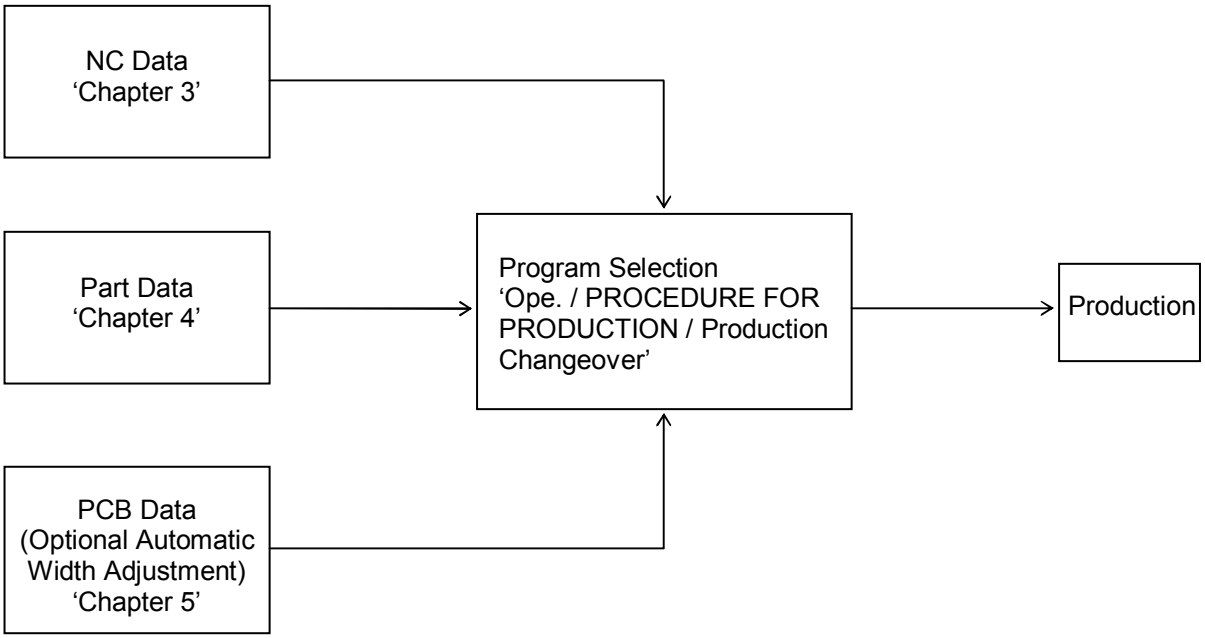
This chapter describes the types and editing procedures of the data and libraries used by the AV131.

2.1. Data Types and Contents

D79PEC-40-190-A0

2.1.1 Outline of the Data

To insert components with the AV131, it is necessary to prepare in advance the NC data that defines the conditions for component insertion and the part data that consists of information related to the parts to be used.



2.1.2 Parameters

The NC data and part data include the following parameters (data items).

Data	Details	
NC Data	X (ABS)	Insertion position
	Y (ABS)	
	Z No.	Component feeder position
	NO INS	Designation of whether or not to insert the part
	ROT	Orientation of part to be inserted
	INSERT WIDTH	Insertion pitch
	RECOG	Designation of whether or not to perform hole recognition
	SKIP STEP	Designation of steps to be skipped
	S&R	Designation of S&R
	BAD MARK	Designation of bad mark (option)
Part Data	COMPONENT COMMENT	
	COMPONENT THICK	
	LEAD DIAMET	
	CHUCK	Designation of chuck pitch (26/52 mm) and centering function
	SPARE	Designation of spare feeder for Z alteration function
PCB Data (Option)	PCB size – Length	Designation of PCB size
	PCB size – Width	
	Hole pitch	Designation of interval between reference holes

3. NC DATA CREATION

This chapter describes the procedure for creating the NC data.

3.1. Setting Items of the NC Data

D79PEC-43-070-A0

3.1.1 Parameters

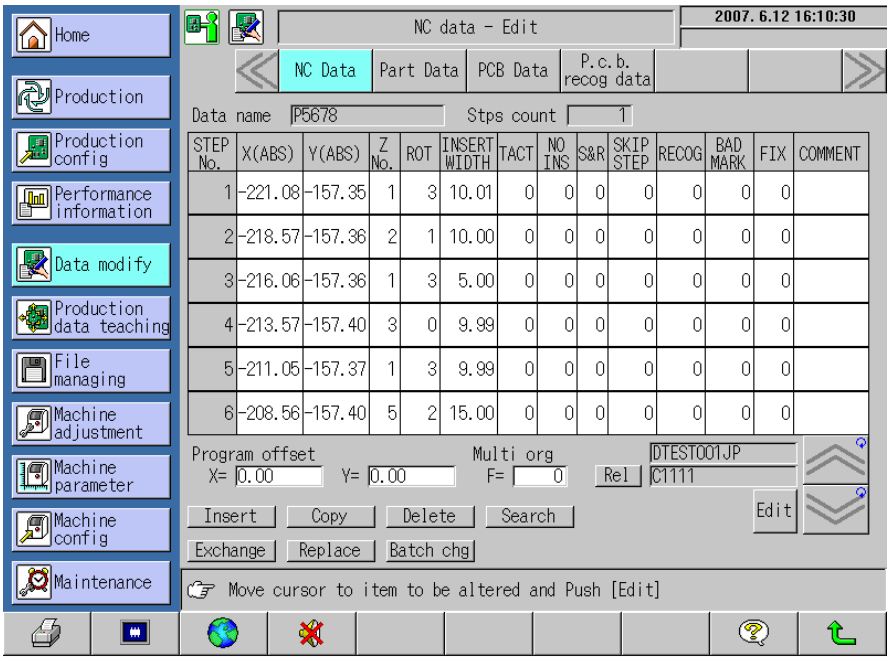
The AV131 inserts parts on PCBs according to data in each step of the NC program.
The machine does not run unless the NC data has been input or selected.
The NC data define where to insert what parts in what order.
The NC data include the following parameters.

Parameter	Input range / Standard Setting	Details
STEP No.	1 to 5000	ID numbers used to distinguish steps of the NC program.
X(ABS) Y(ABS)	X: -9999.99 to 9999.99 (mm) Y: -9999.99 to 9999.99 (mm)	ABS: Part insertion position based on the PCB origin (X-Y coordinates origin). INC: Part insertion position based on the previous insertion position.
Z No.	● 120 inputs 1 to 121: For axial lead parts 121: For jumper wires ● 60 inputs 1 to 60: For axial lead parts 121: For jumper wires	Numbers that designate the position of component feeders on the feeder carriage. 121 designates jumper wire unit.
ROT	0 = 0°, 1 = 90° 2 = 180°, 3 = 270°	Angle of part when inserted on PCB.
INSERT WIDTH	0 to 999.99 (mm)	Pitch data of parts to be inserted Standard setting is 5 to 26 mm.
TACT	Not used	Not used
NO INS	0 = Insertion 1 = No insertion	Codes that designate whether part insertion is executed or not.
S&R	0 = Normal (No repeat) 1 = Step repeat / No rotation 2 = Pattern repeat / No rotation 11 = Step repeat / 90° rotation 12 = Pattern repeat / 90° rotation 21 = Step repeat / 180° rotation 22 = Pattern repeat / 180° rotation 31 = Step repeat / 270° rotation 32 = Pattern repeat / 270° rotation	Codes that designate step repeat or pattern repeat and rotation insertion angle.
SKIP STEP	0 to 9	Designates step to be skipped.
RECOG	0 = Without hole recognition 1 = With hole recognition	Codes that designate whether or not to correct the given hole position (correct the step)
BAD MARK	0 = Without bad mark recognition 1 = With bad mark recognition	Codes that designate whether bad mark detection is executed with the bad mark detection sensor.
FIX	Not used	Not used
COMMENT	Within 8 alphanumeric characters	Any user-set comment for the insertion position (step). (Can be omitted.)
Program offset	X: -999.99 to 999.99 (mm) Y: -999.99 to 999.99 (mm)	The difference between the coordinate origin on the PCB and mechanical origin of the machine.
Multi org	F1: 1 to 120 (Feeder No.)	The origin of the feeder carriage designated by the position of a component feeder.

3.1.2 NC Data Editing Screen/NC Data Sheet

STEP No.	X (ABS)	Y (ABS)	Z No.	ROT	INSERT WIDTH	TACT	NO INS	S&R	SKIP STEP	RECOG	BAD MARK	FIX	COMMENT
1													
2													
3													
4													
5													
6													
7													
8													
9													
10													

The NC data can be input from the NC Data Editing screen. The NC Data Editing screen is shown below. Prepare an NC data sheet completed in advance, and input the data to the parameter on the NC data - Edit screen.



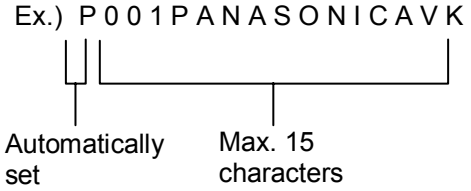
3.1.3 Creating the NC Data

■ Procedure for Creating the NC Data

1. Determine the NC data name (max. 16 characters).

◆ Data name:

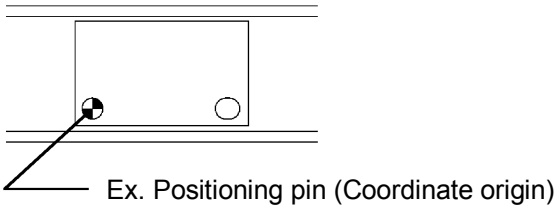
This is the name for distinguishing the NC data. The user may assign the name freely.
The letter 'P' is automatically set as part of the program name.
The rest of the name can be composed of numbers 0 to 9, letters A to Z, plus sign '+', minus sign '-', period '.', and space ' '.



2. Determine the origin of the coordinate on the PCB.

◆ Coordinate Origin:

Point of reference on the PCB from where part insertion position is measured.



=HINT=

The user can determine the origin of the coordinate freely. In this manual, the origin of the coordinate is set on the position of the positioning pin (fixed side).

3. Determine the program offset.

◆ Program offset:

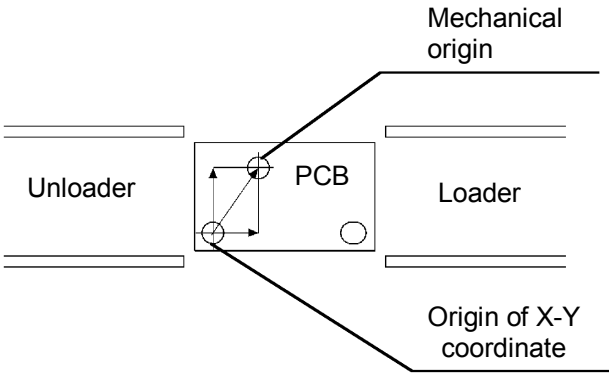
This is the distance from the origin of the coordinate on the PCB to the mechanical origin.

- The mechanical origin never changes.
  Since a different coordinate origin can be established for each NC data, the program offset value must be input for each NC program.

Standard Setting (when the position of the positioning pin [fixed side] is the origin of the coordinate);

	Standard Setting (Approx.)
X	221.00 mm
Y	385.00 mm

Ex.) When the position of the positioning pin (fixed side) is the origin of the coordinate;



=HINT=

The program offset values vary slightly for each machine. To find the exactly correct values, execute a test insertion. Correct the values until the positioning is perfect.

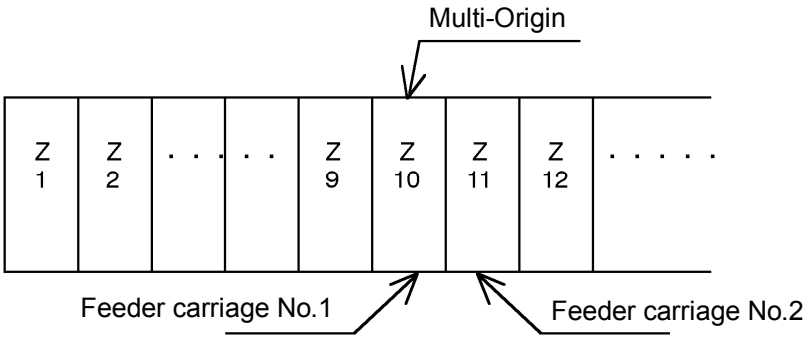
4. Determine the multi-origin (1 to 120 feeders (120 inputs)) / (1 to 60 feeders (60 inputs)).

◆ MULTI ORG:

This is the origin of the feeder carriage (Z) designated by the Z No. of the component feeder.

- The designated Z No. is the reference position (position of feeder carriage No. 1) of the feeder carriage No.
- As this can be designated in each NC data, by setting the feeder with various arrays, production of PCBs can be switched only by changing the designation of the multi-origin.

Ex.) When multi-origin designation is 10;



=REMARKS=

Even if the multi-origin is input, PCBs can be produced based on the data with no multi-origin input.
For the example above, the part data for Z1 and Z2 are required but not for Z10 and Z11.
When feeder carriage action mode is set in exchange connection mode (for 120 inputs) or EOP recovery function is selected, multi origin function cannot be used.

5. Determine the insertion order, and assign the step No. (step 1 to 5000).

◆ STEP No.:

This is the ID No. of NC data steps.

- Determine the insertion order before assigning the step Nos.
- The NC data is executed in the order of the step Nos.

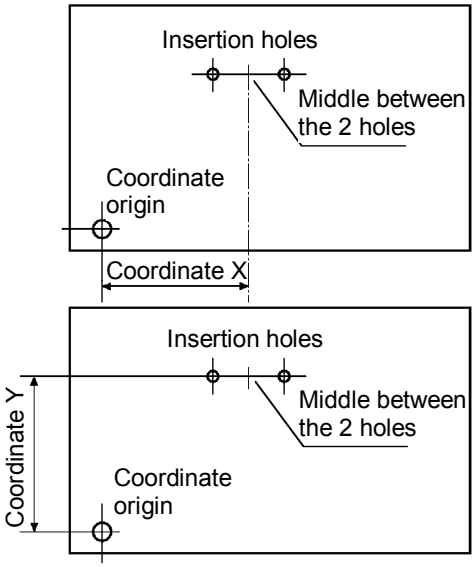
=HINT=

- Arrange the insertion order so that the movement of the X-Y table is minimized.
- Arrange the insertion order so that the movement of the Z axis is minimized.
- When a step for step and repeat is necessary before a step for part insertion, follow ‘Priority of Steps’.

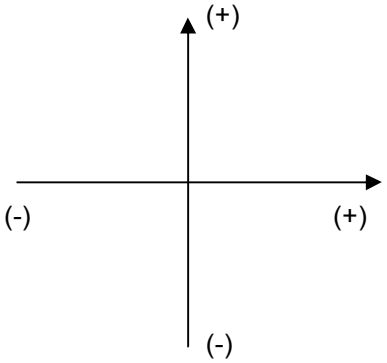
Priority of steps:

(1) S&R step
(2) Optional hole offset designation step
(3) Part insertion step

6. Determine the insertion positions. (-9999.99 to 9999.99 mm for both X and Y)
Measure the distance from the origin of the coordinate, specified in step 2, to the part insertion position, and determine the X and Y coordinates.



=HINT=
X-Y coordinate system



◆ X (ABS):

Distance from coordinate origin to insertion position in the X direction. Unit: mm

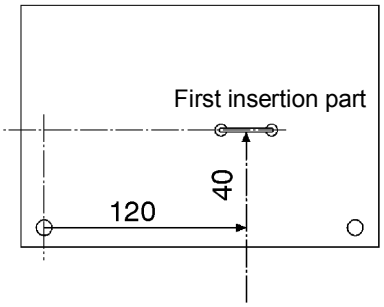
◆ Y (ABS):

Distance from coordinate origin to insertion position in the Y direction. Unit: mm

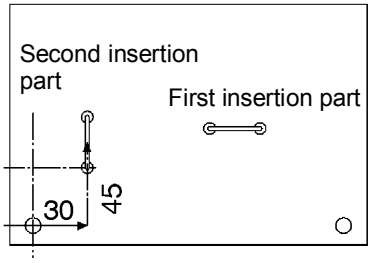
Absolute Method (ABS):

This is the method in which the part insertion position is always measured from the origin of the coordinate.

Ex.)



First Insertion X: 120 mm
Y: 40 mm



Second Insertion X: 30 mm
Y: 45 mm

7. Determine the feeder Nos. (Z1 to Z121) - (120 inputs) / (Z1 to Z60 & Z121) - (60 inputs).

◆ Z No.:

Designate the position where the component feeder of the feeder carriage is set with Z Nos.

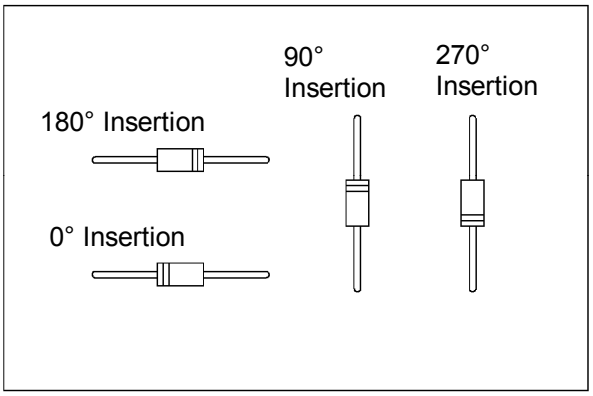
- Determine which part (from which feeder) will be inserted for each insertion step created in step 6.
- Executing of NC program in the semi-auto mode brings the feeder, designated by Z axis No., to the chuck position of the rotary chuck.
- Usually, Z axis No. is the same as Z No. However, when the multi-origin is set at a number other than 1, Z axis No. count starts from the designated Z No., which becomes Z axis No.1.
- Z121 designates the insertion of jumper wire, not any Z axis (component feeders). This designation is not effective whatever the multi-origin may be.

8. Determine the insertion angle (0°= 0, 90°= 1, 180°= 2, 270°= 3).

◆ ROT:

This is the angle of a part being inserted on PCBs.

(Front view)

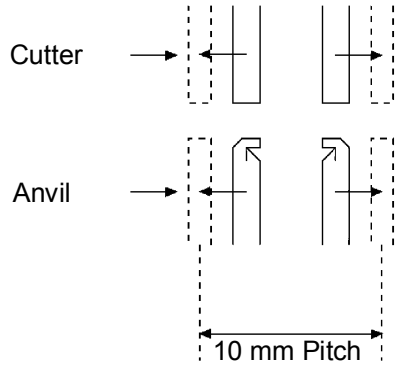


9. Determine the insertion pitch.

◆ INSERT WIDTH:

Designates the insertion pitch width of the cutter and anvil. The feeding length of the jumper wire is also determined by this value.

Ex.) When 10.00 mm is designated as the insertion pitch;



The width of the cutter and anvil is both 5 mm when they are at their origins. If the insertion width is set to 10 mm, the cutter and anvil move 5 mm further apart.

=HINT=

The width of the cutter and anvil is 5 mm when they are at their origins.

10. Designate either Insertion / No insertion.

◆ NO INS:

This is the code whether or not to insert parts.

Code	
0	Insertion
1	No insertion

- For the step designated no insertion, insertion is not executed. However, XY table positioning is executed.

Ex.) In the case that no insertion is designated on the N2 step.

	N1	**N2**	**N3**	**N4**
Head	Dry Run + Dry Run + Insertion	Stop	Dry Run + Dry Run + Insertion	Insertion
XY table	N1 Step Coordinates	N2 Step Coordinates	N3 Step Coordinates	N4 Step Coordinates

11. Designate the S&R.

◆ S&R:

Step repeat and pattern repeat are special parameters that instruct the machine to duplicate the instructions of a base pattern in other sections of the same board (sectioned PCB).

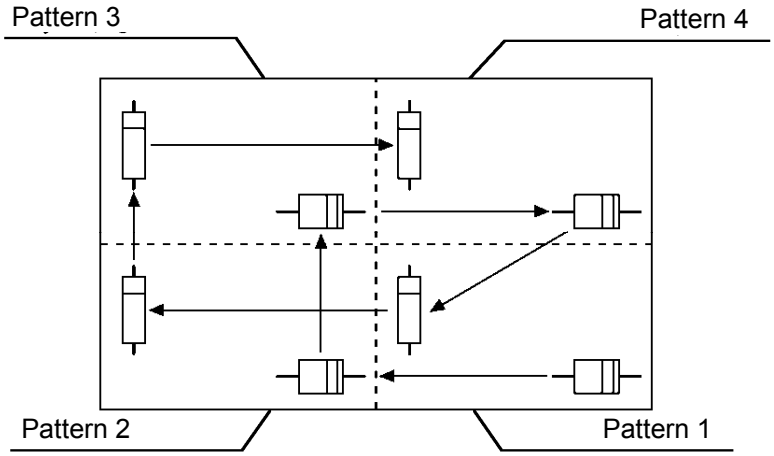
The angle of pattern rotation insertion (for polar coordinate) is also instructed by the code below.

Code	Meaning
0	Normal (No repeat)
1	Step repeat / No rotation
2	Pattern repeat / No rotation
11	Step repeat / 90° rotation
12	Pattern repeat / 90° rotation
21	Step repeat / 180° rotation
22	Pattern repeat / 180° rotation
31	Step repeat / 270° rotation
32	Pattern repeat / 270° rotation

=📖= For details of these settings, refer to ‘Creating the NC Data for Sectioned PCBs’ in this section.

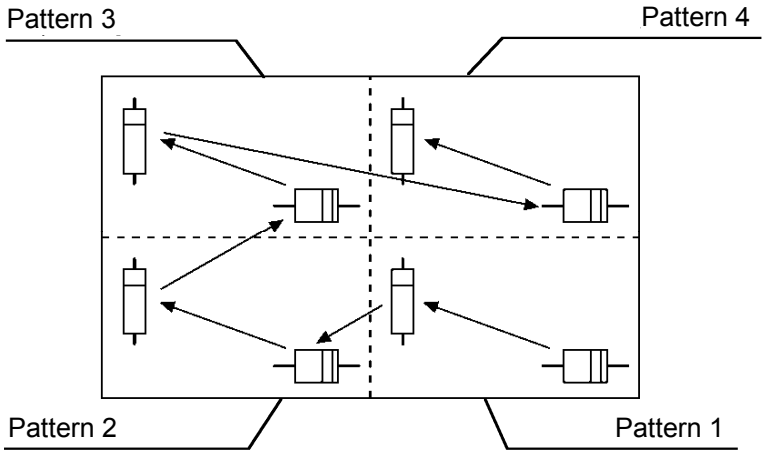
Step Repeat:

The same part is inserted throughout the patterns before proceeding to the next part.



Pattern Repeat

All parts are inserted on one pattern before proceeding to the next pattern.

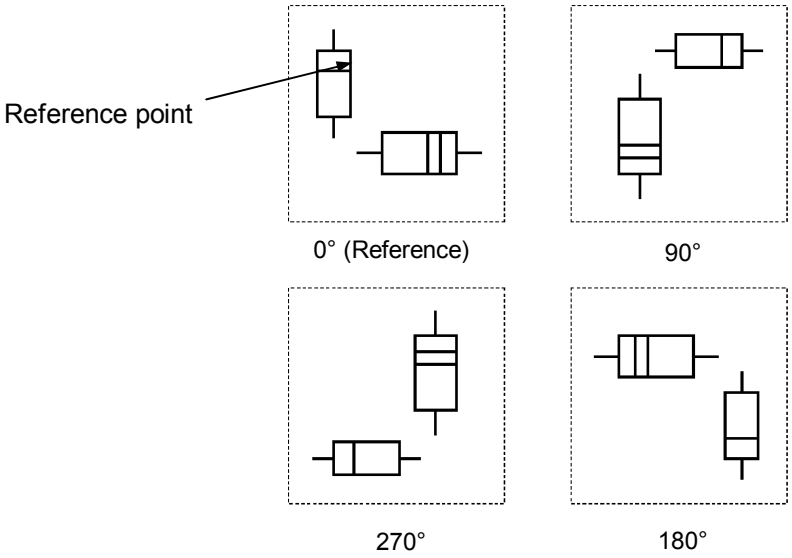


=HINT=

The offset (X, Y positions) of the first pattern is always (0, 0) for step and pattern repeat.

Pattern Rotation Insertion

Insertion pattern is rotated in a clockwise direction. The first insertion pattern is regarded as the reference.



The first insertion point of the first pattern is regarded as the reference point. This reference point cannot be deleted or skipped. If this reference point is to be deleted or skipped, then NC teaching for next new reference point has to be done to register that point as a new reference point, otherwise it will result in insertion to incorrect XY coordinates.

=REMARKS=

To disable insertion to the reference point temporarily, the NO INS code may be used. This will prevent insertion to incorrect XY coordinates and save you from having to perform NC teaching for a new reference point.

12. Input the step to be skipped.

◆ SKIP STEP:

This code is used to specify particular MC data step to be skipped.

- Default is 0. Change the setting if the step is to be skipped.

Code	Meaning
0	Execute the steps.
7	Never executes the steps.
1 – 6 **8, 9**	Codes that identify the steps to be skipped. If he code designated in the Product condition setting is set, the step will be skipped. However, the step will be executed unless the code is designated in the Product condition setting.

13. Designate the hole correction.

◆ RECOG:

This code designates whether or not to correct the given hole position (correct the step).

Code	
0	Without hole recognition
1	With hole recognition

14. Designate the bad mark (option).

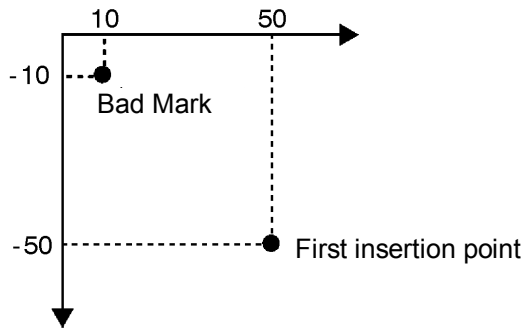
◆ BAD MARK:

Set this code at 1 to enable bad mark detection by the sensor when canceling insertion on that PC board or the pattern concerned.

Code	Meaning
0	Without bad mark recognition
1	With bad mark recognition

Ex.1) With normal NC data, designate the bad mark in the first step.

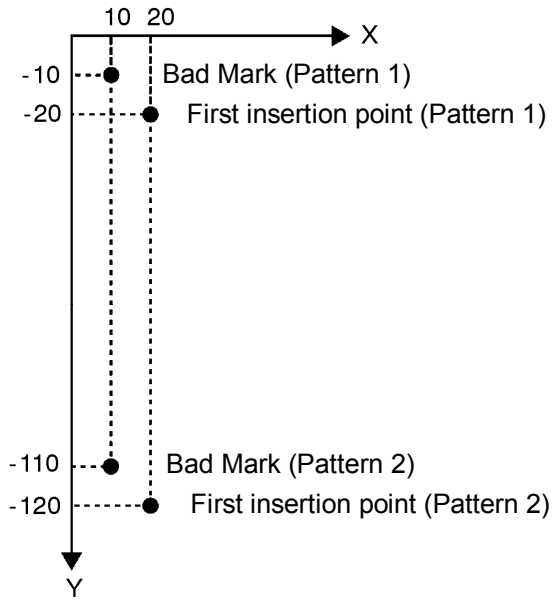
STEP No.	X (ABS)	Y (ABS)	Z No.	ROT	INSERT WIDTH	TACT	NO INS	S&R	SKIP STEP	RECOG	BAD MARK	FIX	COMMENT
1	10	-10	1	0	0	0	0	0	0	0	1	0	
2	50	-50	1	90	5.08	0	0	0	0	0	0	0	



If there is a bad mark in the position (10, -10), the insertion on that PCB will be cancelled.
If no bad mark exists, insertion on that PCB will be carried out.

Ex.2) Designate the BAD MARK code at 1 for the step immediately after the step where S&R is designated.

STEP No.	X (ABS)	Y (ABS)	Z No.	ROT	INSERT WIDTH	TACT	NO INS	S&R	SKIP STEP	RECOG	BAD MARK	FIX	COMMENT
1	0	0	1	0	0	0	0	2	0	0	0	0	
2	0	-100	1	0	0	0	0	2	0	0	0	0	
3	10	-10	1	0	0	0	0	0	0	0	1	0	
4	20	-20	1	90	5.08	0	0	0	0	0	0	0	

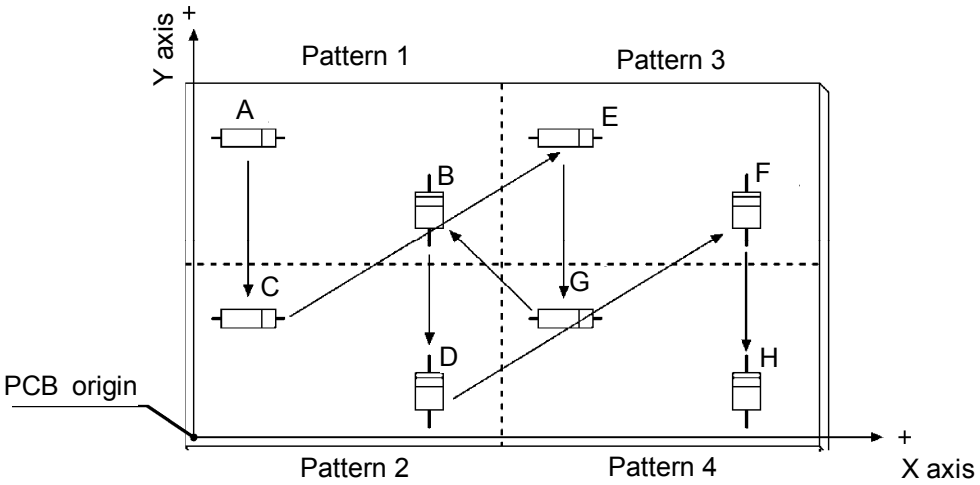


If there is a bad mark in the position (10, -10), insertion on pattern 1 will be cancelled.
If there is a bad mark in the position (10, -110), insertion on pattern 2 will be cancelled.

3.1.4 Creating the NC Data for Sectioned PCBs

■ Step Repeat

Ex.) A program to insert the same part throughout the 4 patterns before proceeding to the next part.

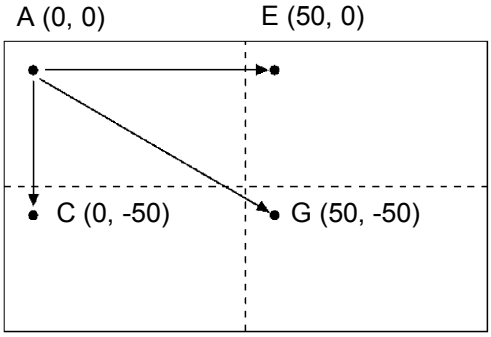


Position	No.	Angle	Pattern
A (Xa=10, Ya=90)	1	0°	Pattern 1
B (Xb=40, Yb=60)	2	90°	
C (Xc=10, Yc=40)	1	0°	Pattern 2
D (Xd=40, Yd=10)	2	90°	
E (Xe=60, Ye=90)	1	0°	Pattern 3
F (Xf=90, Yf=60)	2	90°	
G (Xg=60, Yg=40)	1	0°	Pattern 4
H (Xh=90, Yh=10)	2	90°	

=HINT=

The first insertion pattern is regarded as the reference pattern.
The first insertion point of the first insertion pattern is also regarded as the reference point. This reference pattern and point cannot be deleted or skipped, otherwise it will result in insertion to incorrect XY coordinates.
NC teaching for next new reference pattern or point has to be done to register that pattern or point as a new reference.

1. Determine the offset value of each pattern with reference to pattern 1. (Reference pattern)



Pattern 1 (0, 0)
Pattern 2 (0, -50)
Pattern 3 (50, 0)
Pattern 4 (50, -50)

(1) Input the offset data in X and Y positions of step Nos. 1 to 4.
(2) Input 1 (step repeat) in the S&R designation of step Nos.1 to 4.

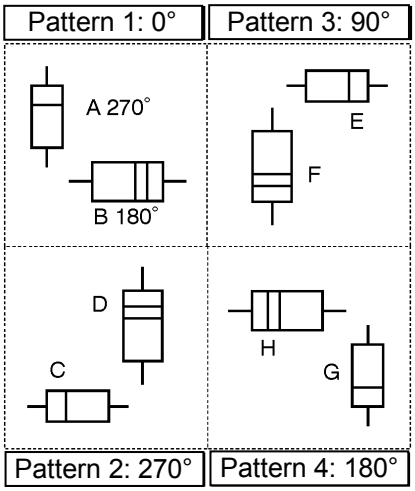
2. Input the X and Y coordinates of points A and B of pattern 1.
(1) Input the coordinate data of points A and B of pattern 1 in step Nos. 5 and 6.
(2) The coordinate data of insertion points on patterns 2 to 4 are not necessary since block Nos. 2 to 4 contain the offset data of these patterns.

STEP No.	X (ABS)	Y (ABS)	Z No.	ROT	INSERT WIDTH	TACT	NO INS	S&R	SKIP STEP	RECOG	BAD MARK	FIX	COMMENT
1	0	0	1	0	0	0	0	1	0	0	0	0	
2	0	-50	1	0	0	0	0	1	0	0	0	0	
3	50	0	1	0	0	0	0	1	0	0	1	0	
4	50	-50	1	0	0	0	0	1	0	0	0	0	
5	10	90	1	0	5.08	0	0	0	0	0	0	0	
6	40	60	1	90	5.08	0	0	0	0	0	0	0	

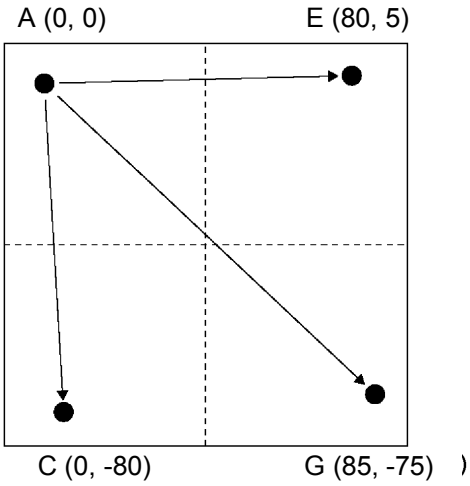
=HINT=
The first insertion pattern is regarded as the reference pattern.
The first insertion point of the first insertion pattern is also regarded as the reference point. This reference pattern and point cannot be deleted or skipped otherwise it will result in incorrect XY coordinates insertion.
NC teaching for next new reference pattern or point has to be done to register that pattern or point as a new reference.

■ Rotation Insertion Pattern (Polar Coordinate Conversion)

Ex.) A program to insert the same part throughout the 4 patterns before proceeding to the next part with step repeat.



1. Determine the offset value of each pattern from the first insertion point (point A) on pattern1.



Pattern 1 (0, 0) Rotation insertion angle 0° (Reference)
Pattern 2 (5, -80) Rotation insertion angle 270°
Pattern 3 (80, 5) Rotation insertion angle 90°
Pattern 4 (85, -75) Rotation insertion angle 180°

(1) Input the offset data in X and Y positions of steps 1 to 4.

STEP No.	X (ABS)	Y (ABS)	Z No.	ROT	INSERT WIDTH	TACT	NO INS	S&R	SKIP STEP	RECOG	BAD MARK	FIX	COMMENT
1	0	0	1	0	0	0	0	1	0	0	0	0	
2	5	-80	1	0	0	0	0	31	0	0	0	0	
3	80	5	1	0	0	0	0	11	0	0	1	0	
4	85	-75	1	0	0	0	0	21	0	0	0	0	
5	10	90	1	270	5.08	0	0	0	0	0	0	0	
6	30	70	2	180	5.08	0	0	0	0	0	0	0	

(2) Input the S&R code corresponding to the desired step mode and rotation in the steps 1 to 4.
Pattern 1: 1
Pattern 2: 31
Pattern 3: 11
Pattern 4: 21

2. Input the X and Y coordinates of points A and B of pattern 1.
(1) Input the coordinate data of points A and B of pattern 1 in step Nos. 5 and 6.
(2) The coordinate data of insertion points on patterns 2 to 4 are not necessary since block Nos. 2 to 4 contain the offset data of these patterns.

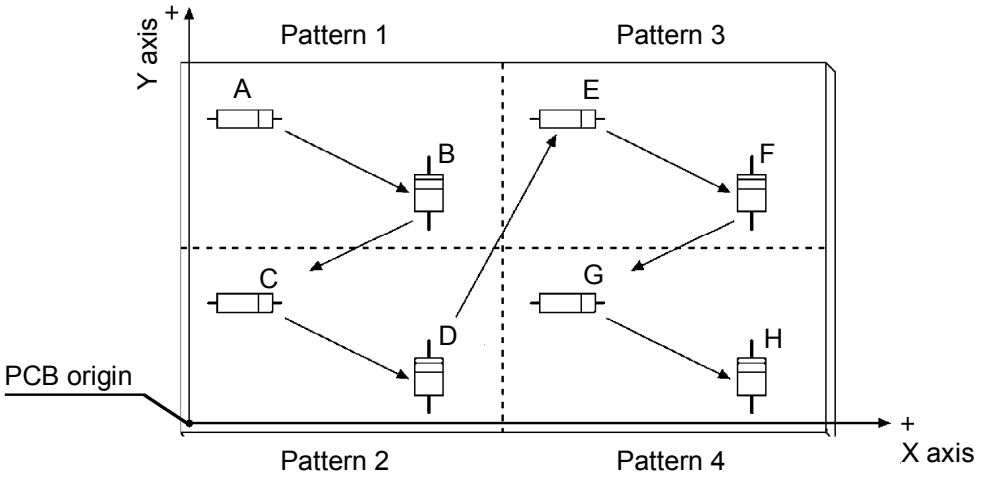
=HINT=

Input 'PC' (Polar coordinate Conversion) at the beginning of COMMENT in the part data. This will convert the pattern rotation 180° to 0° and 270° to 90° automatically, thus reducing possible loss of insertion tact time.

=📖= Refer to 'Prg. / PART DATA CREATION / Setting Items of Part Data / Creating the Part Data'.

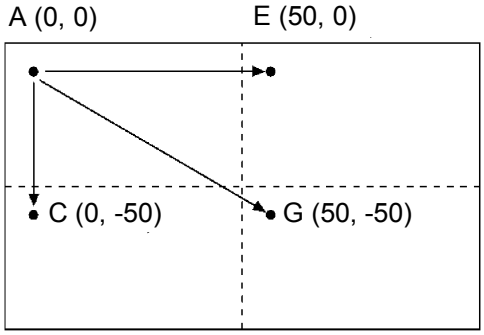
■ Pattern Repeat

Ex.) A program to insert all the parts on one pattern before proceeding to the next pattern of the 4-sectioned PCB.



Position	No.	Angle	Pattern
A (Xa=10, Ya=90)	1	0°	Pattern 1
B (Xb=40, Yb=60)	2	90°	
C (Xc=10, Yc=40)	1	0°	Pattern 2
D (Xd=40, Yd=10)	2	90°	
E (Xe=60, Ye=90)	1	0°	Pattern 3
F (Xf=90, Yf=60)	2	90°	
G (Xg=60, Yg=40)	1	0°	Pattern 4
H (Xh=90, Yh=10)	2	90°	

1. Determine the offset value of each pattern from pattern 1.



Pattern 1 (0, 0)
Pattern 2 (0, -50)
Pattern 3 (50, 0)
Pattern 4 (50, -50)

(1) Input the offset data in X and Y positions of step Nos. 1 to 4.
(2) Input 2 (pattern repeat) in the S&R designation of step Nos.1 to 4.

2. Input the X and Y coordinates of points A and B of pattern 1.
(1) Input the coordinate data of points A and B of pattern 1 in step Nos. 5 to 6.
(2) The coordinate data of insertion points on patterns 2 to 4 are not necessary since step Nos. 2 to 4 contain the offset data of these patterns.

STEP No.	X (ABS)	Y (ABS)	Z No.	ROT	INSERT WIDTH	TACT	NO INS	S&R	SKIP STEP	RECOG	BAD MARK	FIX	COMMENT
1	0	0	1	0	0	0	0	2	0	0	0	0	
2	0	-50	1	0	0	0	0	2	0	0	0	0	
3	50	0	1	0	0	0	0	2	0	0	1	0	
4	50	-50	1	0	0	0	0	2	0	0	0	0	
5	10	90	1	0	5.08	0	0	0	0	0	0	0	
6	40	60	2	90	5.08	0	0	0	0	0	0	0	

3.2. Creating and Editing the NC Data

D79PEC-43-080-A0

3.2.1 Creating the NC Data

This section describes the procedure for manually inputting new NC data. Prepare a completed NC data sheet.

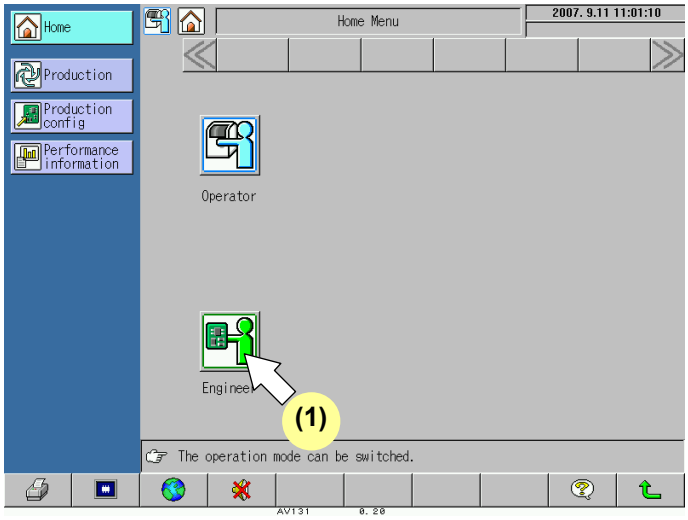
Main Procedure

1	Turn the power ON (“Home Menu” screen).
2	Change to the Operator level.
3	Call up the Data Modify Menu screen.
4	Select the NC data.
5	Call up the NC Data Input screen.
6	Input the program offset.
7	Input the multi-origin.
8	Insert steps.
9	Input in parameters.
10	Complete the NC data input

■ Procedure for Creating the NC Data

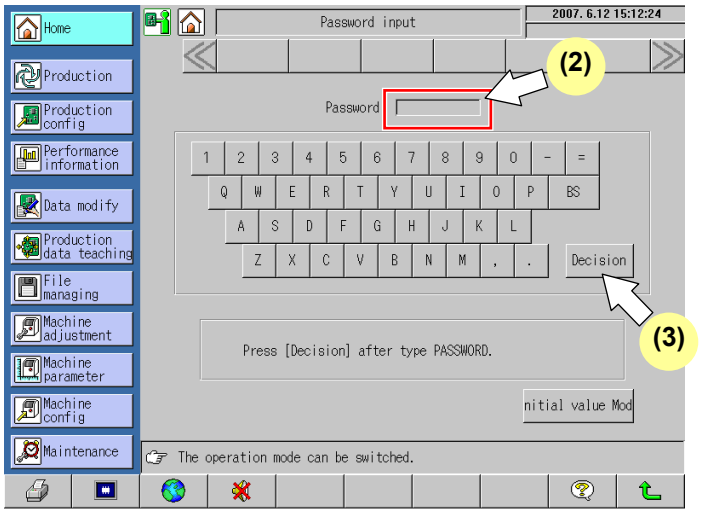
1. Turn the power ON.

- The “Home Menu” screen appears on the display.



2. Change to the Engineer level.

- Change to the Operator level at which data can be modified.
  (1) Press [Engineer] .
  (2) Enter the password.
  (3) Press [Decision] .

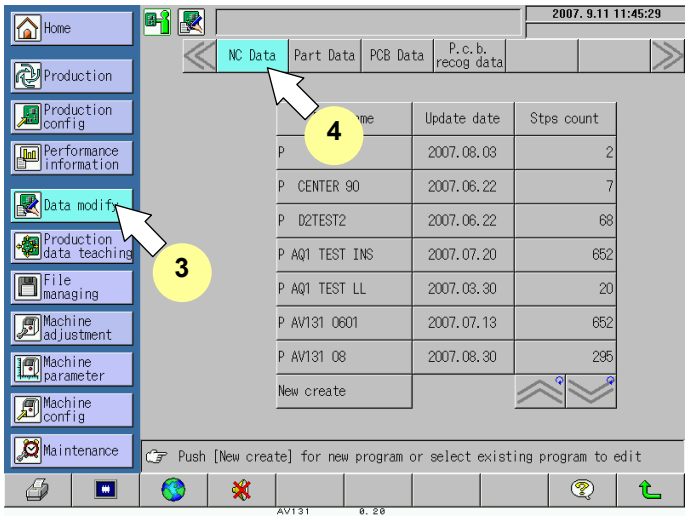


3. Call up the Data Modify screen.

- Press [Data Modify] on the “Home Menu” screen.

4. Select the NC data.

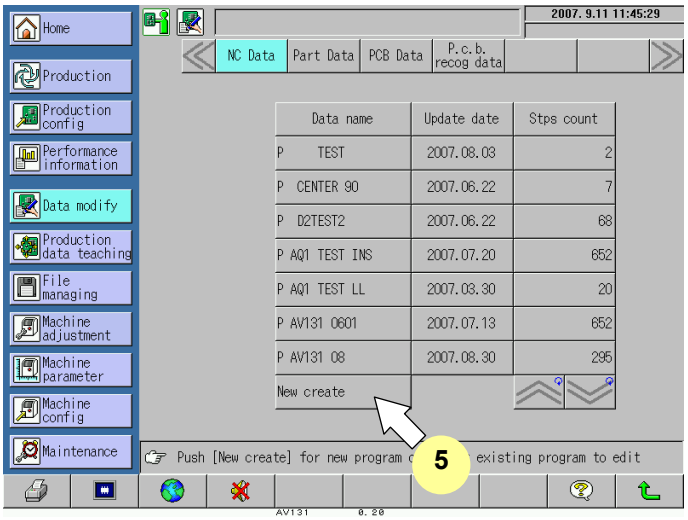
- Press [NC Data] .



5. Create the NC data.

Ex.) Create a new NC data named as “P007”.

- Press [New create] .

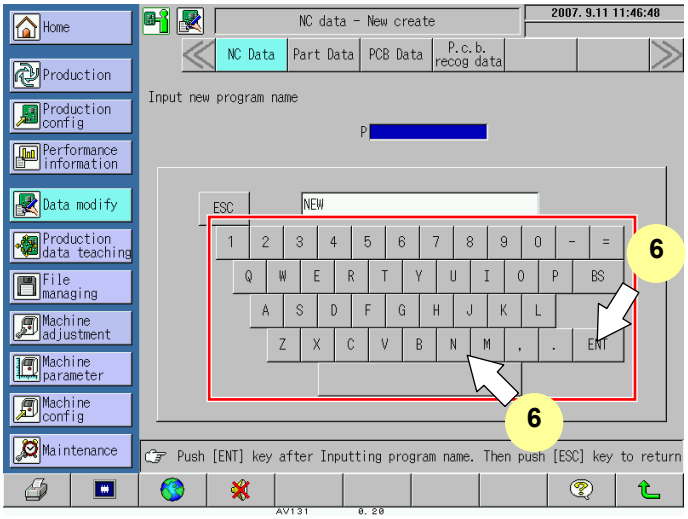


6. Input an NC data name.

- Input an NC data name.
  Ex.) Input “007”, then press [ENT] to set the data.

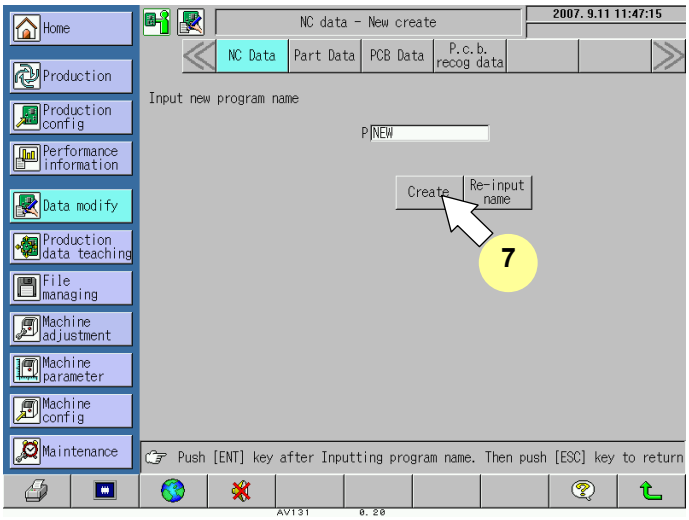
=HINT=

The input window closes automatically.



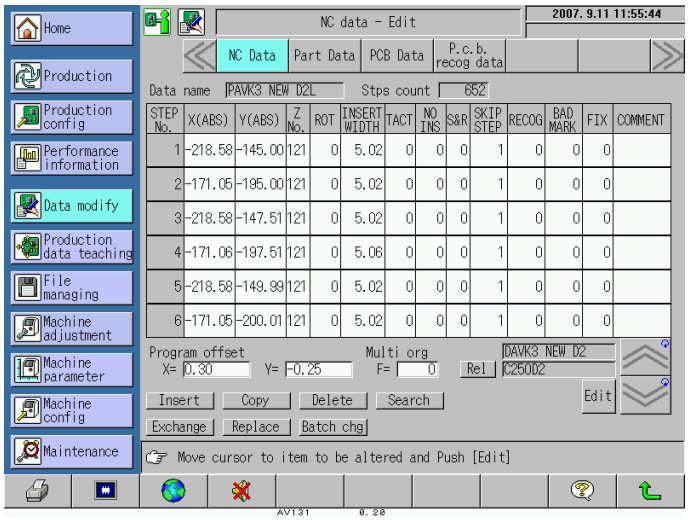
7. Call up the NC Data Input screen.

- Pressing [Create] to switch to the NC Data Input screen.



8. Call up the NC Data Input screen.

- The display shows the NC Data Input screen is selected in step 5. above.

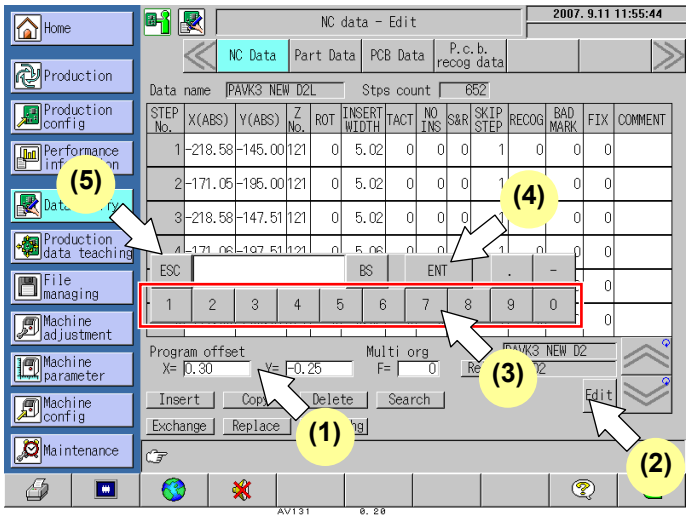


9. Input the program offset.

- Input the program offset.
  Input the offset for the X and Y axes in order.

  (1) Press the input field of X (Y input field for Y axis).
  (2) Press [Edit] .
  Numeric keypad appears.
  (3) Press the number keys to input the offset value.
  (4) Press [ENT] to register the data.
  (5) To finish the input, press [ESC] to close the numeric keypad.

  Repeat the steps (1) to (4) to input for the Y axis.

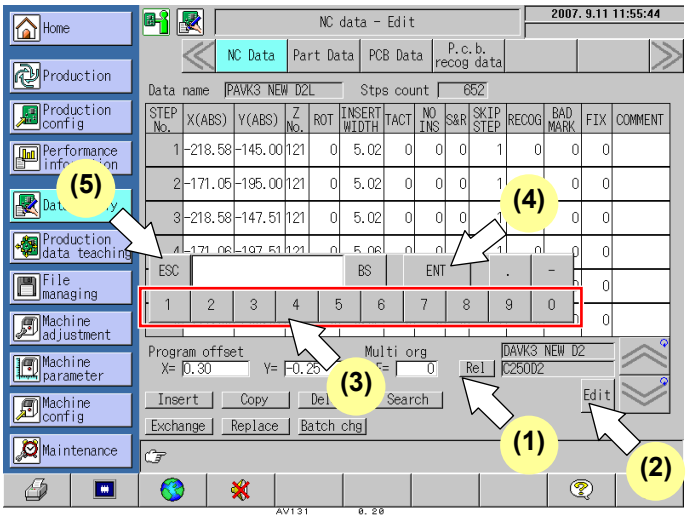


10. Input the multi-origin.

- Input the multi-origin value.

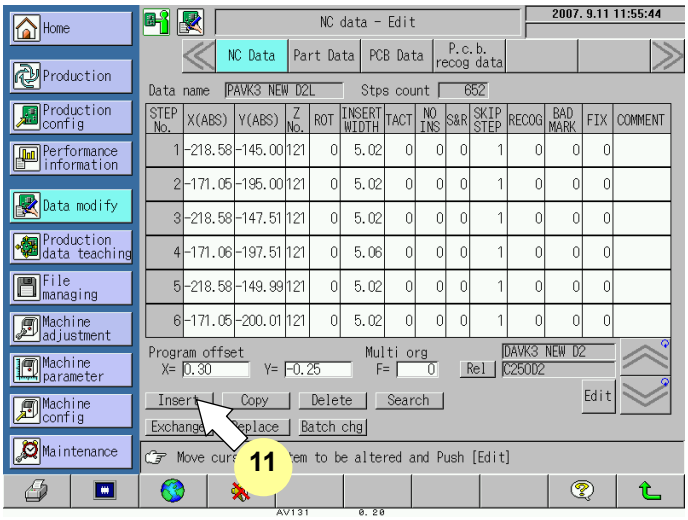
  (1) Press the multi-origin field.
  (2) Press [Edit] .
  Numeric keypad appears.
  (3) Enter the multi-origin value via the keypad.
  (4) Press [ENT] to register the data.
  (5) Press [ESC] to close the numeric keypad.

  =HINT=
  When the multi-origin function is not used, input '0' or '1'.



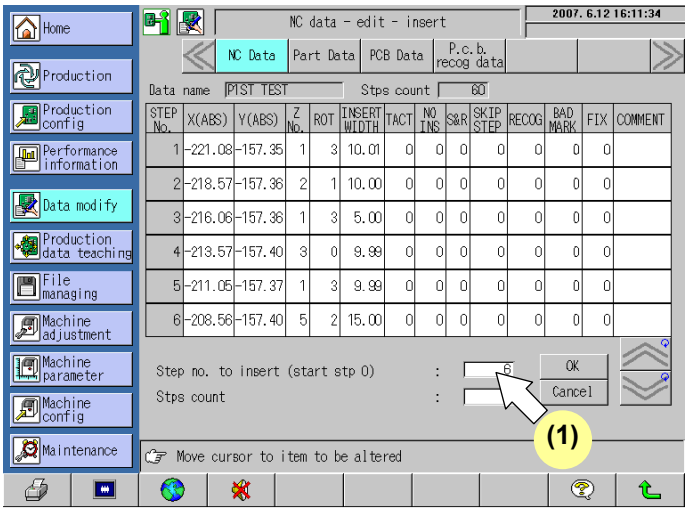
11. Insert steps.

- Press [Insert] .



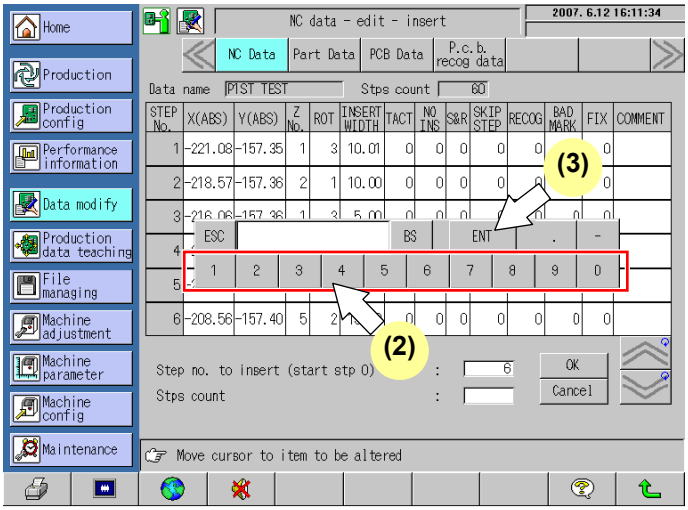
- The edit dialog for insert appears.

  (1) Press the input field of step No. Numeric keypad appears.



  (2) Enter the step No. at which you wish to insert new steps.

  (3) Press [ENT] to register the data.



- Input the number of steps to be inserted.

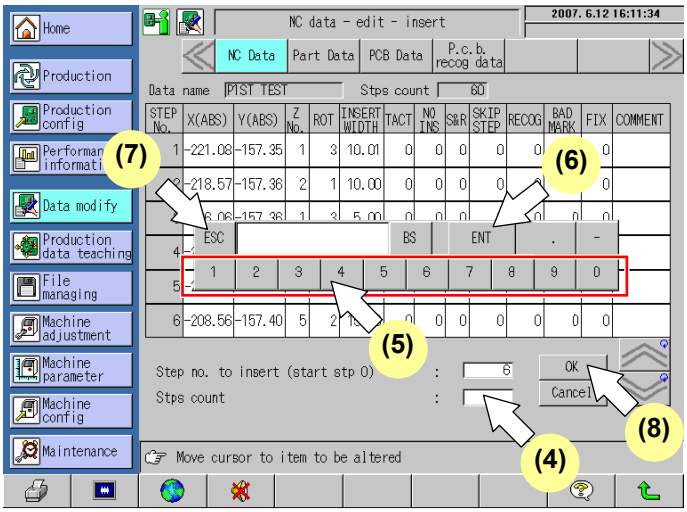
  (4) Press the input field of step count.

  (5) Input the number of steps to insert.

  (6) Press [ENT] to register the data.

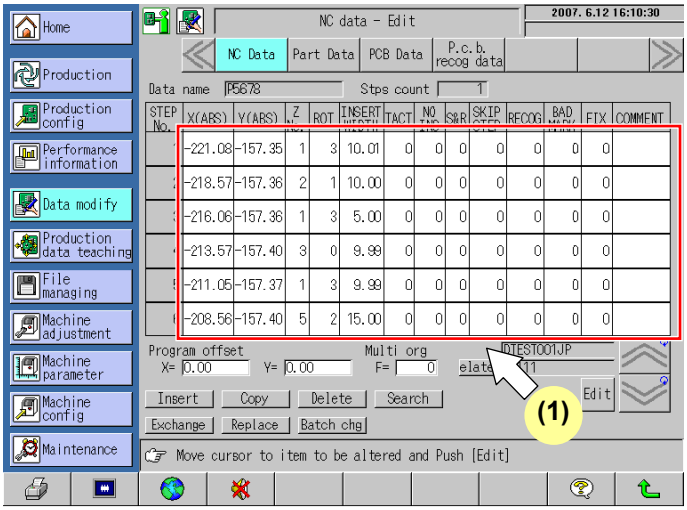
  (7) Press [ESC] to close the keypad window.

  (8) Press [OK] .
  New steps as many as the specified step count are inserted to the step specified at (2).



12. Input in parameters.

(1) Press the parameter input field.
Numeric keypad appears.

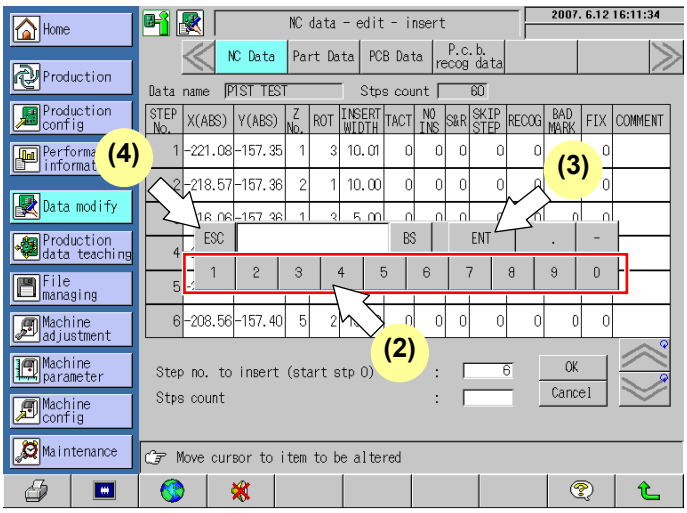


(2) Input data for the parameters of each step.

(3) Press [ENT] to register the data.

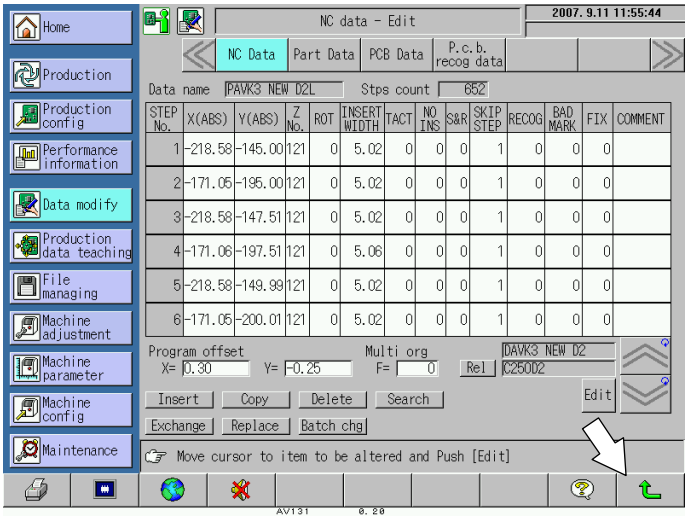
- Repeat the procedure from (1) to continue to input the parameters.

(4) Press [ESC] to close the keypad window.



13. Complete the NC data input.

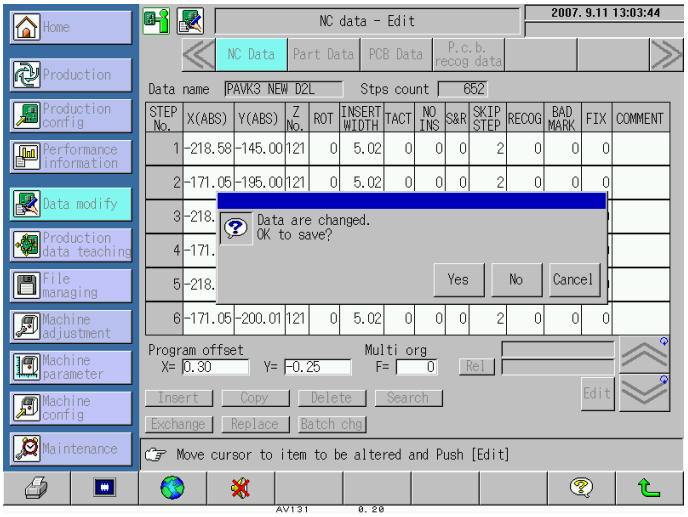
- After all the parameters are input, press .
  NC Data Selection screen is displayed again.



14. Confirm the save.

The confirmation dialog appears. Press the desired button.

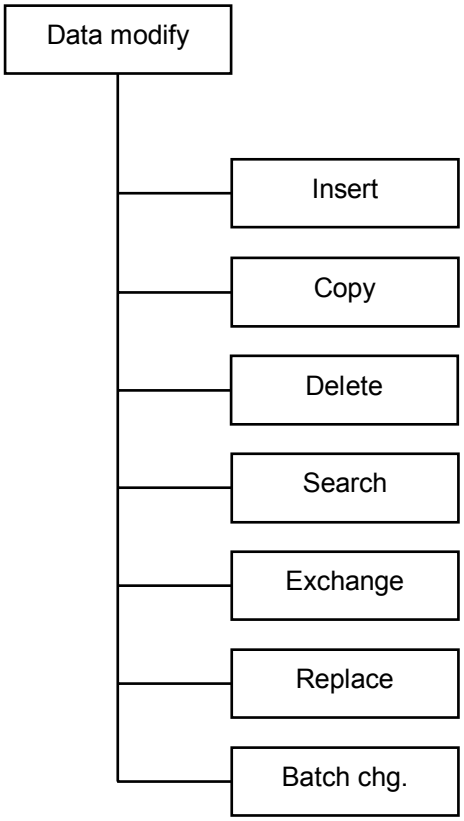
- [Yes]
  Pressing this button saves the data and closes the dialog.
- [No]
  Pressing this button discards the changes you made and closes the dialog.
- [Cancel]
  Pressing this button continues the editing.

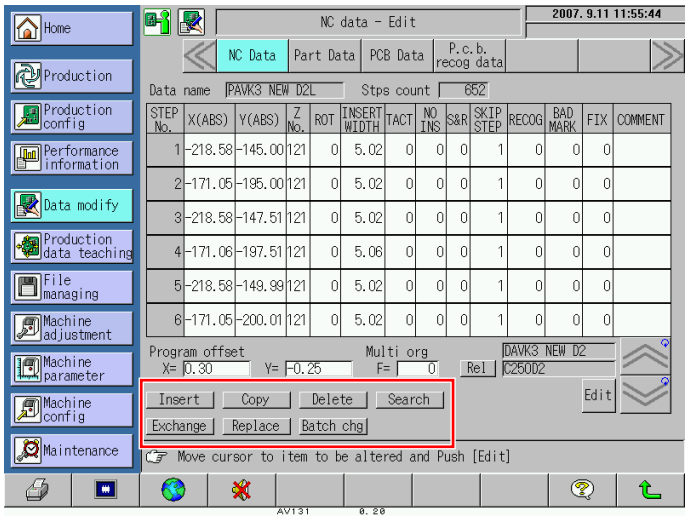


3.2.2 Editing the NC Data

This section describes the editing functions available for editing the NC data. The data editing functions include ‘insert’, ‘copy’, ‘delete’, ‘search’, ‘exchange’, ‘replace’, and ‘batch change’.

- Access the NC data - Edit screen via the Data modify menu.

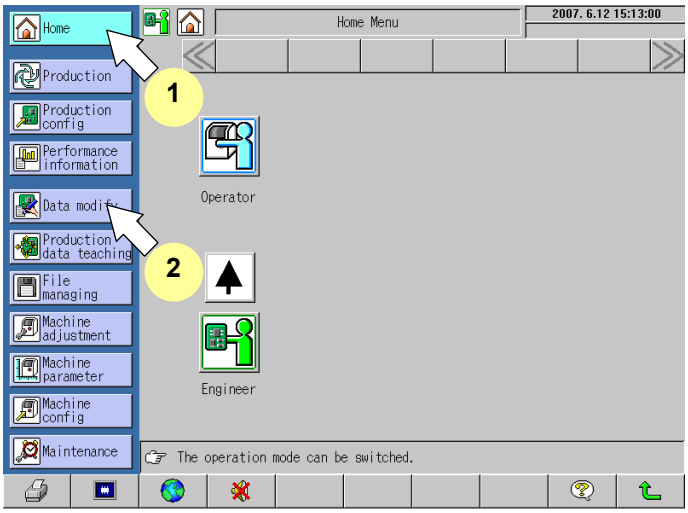




■ Procedure for Editing the NC Data

1. Call up the “Home Menu” screen.
- Press [Home] .

2. Call up the “Data Modify Menu” screen.
- Press [Data modify] .

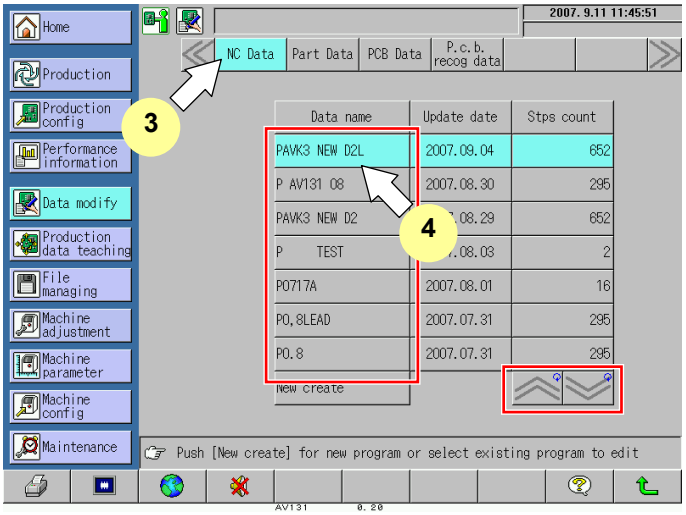


3. Call up the “NC Data Modify Menu” screen.
- Press [NC Data] .

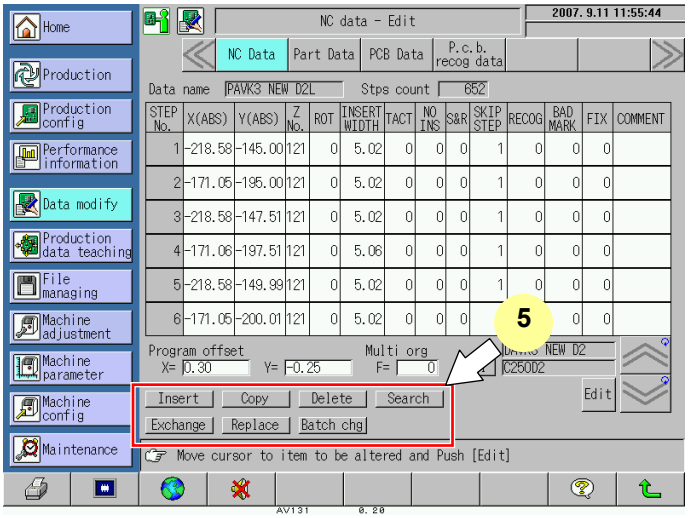
4. Designate the NC data.
Ex.) Press “P00007”.
- Select the data to modify from the list of data names displayed on the screen.

=HINT=

Use the to scroll through the list of data displayed on the screen.



5. Select the editing functions.
- Select the desired function.



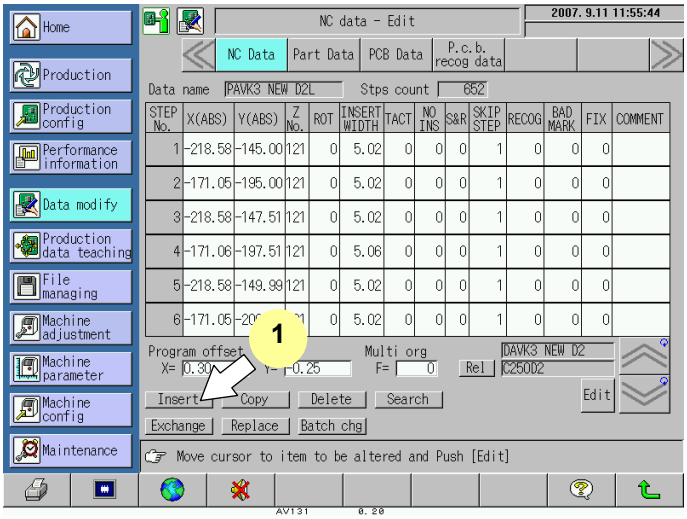
◆ Insertion Procedure

This is used to add the NC data steps to the following positions:

- Between steps of the NC data
- After the last step of the NC data
- Before the first step of the NC data

1. Select the insertion function.

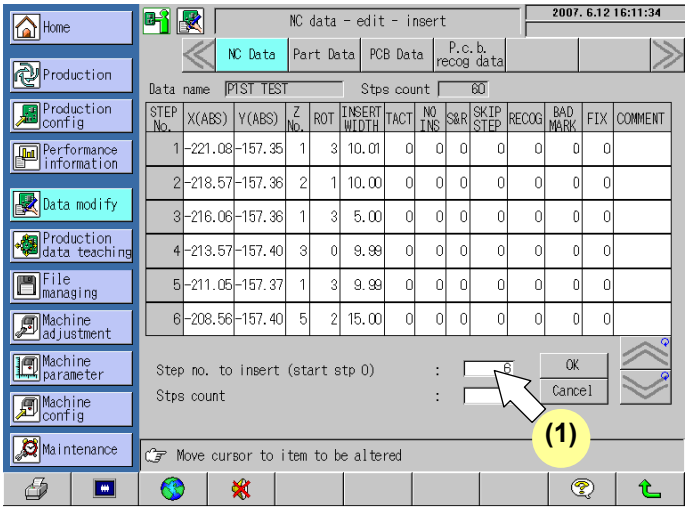
- Press [Insert] .



2. Designate the position and number of steps to be inserted.

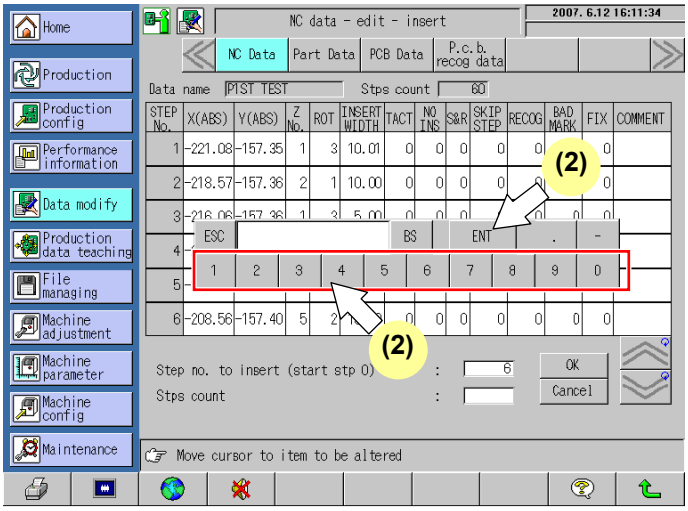
Ex.) To add nine steps between steps 3 and 4;

(1) Press the input field of step No.
Numeric keypad appears.



(2) Input the step No. after which new steps are to be inserted.

Ex.) Press '3', and [ENT] to set the data.

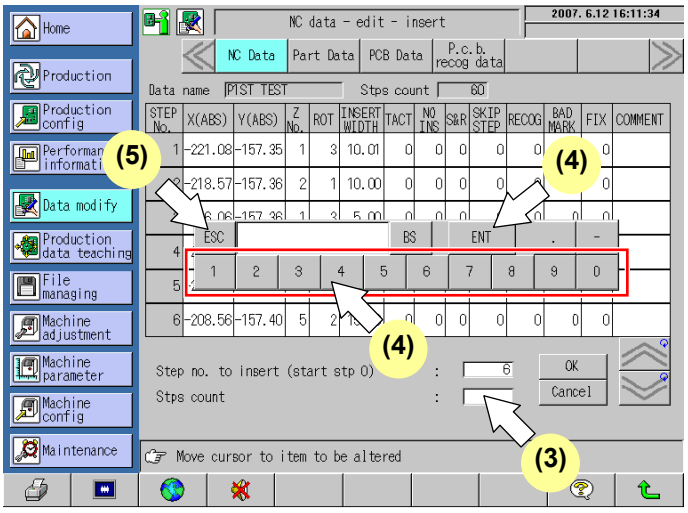


(3) Press the input field of steps count.

(4) Input the number of steps to insert.
Ex.) Press '9', and [ENT] to set the data.

(5) Press [ESC] to close the keypad window.

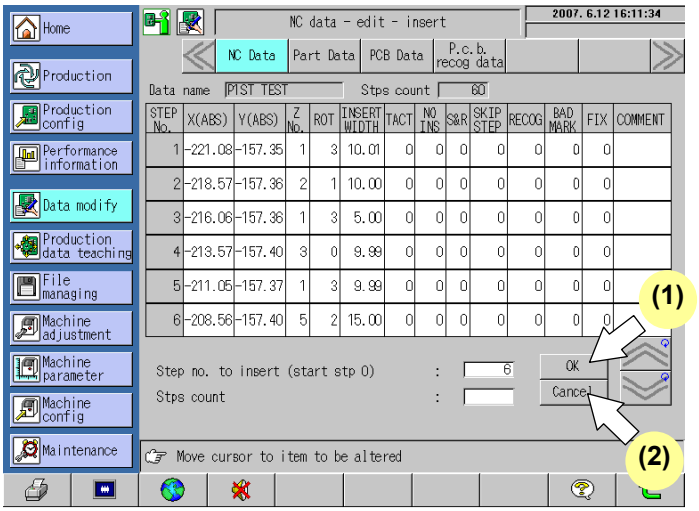
=HINT=
To add steps to the top of the existing steps, designate '0' for the insertion position.



3. Execute insertion.

(1) Press [OK] to execute insertion.
The screen returns to the NC data – Edit screen automatically.
Ex.) '9' steps are inserted between step '3' and step '4'.

(2) Press [Cancel] to cancel the insertion.
The screen returns to the NC data – Edit screen automatically.

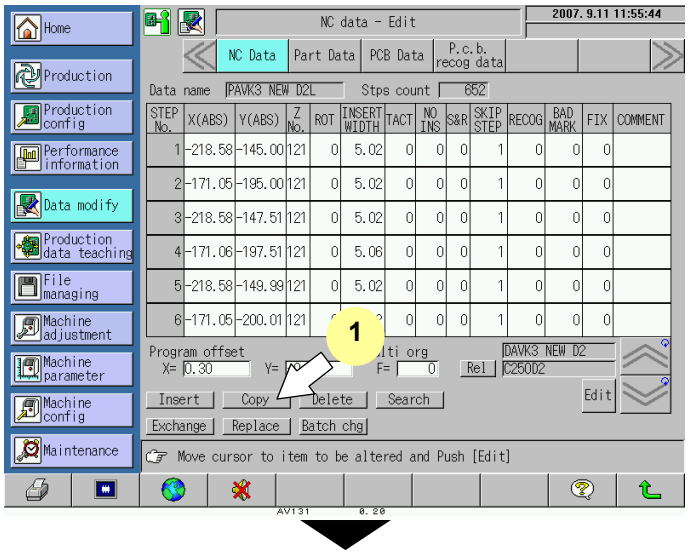


◆ Copy Procedure

This is used to copy the data in the range of designated steps to the position after the target step.

1. Select the copy function.

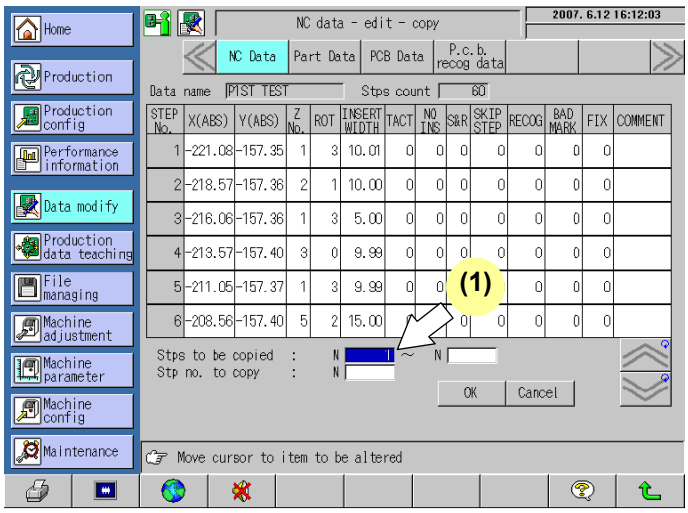
- Press [Copy] .



2. Designate the source and the target steps.

Ex.) To copy steps N1 through N6, to the position after N8;

(1) Press the input field of steps to be copied.
Numeric keypad appears.

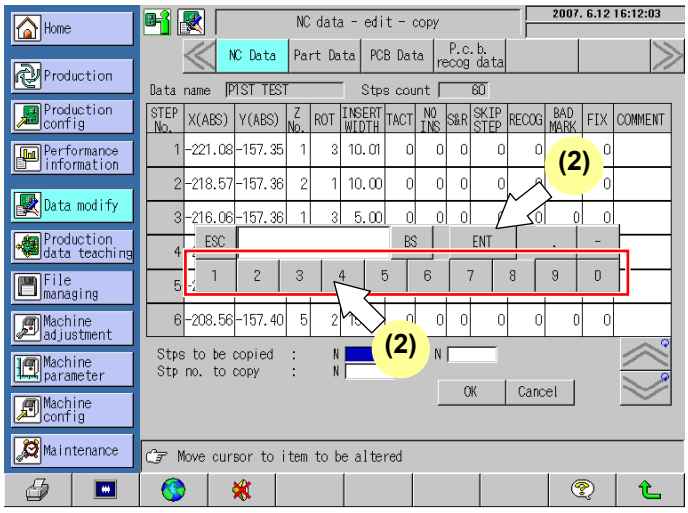


(2) Input the first and last step Nos. of the source range.

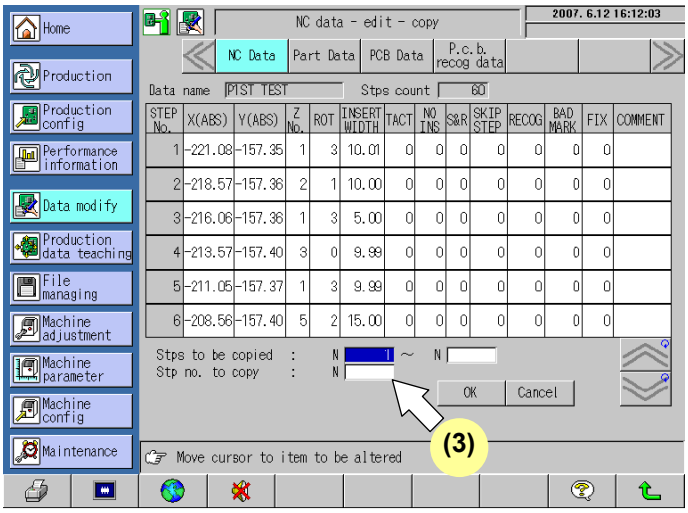
Ex.) Press '1' and [ENT] to set the first step.
Press '6' and [ENT] to set the last step.

=HINT=

Designate the range of the step numbers in ascending order.

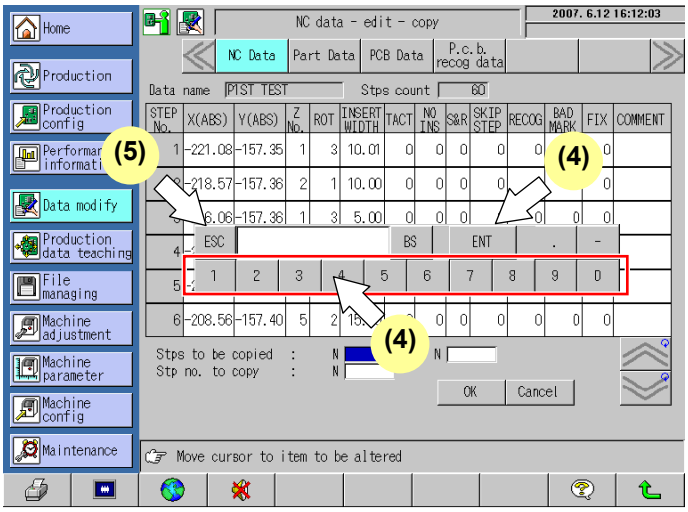


(3) Press the input field of the step No. to copy.



(4) Input the target step No.
Ex.) Press '8', and [ENT] to set the data.

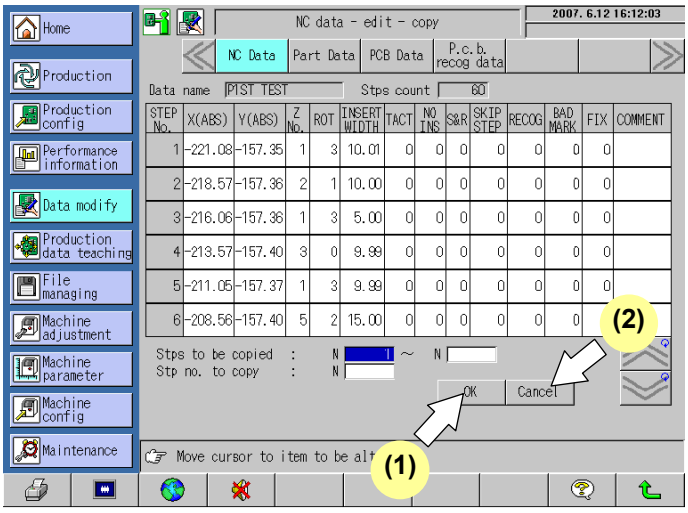
(5) Press [ESC] to close the keypad window.



3. Execute copying.

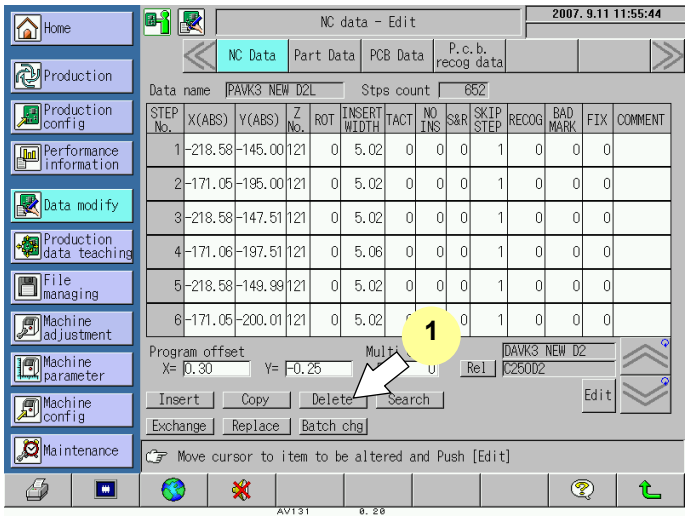
(1) Press [OK] to execute copying.
The screen returns to the NC data – Edit screen automatically.
Ex.) The contents of steps N1 to N6 are copied after step N8.

(2) Press [Cancel] to cancel the copying.
The screen returns to the NC data – Edit screen automatically.



◆ Deletion Procedure

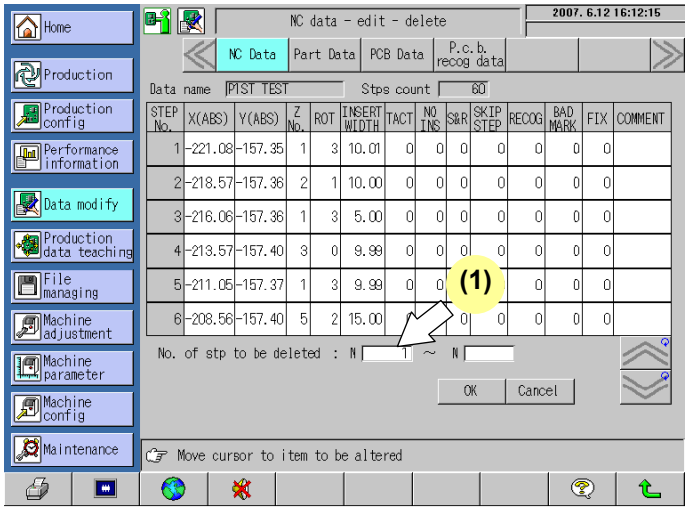
1. This is used to delete unnecessary steps.
- Press [Delete] .



2. Input the range of steps to be deleted.

Ex.) To delete steps N1 to N5;

(1) Press the input field of step range to be deleted.
Numeric keypad appears.



(2) Input the first and last steps of the target range.

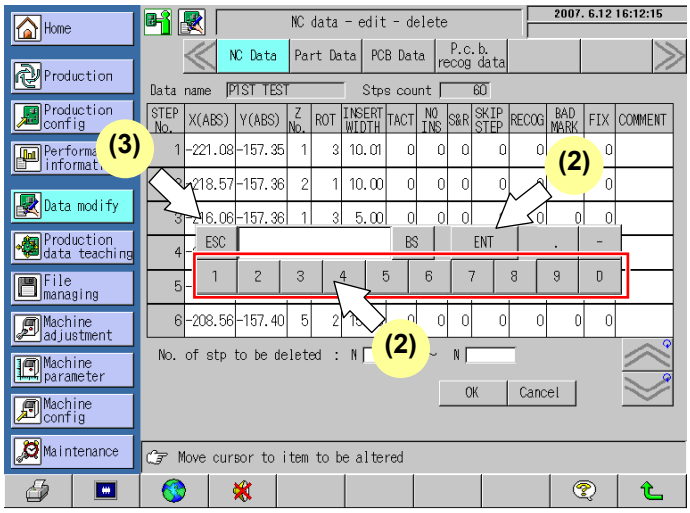
Ex.) Press '1', and [ENT] to set the data.
Press '5', and [ENT] to set the data.

=HINT=

When deleting a single step;

Ex.) To delete step 3, input 3 for the first step and leave the last step blank, or input 3 twice.

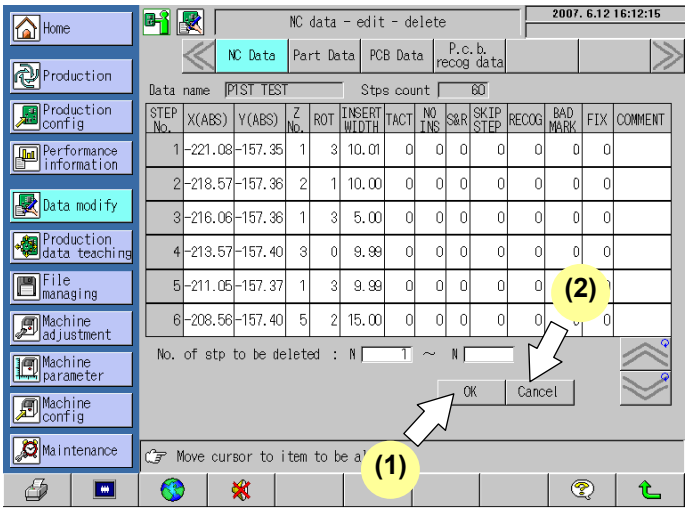
(3) Press [ESC] to close the keypad window.



3. Execute deletion.

(1) Press [OK] to execute deletion.
The screen returns to the NC data – Edit screen automatically.
Ex.) Steps N1 to N5 are deleted.

(2) Press [Cancel] to cancel deletion.
The screen returns to the NC data – Edit screen automatically.

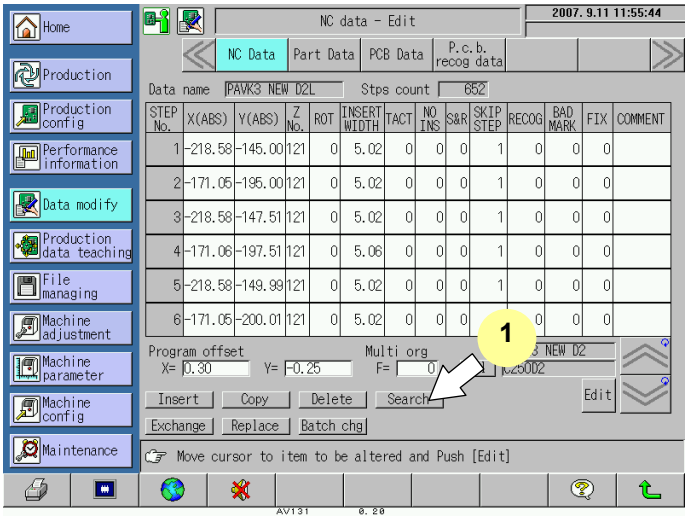


◆ Search Procedure

This function is used to find the step to be changed when changing the NC data etc.

1. Select the search function.

- Press [Search] .

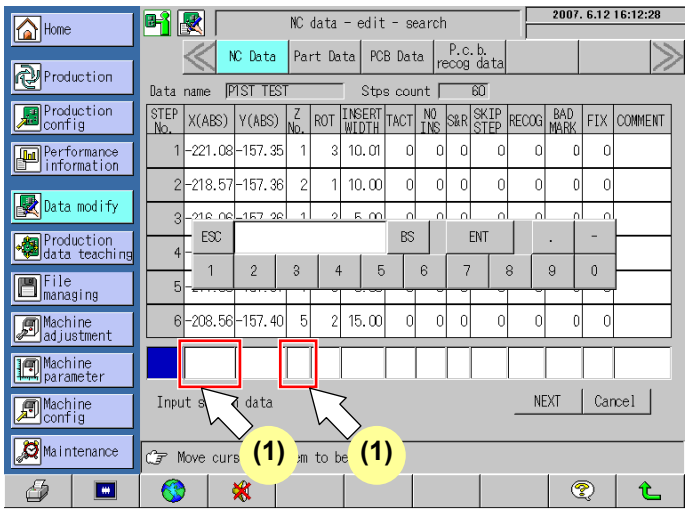


2. Input the data to be searched for.

- Input the data to be searched in the applicable field.

Ex.) To search for the step of coordinate X of 10.00 mm and Z No. 1;

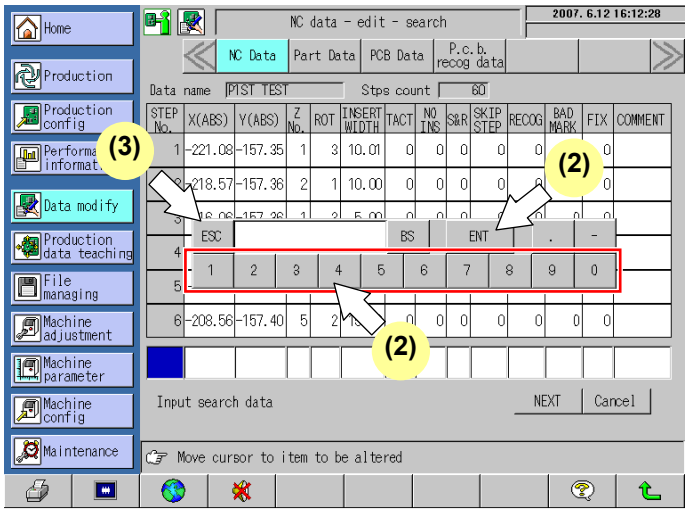
(1) Press the input field of a parameter.
Numeric keypad appears.



(2) Input the data to be searched for in parameter.
Ex.) Designate X(ABS) field and input '10.00', and press [ENT] to set the data.
Designate Z No. field and input '1', and press [ENT] to set the data.

=HINT=
This function searches for the data that match all conditions specified.

(3) Press [ESC] to close the keypad window.

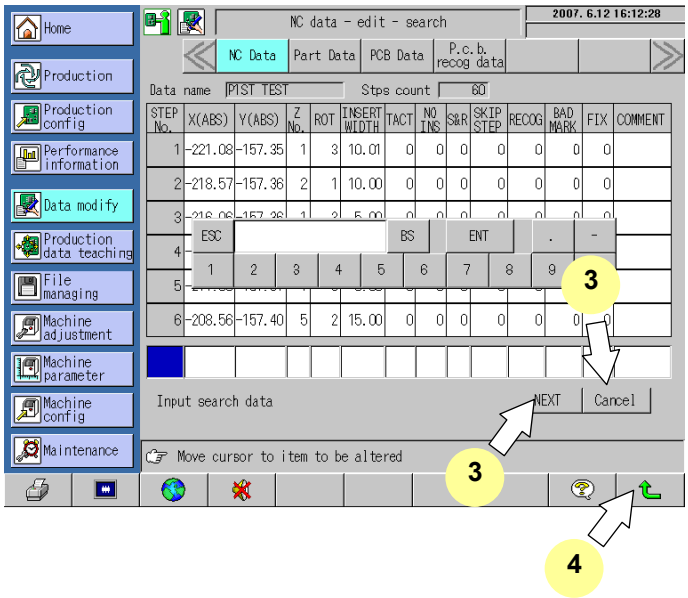


3. Start searching.

- Press [NEXT] to start searching.
  (Keep pressing [NEXT] to search for the another match.)
- If there is no match, the guidance window appears to indicate that 'Searching have reached to the end.'
- Press [Cancel] to cancel searching.

4. Finish searching.

- Press to finish searching.
  The screen returns to the NC data – Edit screen.

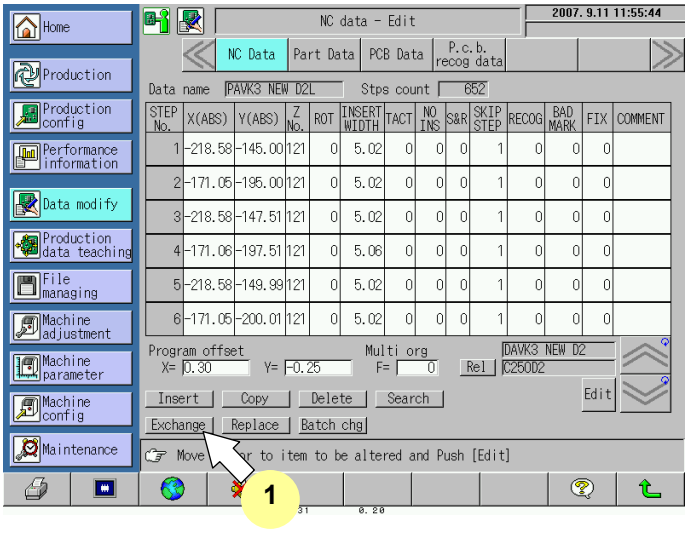


◆ Exchange Procedure

This is used to exchange data between two designated steps.

1. Select the exchange function.

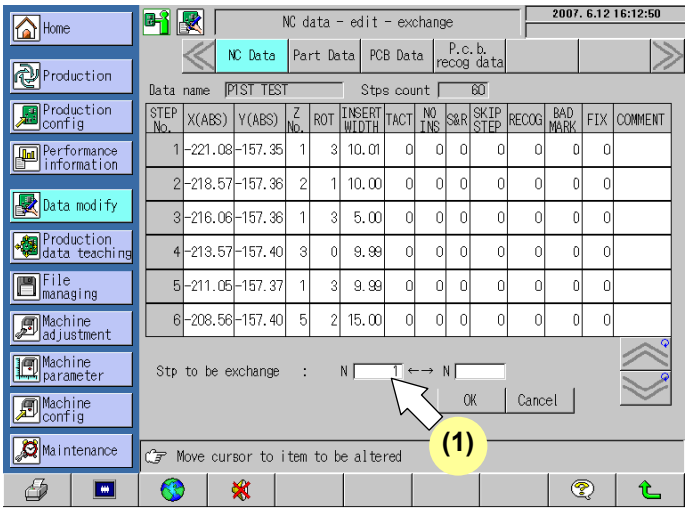
- Press [Exchange] .



2. Input steps to be exchanged.

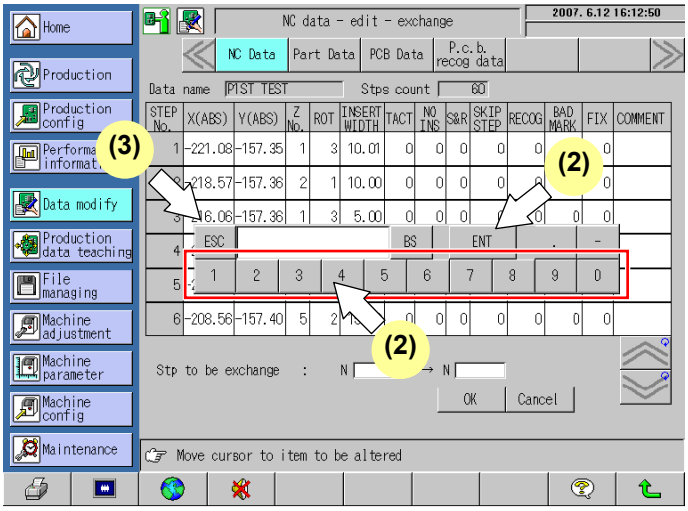
Ex.) To exchange step N1 and step N2;

(1) Press the input field of step to be exchanged.
Numeric keypad appears.



(2) Input the steps to be exchanged.
Ex.) Press '1', and [ENT] to set the data.
Press '2', and [ENT] to set the data.

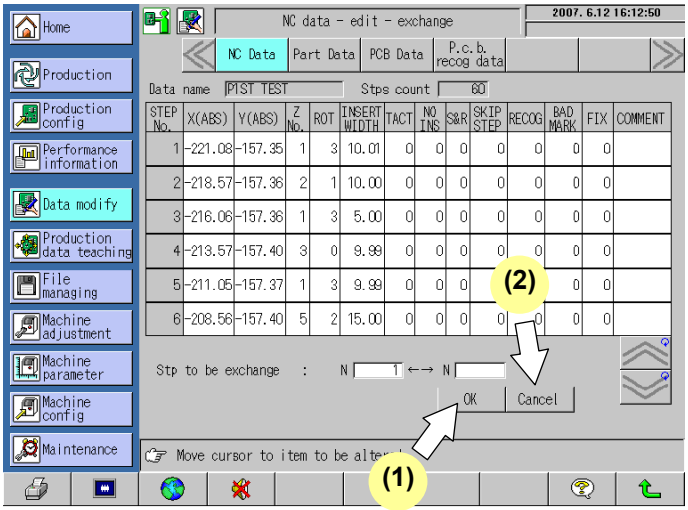
(3) Press [ESC] to close the keypad window.



3. Execute exchanging.

(1) Press [OK] to execute exchanging.
The screen returns to the NC data – Edit screen automatically.
Ex.) N1 and N2 are exchanged.

(2) Press [Cancel] to cancel exchanging.
The screen returns to the NC data – Edit screen automatically.

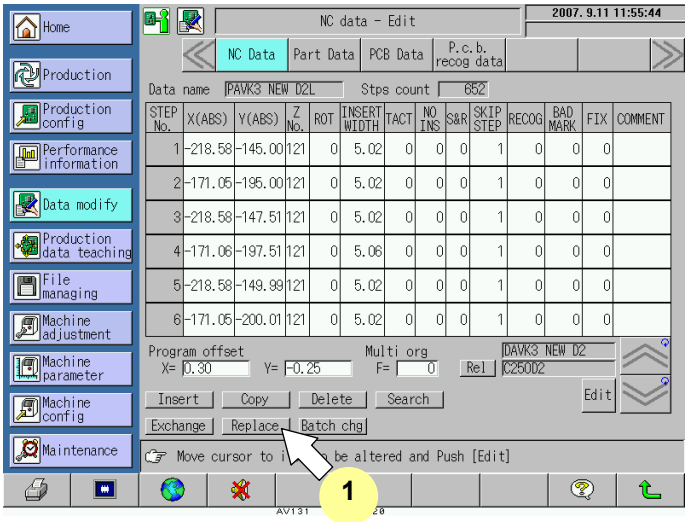


◆ Replace Procedure

This function replaces the designated parameter data with new data.
Use it when applying the change of particular parameter to the entire NC data.

1. Select the replace function.

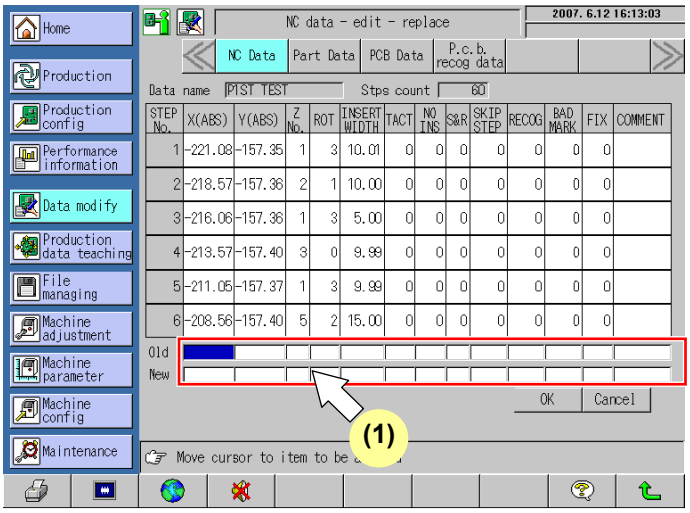
- Press [Replace] .



2. Input the data to be replaced.

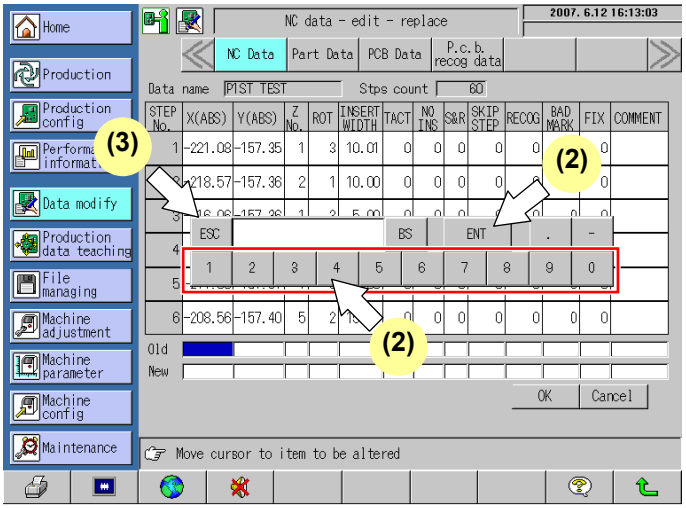
Ex.) To replace all the Z Nos. of the steps with Z No. 1 to No. 5;

(1) Press the input field of Old / New data.
Numeric keypad appears.

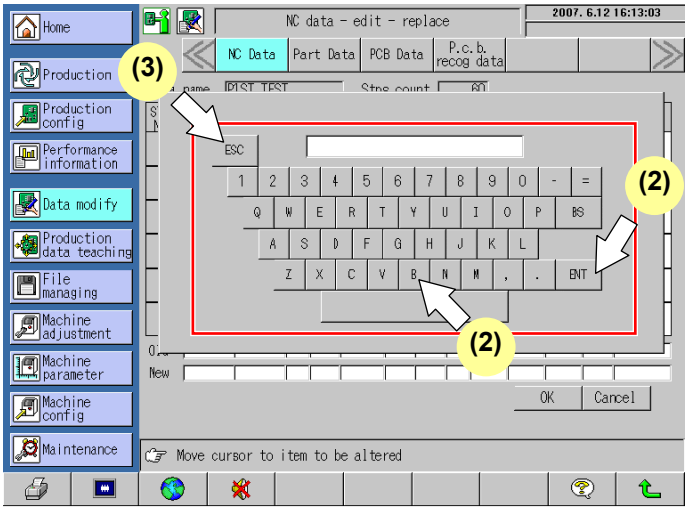


(2) Designate the old/new data.
Ex.) Designate the Old field under Z No., input '1', then press [ENT] to set the data.
Designate the New field under Z No., input '5', then press [ENT] to set the data.

(3) Press [ESC] to close the keypad window.



When inputting numeric

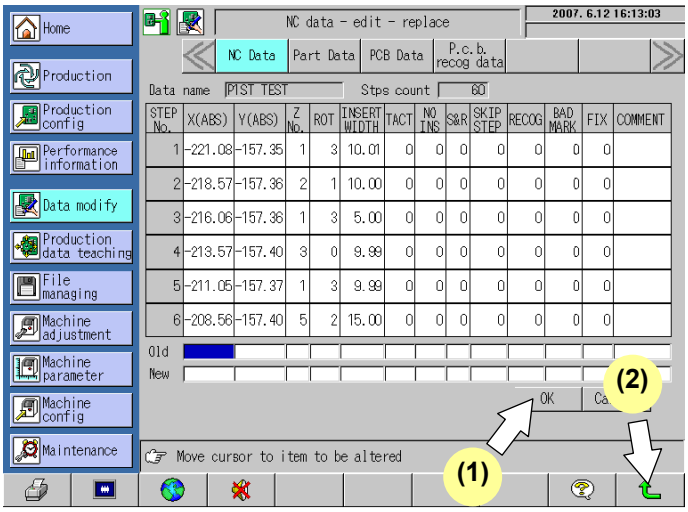


When inputting letters and symbols

3. Execute replacement.

(1) Press [OK] to execute replacement.
The screen returns to the NC data – Edit screen automatically.

(2) Press [Cancel] to cancel replacement.
The screen returns to the NC data – Edit screen automatically.

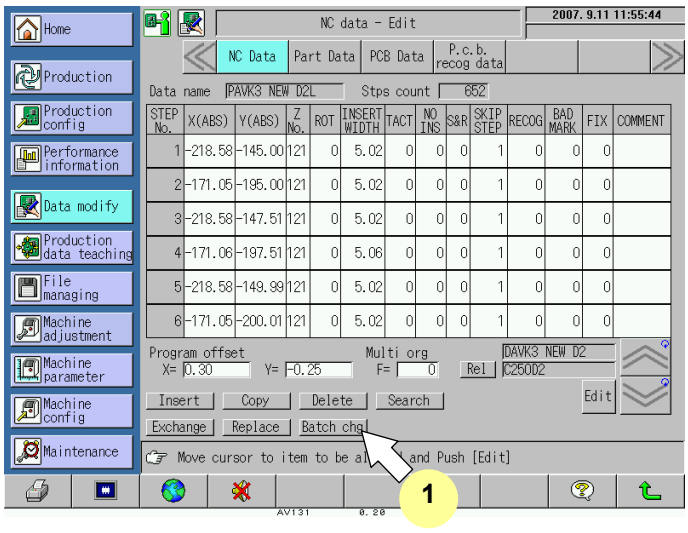


◆ Batch Change Procedure

This function changes the value of the specified parameter within the designated range all at once.
Use it when changing the specified parameter of NC data within the consecutive steps at a time.

1. Select the batch change function.

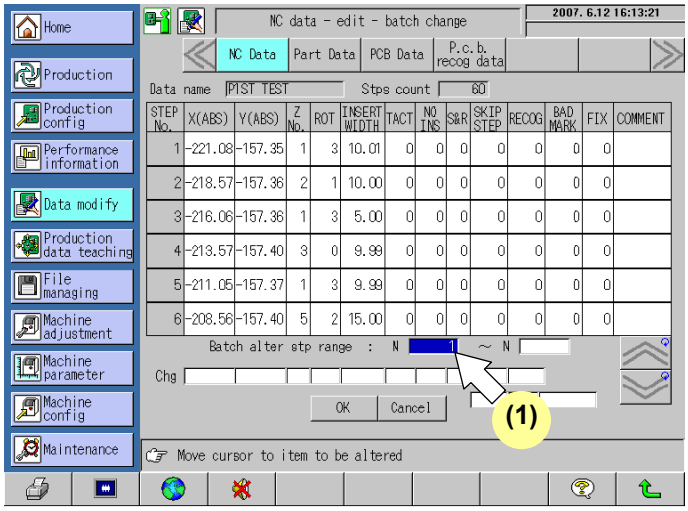
- Press [Batch chg] .



2. Designate the range of target steps and data to be changed.

Ex.) To change all INSERT WIDTH data between N5 and N6 steps to 10.08;

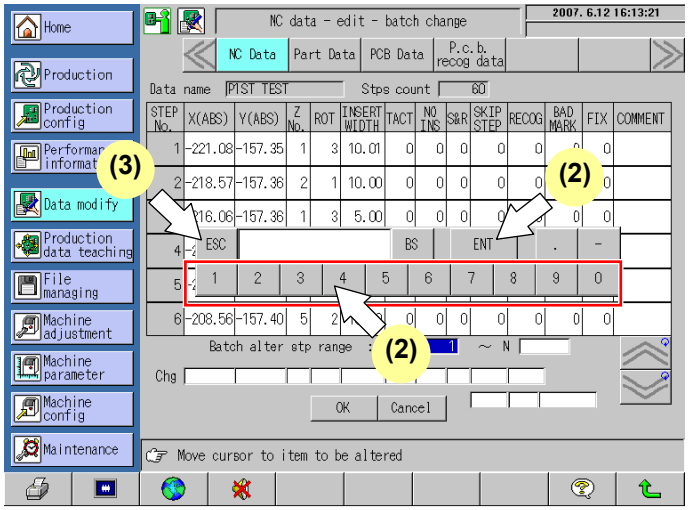
(1) Press the input field of batch alter step range.
Numeric keypad appears.



(2) Designate the data to be changed.

Ex.) Input the range of batch change.
Press '5', and [ENT] to set the first step.
Press '6', and [ENT] to set the last step.
Input the data to be changed.
Designate the INSERT WIDTH field to be changed. Input "1008", and press [ENT] to set the data.

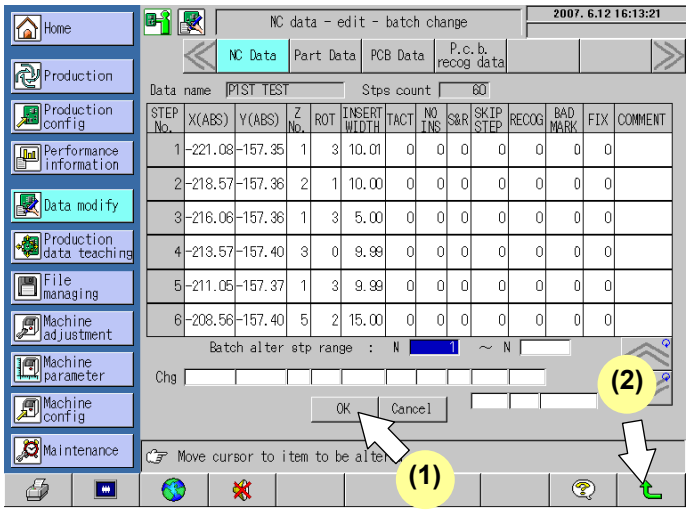
(3) Press [ESC] to close the keypad window.



3. Execute batch change.

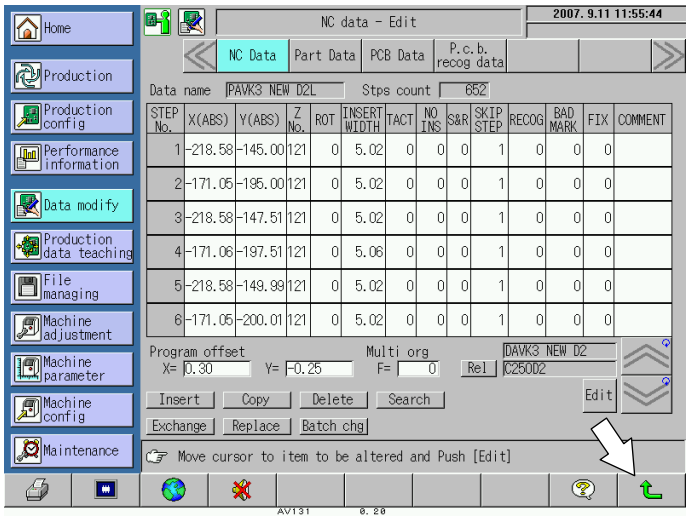
(1) Press [OK] to execute batch change.
The screen returns to the NC data – Edit screen automatically.

(2) Press [Cancel] to cancel batch change.
The screen returns to the NC data – Edit screen automatically.



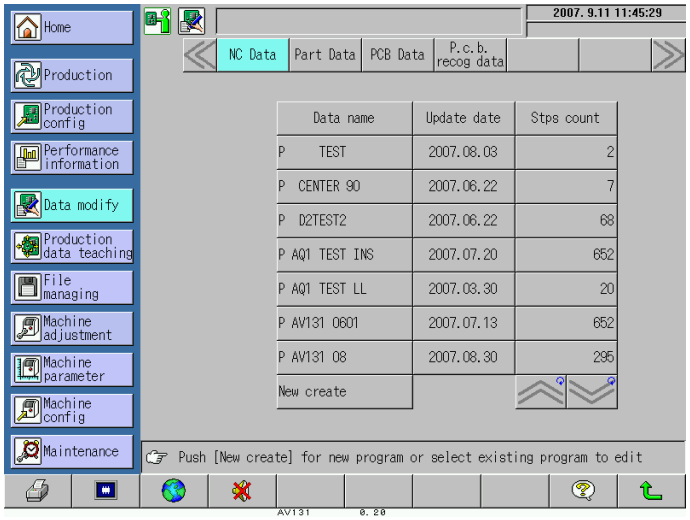
◆ Exiting the Editing Mode

1. Press to exit the editing mode.



2. Confirm the save.
The confirmation dialog appears. Press the desired button.

- [Yes]
  Pressing this button saves the data and closes the dialog.
- [No]
  Pressing this button discards the changes you made and closes the dialog.
- [Cancel]
  Pressing this button continues the editing.



4. PART DATA CREATION

This chapter describes the procedure for creating the part data.

4.1. Setting Items of the Part Data

D79PEC-4R-040-A0

4.1.1 Parameters

The part data is the data that indicates the types of parts and their arrays set on the feeder carriage (Z axis). The part data includes the following parameters.

Parameter	Input range / Standard Setting	Details
Z No.	● 120 inputs 1 to 121: For axial lead parts 121: For jumper wires ● 60 inputs 1 to 60: For axial lead parts 121: For jumper wires	Designate the location of feeder set on the feeder carriage.
COMMENT	Within 16 alphanumeric characters	Name of the part
THICK	0 to 9.99 (mm)	Diameter of the part The same as lead diameter for jumper wire.
LEAD	0 to 9.99 (mm)	Lead diameter of the part
INS LEVEL	Not used	Not used with this machine.
SPARE	1 to 120	Sets spare feeder for the Z alteration function.
CHUCK	0 = 26 mm pitch Centering function disabled 1 = 52 mm pitch Centering function disabled 2 = 26 mm pitch Centering function enabled 3 = 52 mm pitch Centering function enabled	Designates the pitch (26 mm/52 mm) of the rotary chuck that picks up parts from feeders, and also whether the centering function is enabled for the parts or not.
NO. OF LEAD	Not used	Not used with this machine.
LARGE PARTS	Not used	Not used with this machine.
V-CUT	Not used	Not used with this machine.
PARTS WIDTH	Not used	Not used with this machine.
PARTS LENGTH	Not used	Not used with this machine.

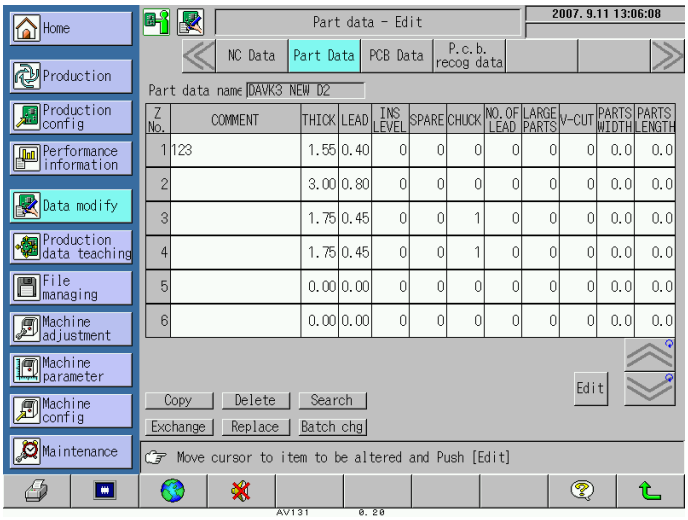
4.1.2 Part Data Edit Screen/Part Data Sheet

■ Example of Part Data Sheet

PART DATA: D

Z No.	COMMENT	THICK	LEAD	INS LEVEL	SPARE	CHUCK	NO.OF LEAD	LARGE PARTS	V-CUT	PARTS WIDTH	PARTS LENGTH
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											

The part data can be input from the Part Data Edit screen. The Part Data Edit screen is shown below. Prepare a part data sheet completed in advance, and input the data to the parameter on the Part data - Edit screen.



4.1.3 Creating the Part Data

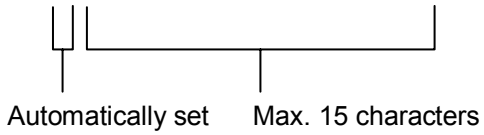
■ Procedure for Creating the Part Data

1. Determine the part data name (max. 16 characters).

◆ Part data name:

This is the name for distinguishing the part data from one another. The user may assign the name freely.
The letter 'D' is automatically set as part of the data name. The rest of the name can be composed of numbers 0 to 9, letters A to Z, plus sign '+', minus sign '-', period '.', and space ' '.

Ex.) D 0 0 1 P A N A S O N I C A V K



2. Determine the feeder No.

◆ Z No.:

This indicates the position on the feeder carriage (Z axis) where parts are set.

- The numbers must match those in the NC data.
- The feeder no. of the jumper wire is 121.

3. Input a component comment.

◆ COMMENT:

Input a name etc. of the part set in the applicable feeder No.

- When creating NC data involving step repeat or pattern repeat, input a comment beginning with "PC". This will convert the insertion direction 180° to 0° and 270° to 90° automatically, thus reducing possible loss of insertion tact time.

  =REMARKS=

  Do not use a comment beginning with "PC" for parts with polarity, or the parts will be inserted with reverse polarity.
- The user can input the comment freely using the numbers 0 to 9 and letters A to Z within 16 characters.

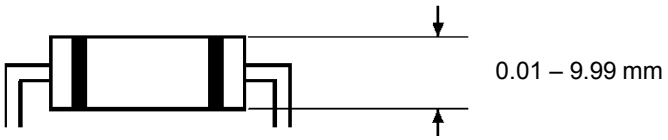
  Ex.) ABC 123

4. Measure the part thickness.

◆ THICK:

Thickness of the part (Unit: mm)

	Input Range	Unit
THICK	0.01 to 9.99	mm

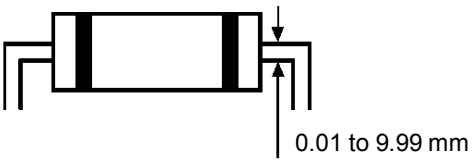


5. Measure the lead diameter.

◆ LEAD:

Diameter of the lead (Unit: mm)

	Input Range	Unit
LEAD	0.01 to 9.99	mm



6. Designate a spare feeder.

- Z Alteration:
  Prepare spare feeders with parts which are used most often, and set the spare feeders in positions other than the ones designated in the NC data. This function, when the master feeders are exhausted of parts, can make production continue without interruption by registering the Z Nos. of spare component feeders in the part data.

◆ SPARE:

Input the Z Nos. of the spare feeders so that the Z alteration function can be utilized.
Ex.) To set the spare feeders for the feeder 1 to the positions 10 and 11.

Z NO.	----------	SPARE
1		10
.		.
.		.
10		11
11		

=HINT=
Input THICK and LEAD data also for spare parts.
When the part supply mode is set in exchange connection mode or EOP recovery function is selected, Z alteration function cannot be used.

7. Determine the chuck pitch.

◆ CHUCK:

=📖= Designate the chuck pitch and centering function with a code. For the centering function, refer to ‘Ope. / DESCRIPTION / Mechanical Movement / Insertion’.

Code	Chuck Pitch	Centering Function
0	26 mm	Disabled
1	52 mm	Disabled
2	26 mm	Enabled
3	52 mm	Enabled

8. Determine the No. of leads to insert.

◆ NO. OF LEAD:

Not used with this machine.

9. Designate large parts.

◆ LARGE PARTS:

Not used with this machine.

10. Designate the V-cut modification.

◆ V-CUT:

Not used with this machine.

11. Measure the part width.

◆ PARTS WIDTH:

Not used with this machine.

12. Measure the part length.

◆ PARTS LENGTH:

Not used with this machine.

4.2. Creating and Editing the Part Data

D79PEC-4R-050-A0

4.2.1 Part Data Input

This section describes the procedures for manually inputting new part data. Prepare a complete part data sheet.

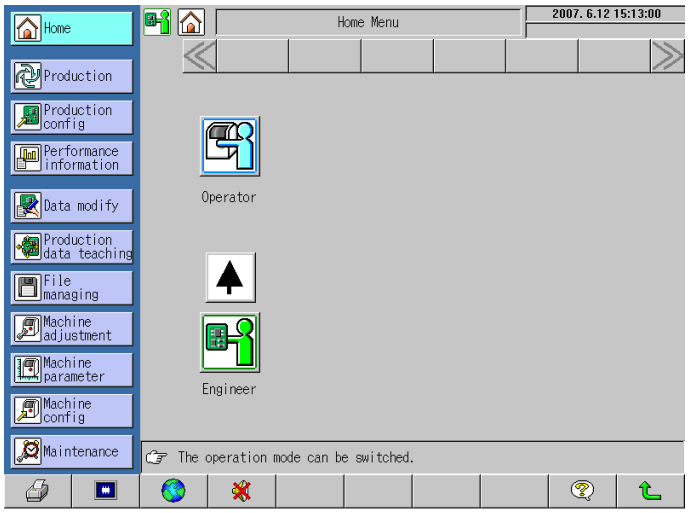
Main Procedure

1	Turn the power ON (“Home Menu” screen).
2	Change to the Operator level.
3	Call up the Data Modify Menu screen.
4	Select Part data.
5	Create a new part data.
6	Input a Part data name.
7	Call up the Part Data Input screen.
8	Input a part comment.
9	Input parameters.
10	Complete the part data input.

■ Procedure for Creating the Part Data

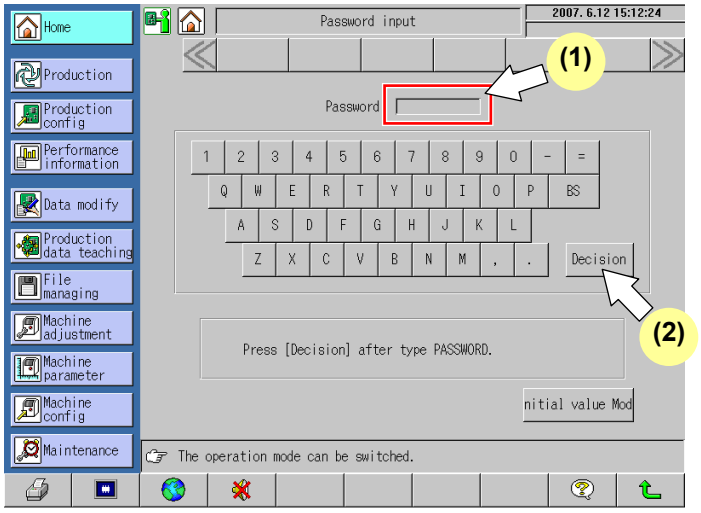
15. Turn the power ON.

- "Home Menu" screen appears on the display.



16. Change to the Operator level.

- Change to the Operator level at which data can be modified.
  (1) Enter the password.
  (2) Press [Decision] .

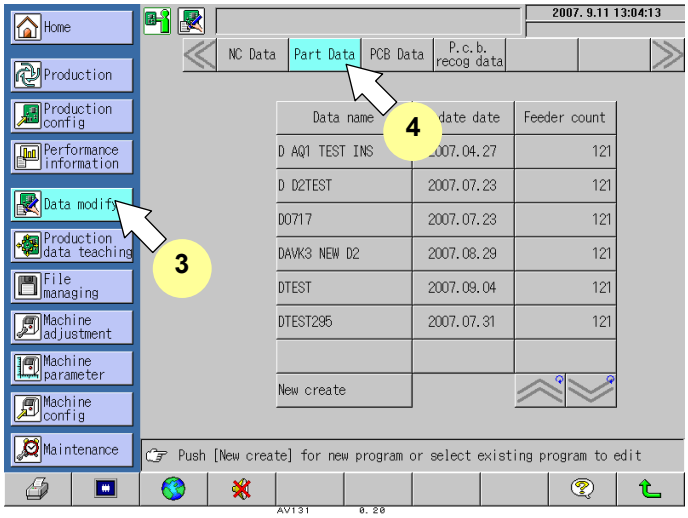


17. Call up the Data Modify Menu screen.

- Press [Data Modify] on the "Home Menu" screen.

18. Select part data.

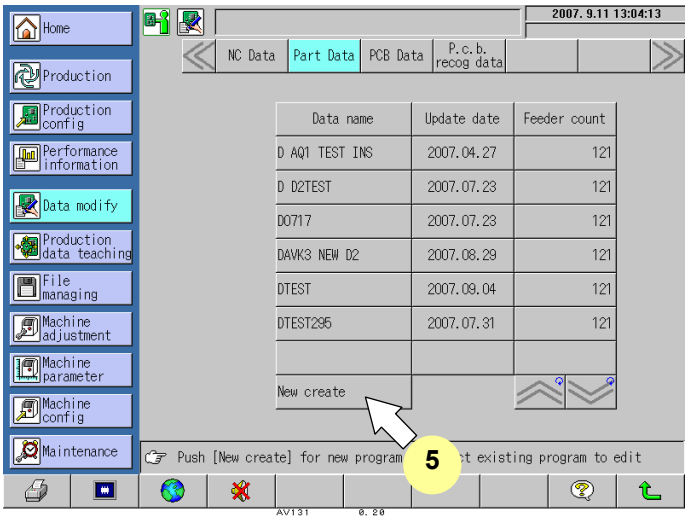
- Press [Part Data] .



19. Create a new part data.

Ex.) Create a part data named as ‘D007’.

- Press [New Create] .

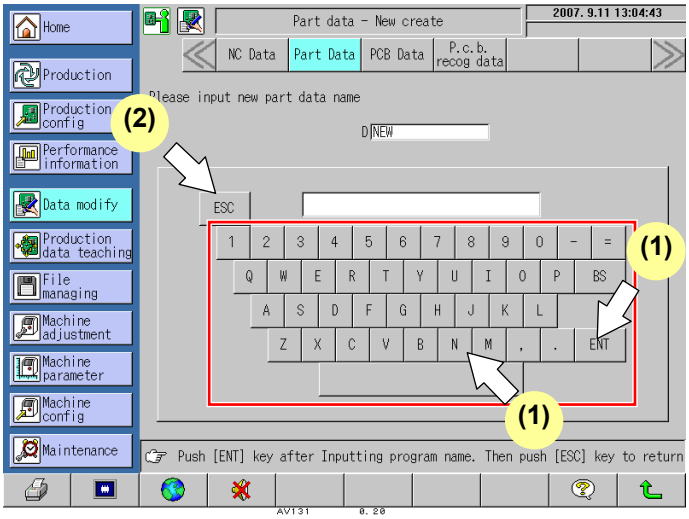


20. Input a part data name.

(1) Input a part data name.

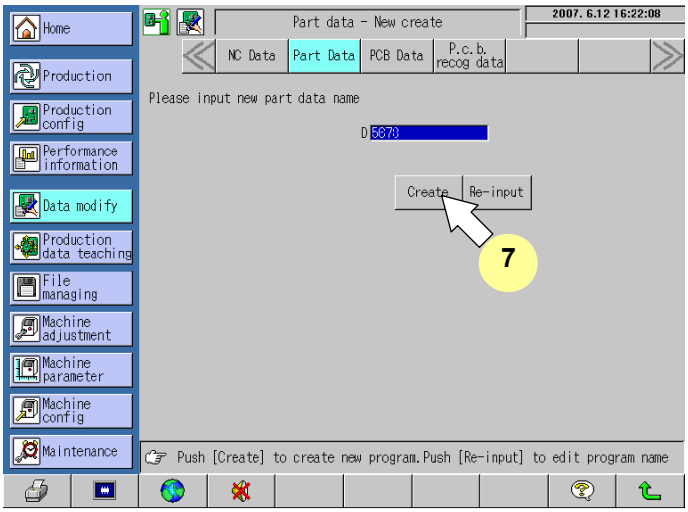
Ex.) Input ‘007’, and press [ENT] to set the name.

(2) Press [ESC] to close the input window.



21. Call up the Part Data Input screen.

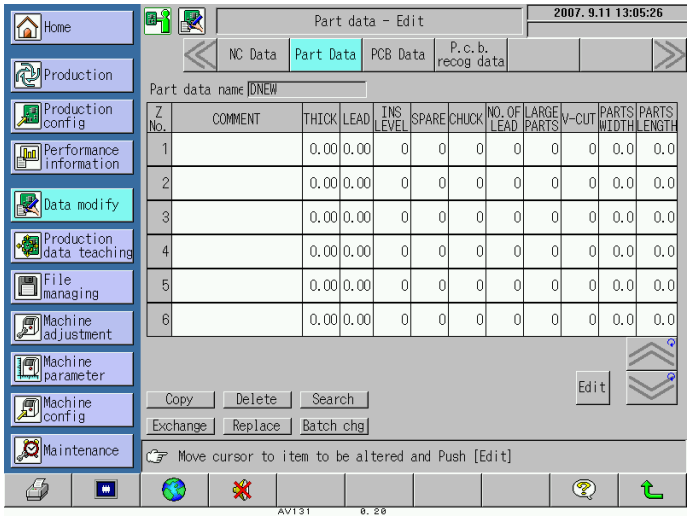
- Press [Create] to switch to the Part Data Input screen.



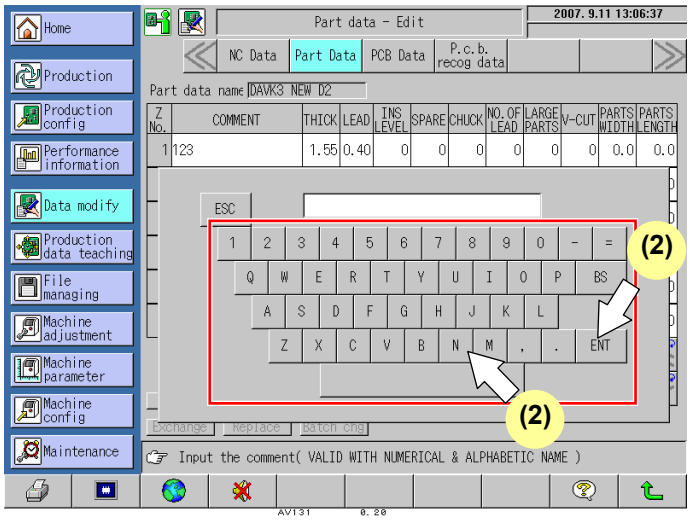
22. Input a part comment.

Ex.) To input ABC;

(1) Press [Edit] .
The input window appears.



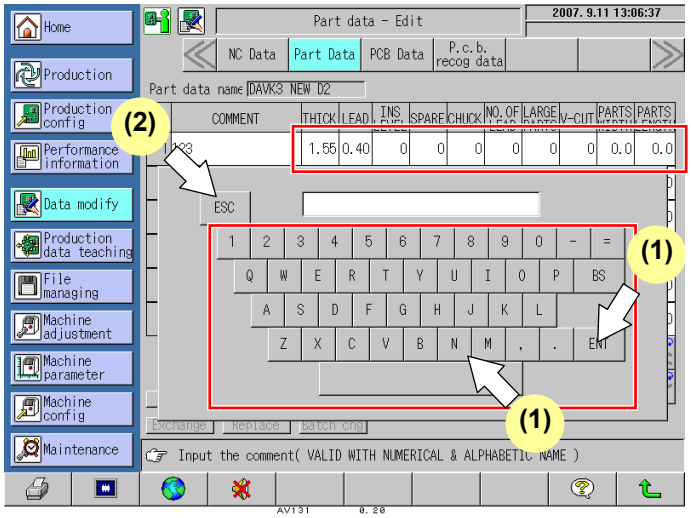
(2) Input a comment
Ex.) Input 'ABC', and press [ENT] to set the data.



23. Input parameters.

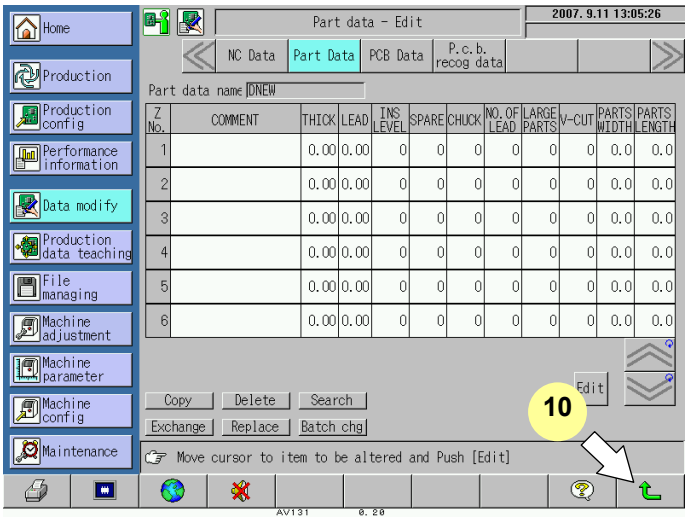
(1) Input data for each parameter.
Ex.) THICK: 1.80 mm
LEAD: 0.60 mm
SPARE: 2
Press [ENT] after each time you input data for a parameter.

(2) Press [ESC] to close the input window.



24. Complete the part data input.

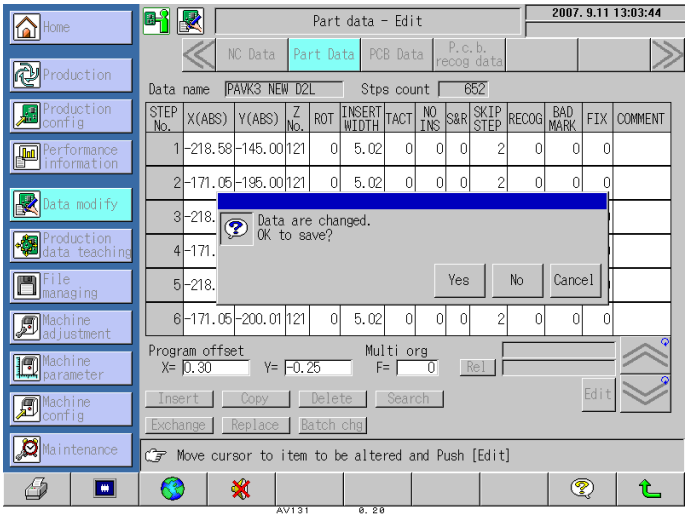
- After all the parameters are input, press .
  Part Data Selection screen is displayed again.



25. Confirm the save.

The confirmation dialog appears. Press the desired button.

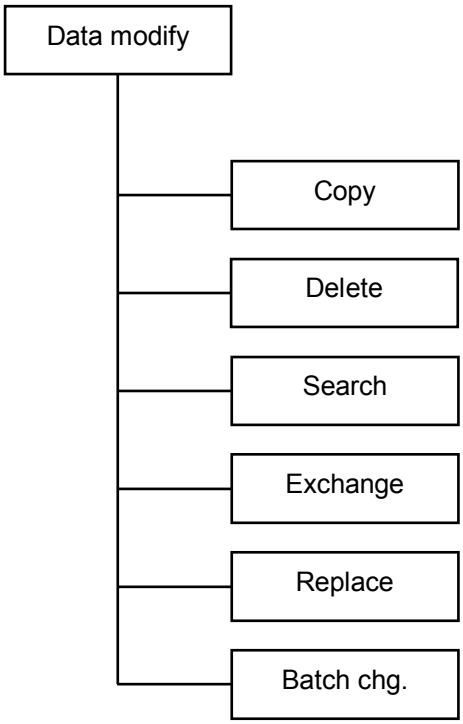
- [Yes]
  Pressing this button saves the data and closes the dialog.
- [No]
  Pressing this button discards the changes you made and closes the dialog.
- [Cancel]
  Pressing this button continues the editing.

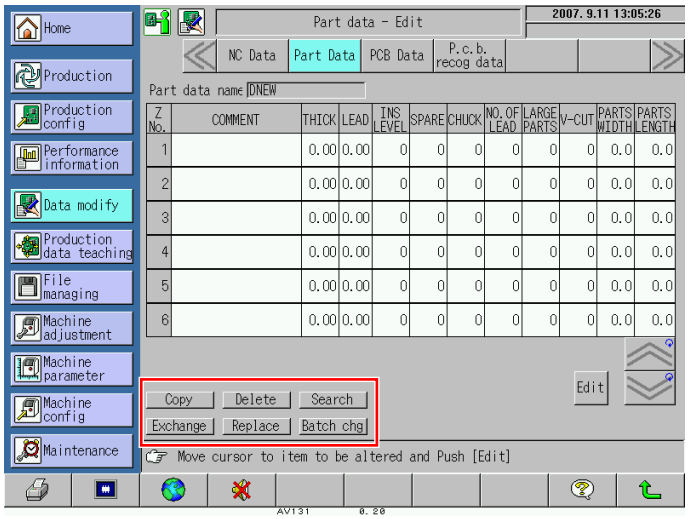


4.2.2 Editing the Part Data

This section describes the functions available for editing the part data.
The editing functions include ‘copy’, ‘delete’, ‘search’, ‘exchange’, ‘replace’, and ‘batch alteration’.

- Access the Part data - Edit screen via the Data modify menu.

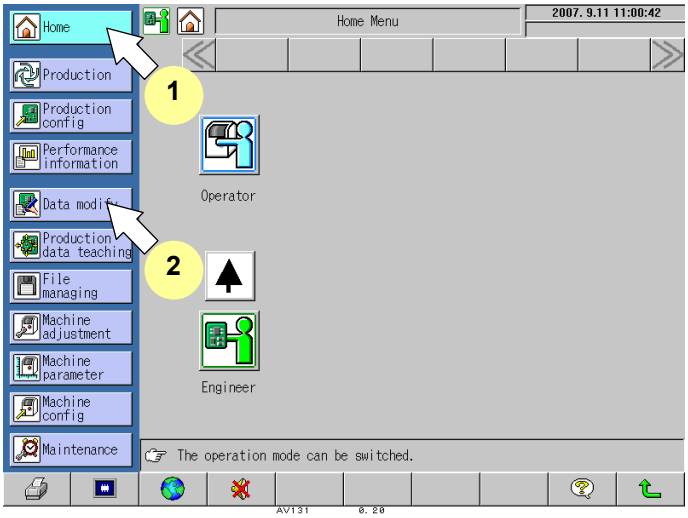




■ Procedure for Editing the Part Data

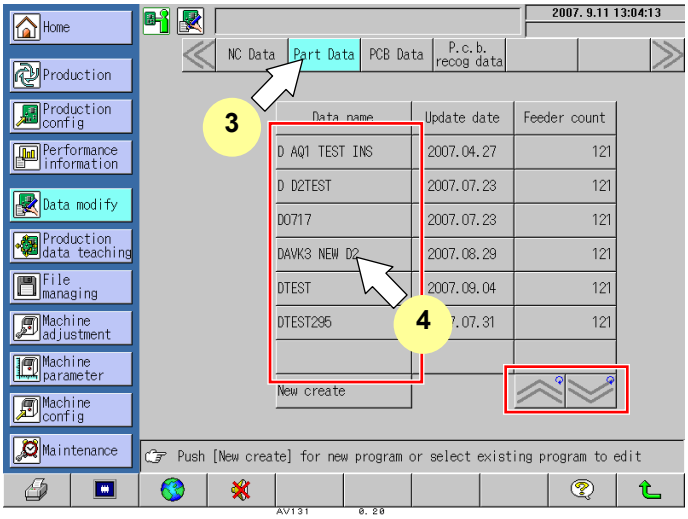
1. Call up the “Home Menu” screen.
- Press [Home] .

2. Call up the Data Modify Menu screen.
- Press [Data modify] .

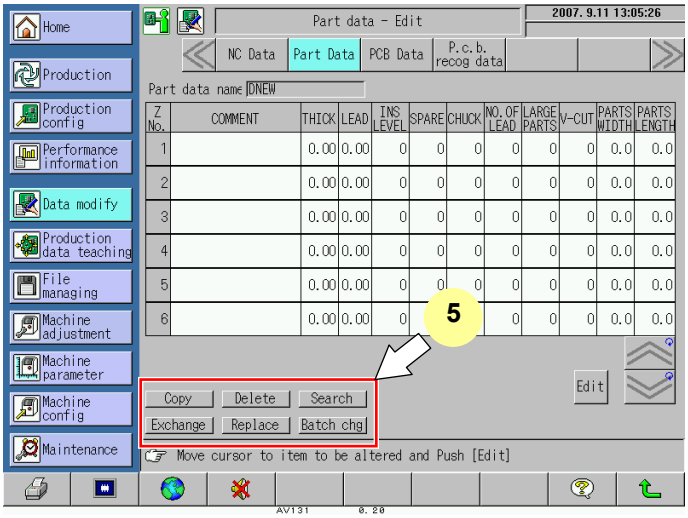


3. Call up the Part Data Edit screen.
- Press [Part Data] .

4. Designate a part data.
Ex.) Press ‘DAVK3 NEW D2’ .
- Select the data to modify from the list of data names displayed on the screen.
  =HINT=
  Use the to scroll through the list of data displayed on the screen.



5. Select the editing functions.
- Select the desired function.

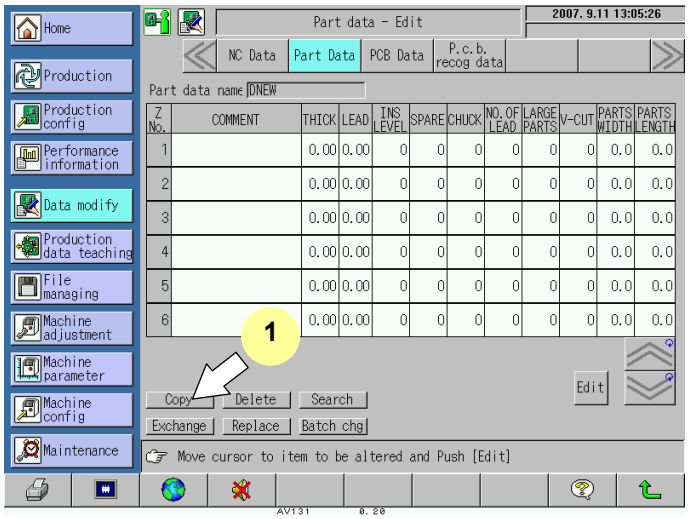


◆ Copy Procedure

This is used to copy the data in the range of designated Z numbers to the position after the target Z number.

1. Select the copy function.

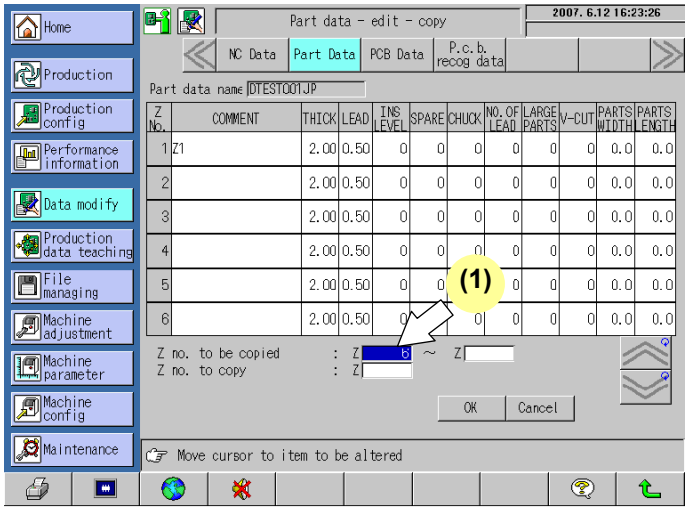
- Press [Copy] .



2. Designate the source range and the target Z No.

Ex.) To copy the contents of feeders Z9 and Z10 to the position after Z11;

(1) Press the input field of Z no. to be copied.
Numeric keypad appears.

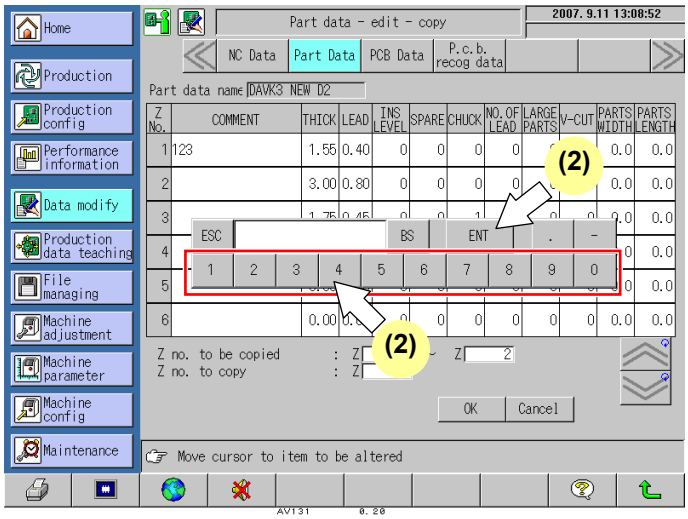


(2) Input the first and last Z Nos. of the source range.

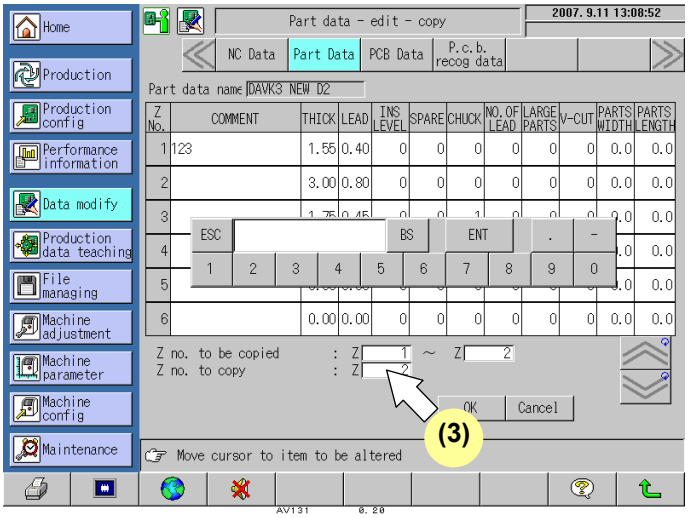
Ex.) Press '9' and [ENT] to set the first Z No.
Press '10' and [ENT] to set the last Z No.

=HINT=

Designate the range of the Z numbers in ascending order.

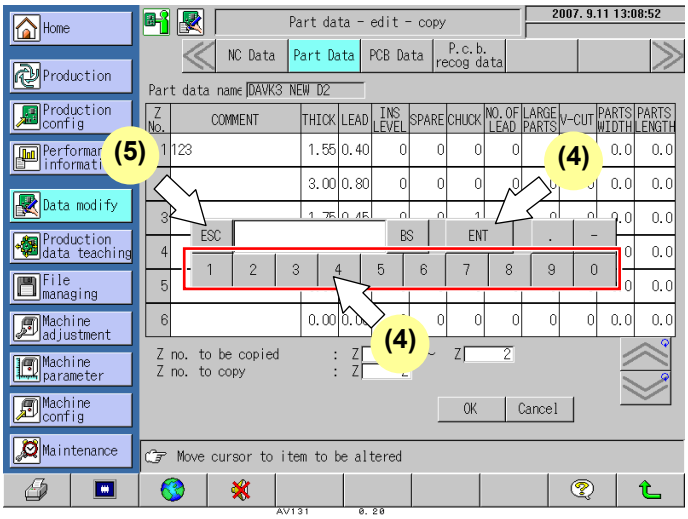


(3) Press the input field of Z No. to copy.



(4) Input the target Z No.
Ex.) Press '11', and [ENT] to set the data.

(5) Press [ESC] to close the keypad window.

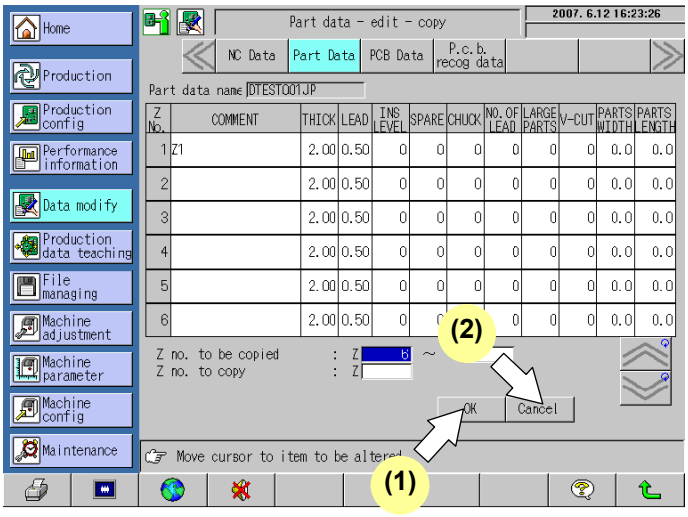


3. Execute copying.

(1) Press [OK] to execute copying.
The screen returns to the NC data – Edit screen automatically.

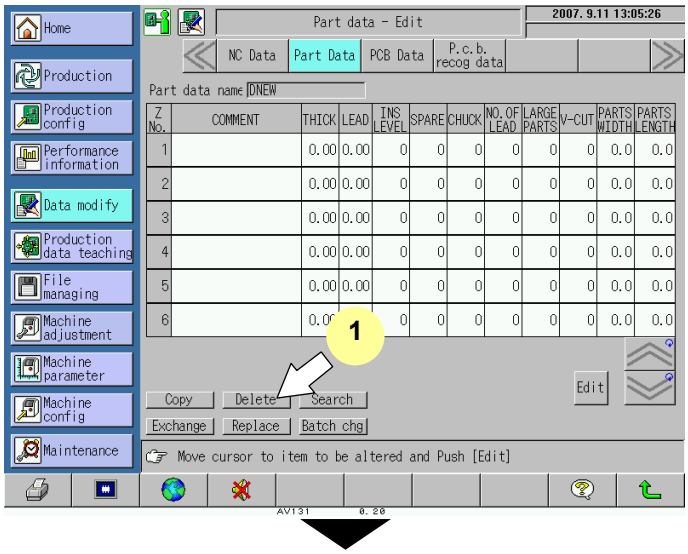
Ex.) The data from feeders Z9 to Z10 is copied after feeder Z11.

(2) Press [Cancel] to cancel copying.
The screen then turns back to Data Modify Menu automatically.



◆ Deletion Procedure

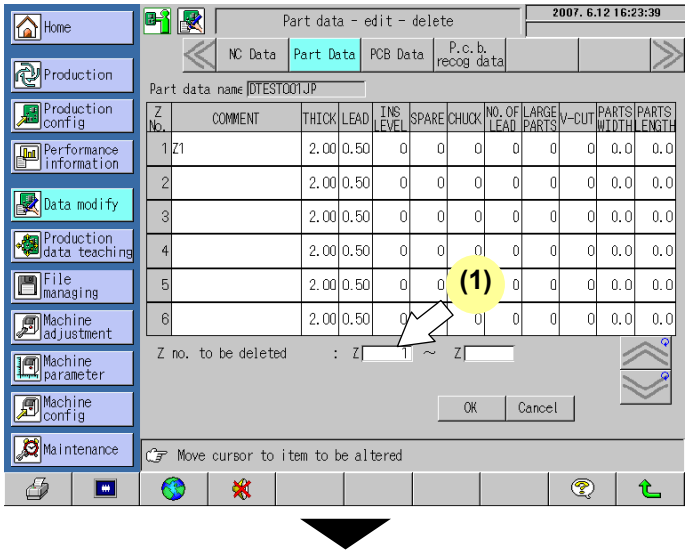
1. This is used to delete unnecessary feeders.
- Press [Delete] .



2. Input the Z Nos. range to be deleted.

Ex.) To delete feeders Z1 to Z5;

(1) Press the input field of Z No. to be deleted.
Numeric keypad appears.



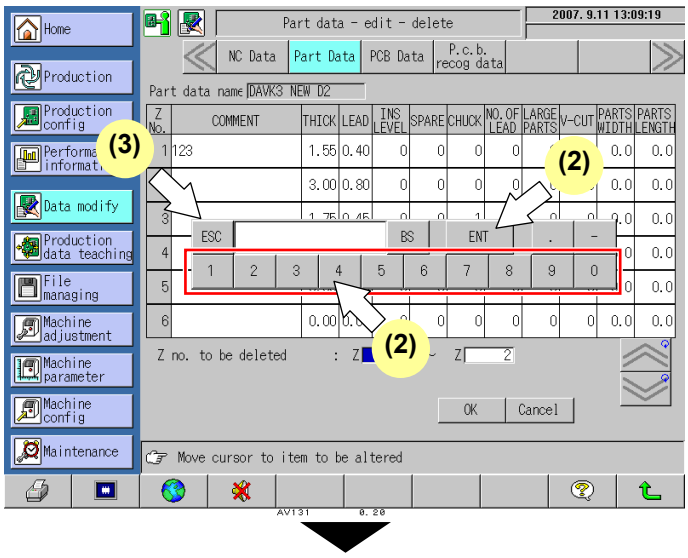
(2) Input the first and last Z Nos. of the target range.

Ex.) Press '1' and [ENT] to set the first Z No.
Press '5' and [ENT] to set the last Z No.

=HINT=

When deleting only one feeder, input the same Z No. in both input fields.

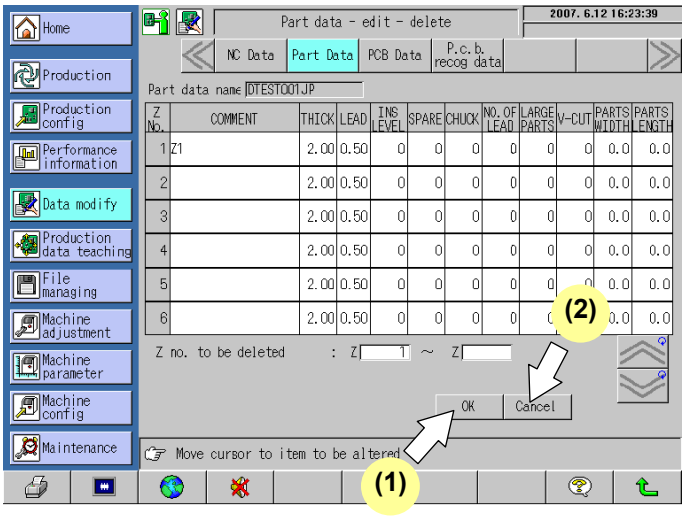
(3) Press [ESC] to close the keypad window.



3. Execute deletion.

(1) Press [OK] to execute deletion.
The screen returns to the NC data – Edit screen automatically.
Ex.) Feeders Z1 to Z5 are deleted.

(2) To cancel deletion, press [Cancel] .
The screen returns to the NC data – Edit screen automatically.

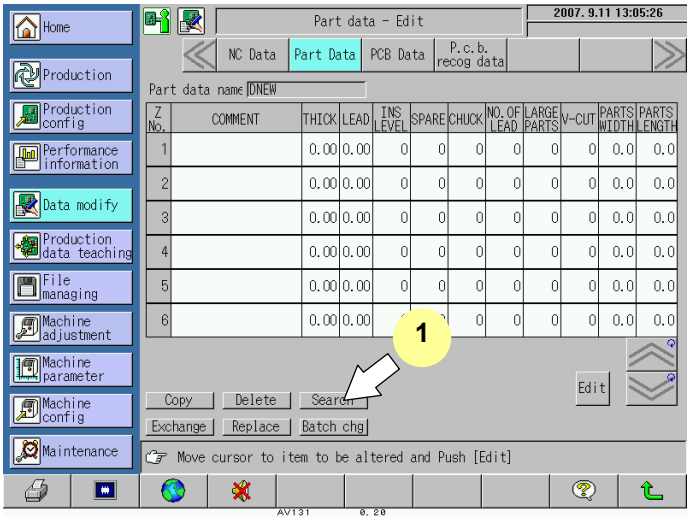


◆ Search Procedure

This function is used to find the data to be changed when changing the part data.

1. Select the search function.

- Press [Search] .

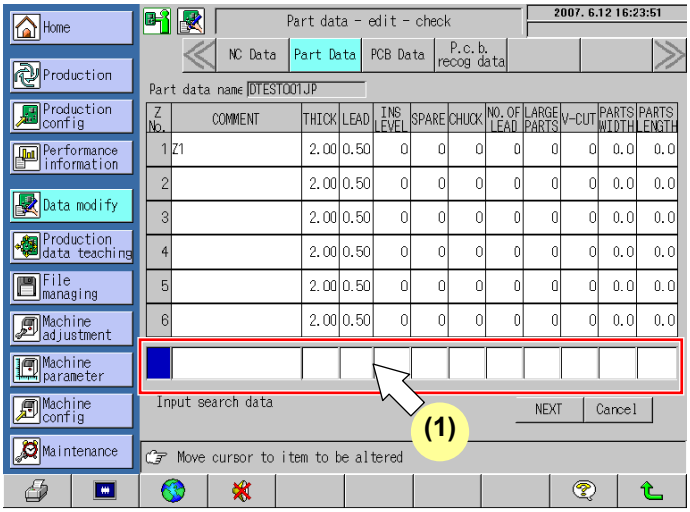


2. Input the data to be searched for.

- Input the data to be searched in the applicable field.

Ex.) To search for the part whose thickness is 1.80 mm and lead diameter is 0.6 mm;

(1) Press the input field of a parameter.
Numeric keypad appears.



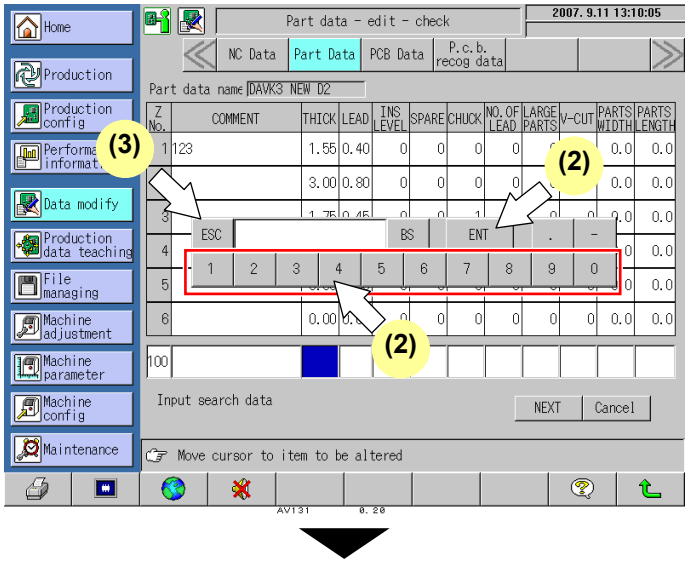
(2) Input the data to be searched in the applicable fields.

Ex.) Designate the THICK field, input '180', and press [ENT] to set the data.
Designate the LEAD field, input '060', and press [ENT] to set the data.

=HINT=

This function searches for the data that match all conditions specified.

(3) Press [ESC] to close the keypad window.

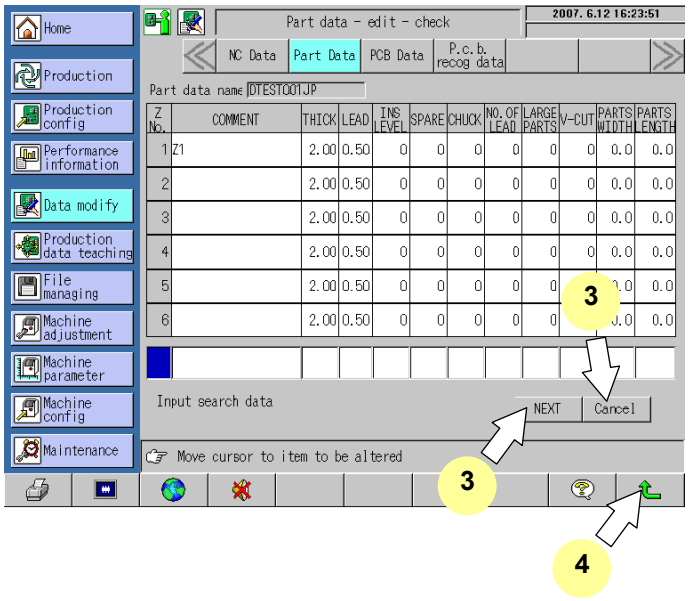


3. Start searching.

- Keep pressing [NEXT] to start searching. (Press [NEXT] to search for another match.)
- The list reappears beginning with the match found.
- If there is no match, the guidance window appears to indicate that 'Searching have reached to the end'.
- Press [Cancel] to cancel searching.

4. Finish searching.

- Press to finish searching.
  The screen to select the functions is displayed again.

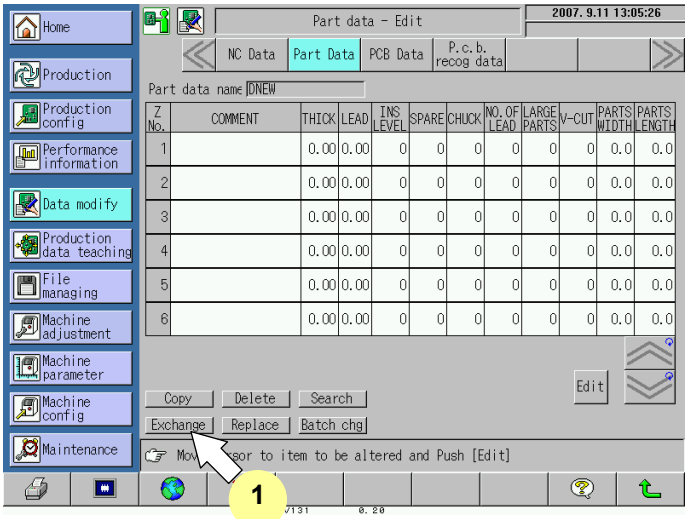


◆ Exchange Procedure

This is used to exchange data between the two designated feeders.

1. Select the exchange function.

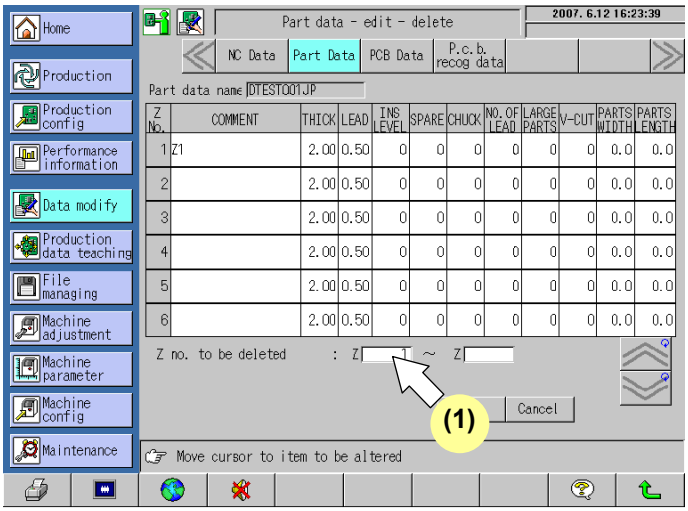
- Press [Exchange] .



2. Input feeders to be exchanged.

Ex.) To exchange feeder Z1 and Z6;

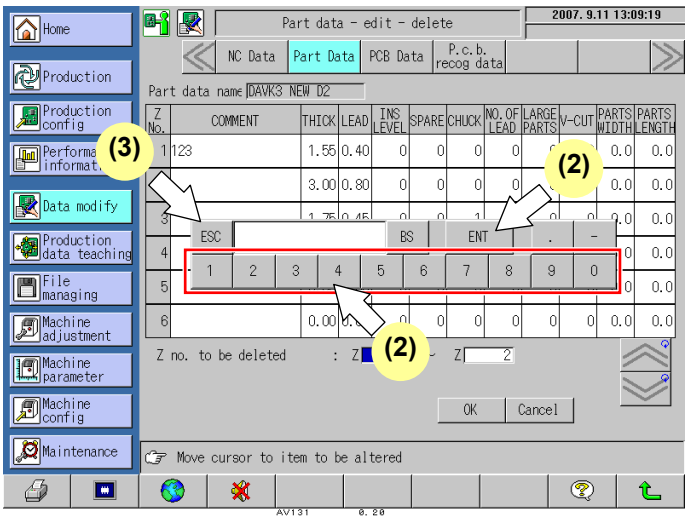
(1) Press the input field of Z No. to be exchanged.
Numeric keypad appears.



(2) Input the Z Nos. to exchange.

Ex.) Press '1', and [ENT] to set the data.
Press '6', and [ENT] to set the data.

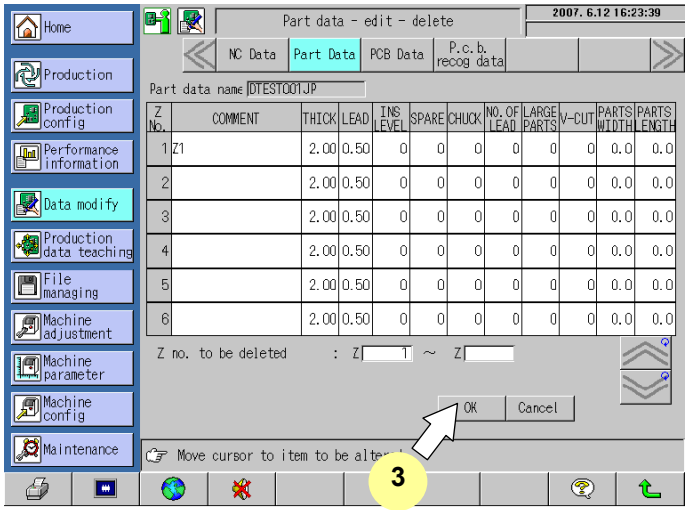
(3) Press [ESC] to close the keypad window.



3. Execute exchange.

(1) Press [OK] to execute exchange.
The screen returns to the NC data – Edit screen automatically.
Ex.) Feeders Z1 and Z6 are exchanged.

(2) Press [Cancel] to cancel exchange.
The screen returns to the NC data – Edit screen automatically.

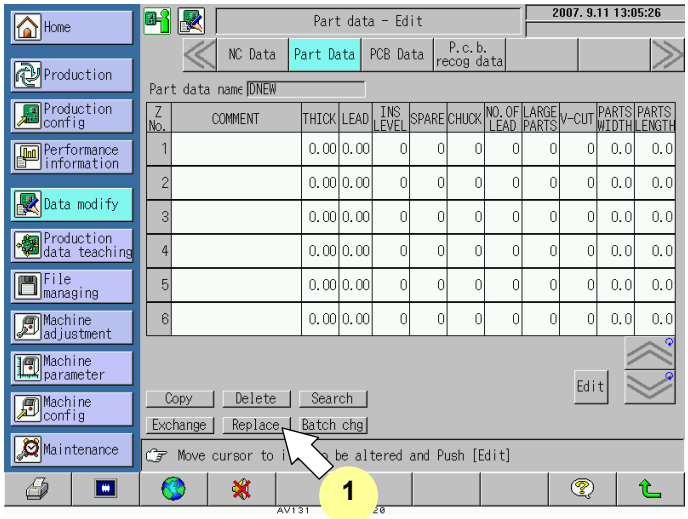


◆ Replacement Procedure

This function replaces the designated parameter data with new data.
Use it when applying the change of particular parameter to the entire part data.

1. Select the replace function.

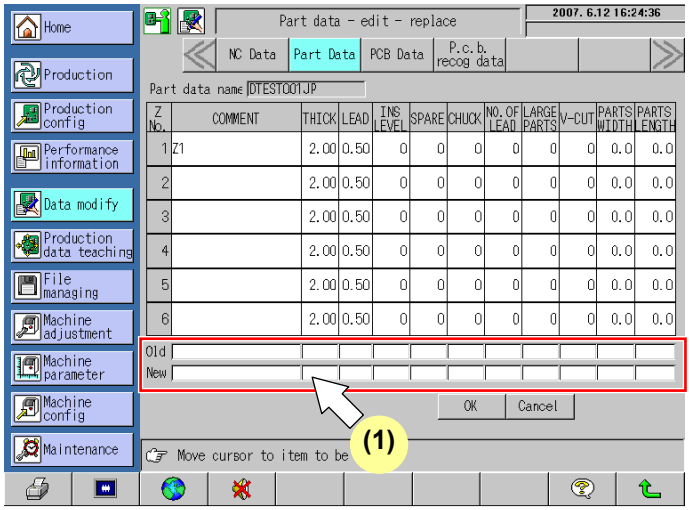
- Press [Replace] .



2. Input the data to be replaced.

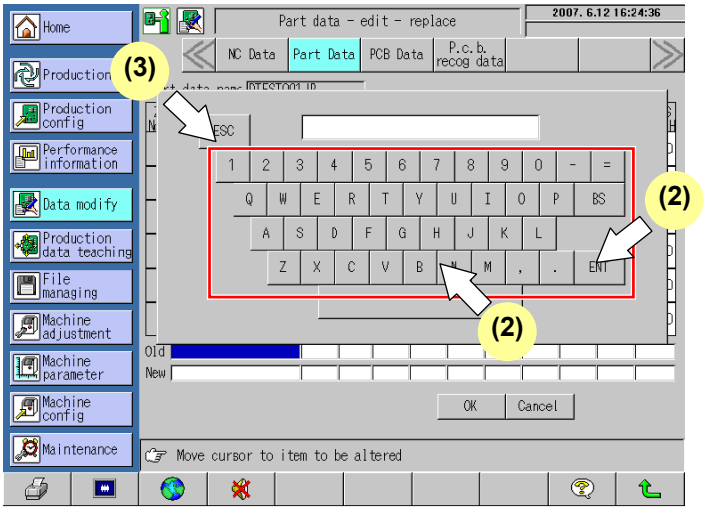
Ex.) To replace all the lead diameter of the parts having the comment of 'R1438008701' with 0.65 mm.

(1) Press the input field of Old / New data.
Numeric keypad appears.



(2) Designate the old/new data.
Ex.) Designate the Old field under "COMMENT", and input 'R1438008701', then press [ENT] to set the data.
Designate the New field under "LEAD", and input '065', then press [ENT] to set the data.

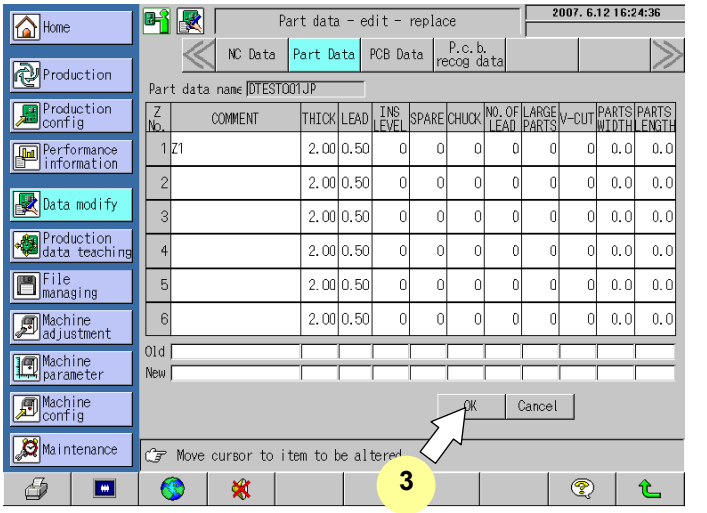
(3) Press [ESC] to close the keypad window.



3. Execute replacement.

(1) Press [OK] to execute replacement.
The screen returns to the NC data – Edit screen automatically.

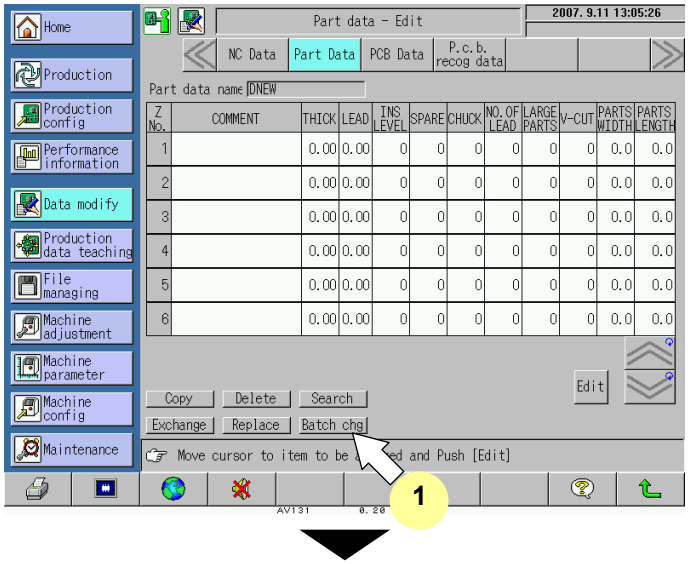
(2) Press [Cancel] to cancel replacement.
The screen returns to the NC data – Edit screen automatically.



◆ Batch Change Procedure

This function changes the value of the specified parameter within the designated range all at once.
Use it when changing the specified parameter of part data within the consecutive Z numbers at a time.

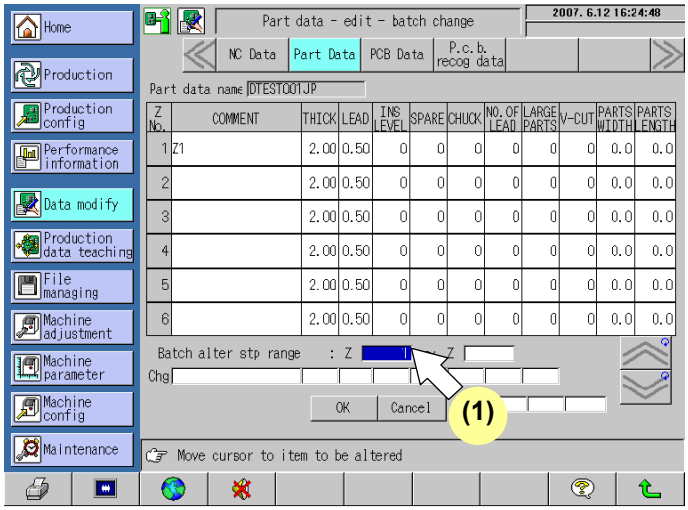
1. Select the batch change function.
- Press [Batch chg] .



2. Designate the Z No. range and data to be changed.

Ex.) To change all of the LEAD data between feeders Z3 and Z4 to 0.75 mm;

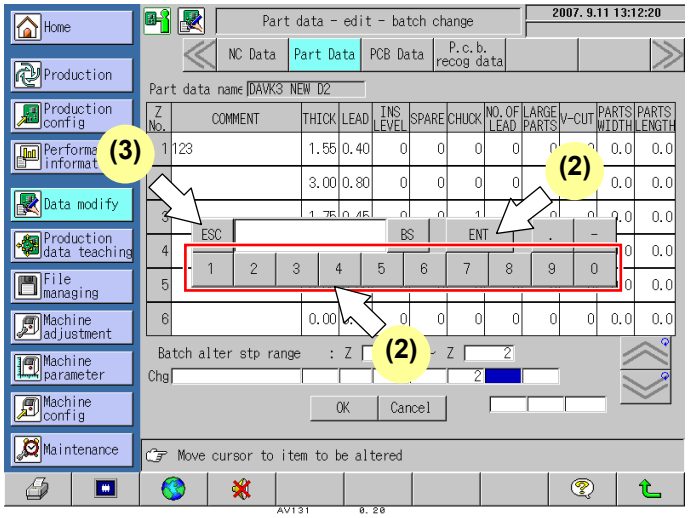
(1) Press the input field of Batch alter step range.
Numeric keypad appears.



(2) Designate the data to be changed.

Ex.) Input the Z No. range.
Press '3', and [ENT] to set the first Z No.
Press '4', and [ENT] to set the last Z No.
Input the data to be changed.
Designate the "LEAD" field to be changed. Input '075', and press [ENT] to set the data.

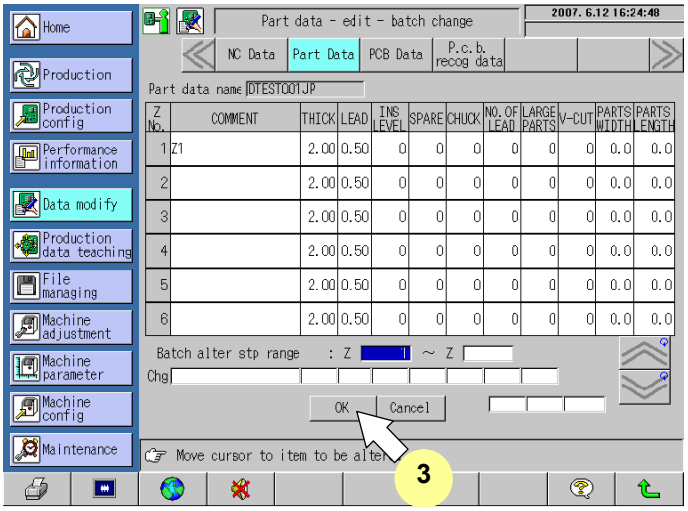
(3) Press [ESC] to close the keypad window.



3. Execute batch change.

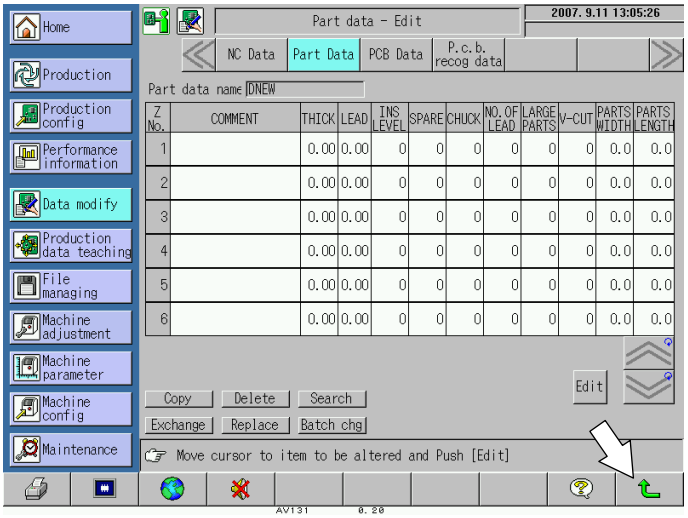
(1) Press [OK] to execute batch exchange.
The screen returns to the NC data – Edit screen automatically.

(2) Press [Cancel] to cancel batch exchange.
The screen returns to the NC data – Edit screen automatically.



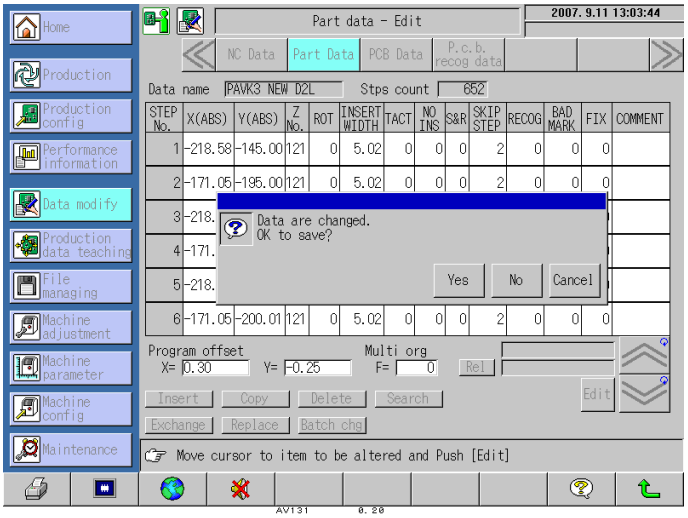
◆ Exit Editing Mode

1. Press to exit the editing mode.



2. Confirm the save.
The confirmation dialog appears. Press the desired button.

- [Yes]
  Pressing this button saves the data and closes the dialog.
- [No]
  Pressing this button discards the changes you made and closes the dialog.
- [Cancel]
  Pressing this button continues the editing.



5. P.C.B. DATA CREATION

This chapter describes the procedure for creating the PC board data.

5.1. Setting Items of the PCB Data

D79PEC-41-030-A0

5.1.1 Parameters

The PCB data is a set of data relevant to the dimensions of a PCB. During production, the PCB transfer rails and positioning pins are positioned based on this data.
The PCB. data includes the following parameters (data items).

Parameter	Input range / Standard Setting	Details
Length	0.00 to 999.99 (mm)	The length of PCBs in the horizontal direction
Width	0.00 to 999.99 (mm)	The length of PCBs in the vertical direction
Hole pitch	0.00 to 999.99 (mm)	Interval between the reference holes on PCBs

=REMARKS=
When the automatic width adjustment function (option) is not used, the PCB data is not required.

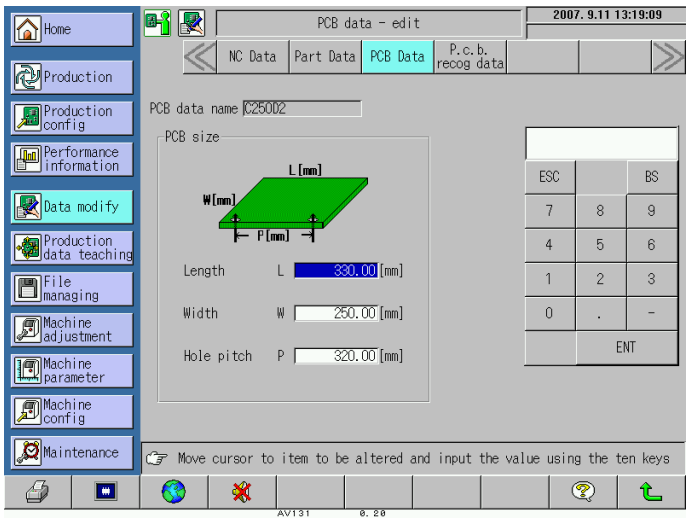
5.1.2 PCB Data Edit Screen/PCB Data Sheet

■ Sample PCB Data Sheet

PCB data name _____

PCB size (mm)	
Length	
Width	
Hole pitch	

The PCB data can be input from the PCB Data Edit screen. The PCB Data Edit screen is shown below.
Prepare a completed PCB data sheet first, and then input those data actually into each parameter of the PCB Data Edit screen.



5.1.3 Creating the PCB Data

■ Procedure for Creating the PCB Data

1. Determine the PCB data name (max. 16 characters).

◆ PCB data name:

This is the name for distinguishing PCB data from one another. The user may assign the name freely. The letter 'C' is automatically set as part of the data name. The rest of the name can be composed of numbers 0 to 9, letters A to Z, plus sign '+', minus sign '-' , period '.', and space ' '.

Ex.) C 0 0 1 P A N A S O N I C A V K

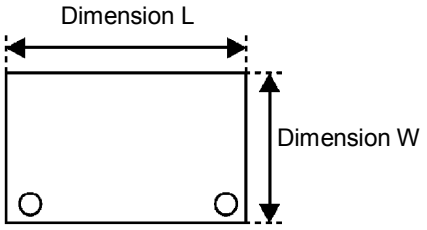
Automatically set Max. 15 characters

2. Determine the PCB dimensions (0.00 to 999.99 mm for both X and Y).

◆ PCB size:

The PCB dimensions are measured in the L and W directions.

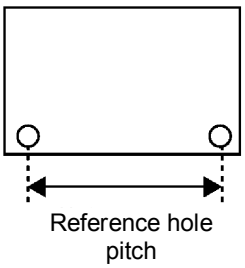
- L: left to right direction seen from the front
- W: front to back direction seen from the front



3. Determine the reference hole pitch (0.00 to 999.99 mm).

◆ Hole pitch:

Measure the reference hole pitch in the X direction of the PCB (the center-to-center distance between the reference holes).



5.2. Creating and Editing the New PCB Data

D79PEC-41-060-A0

5.2.1 Creating the PCB Data

This section describes the procedures for manually inputting new PCB data. Prepare a completed OCB data.

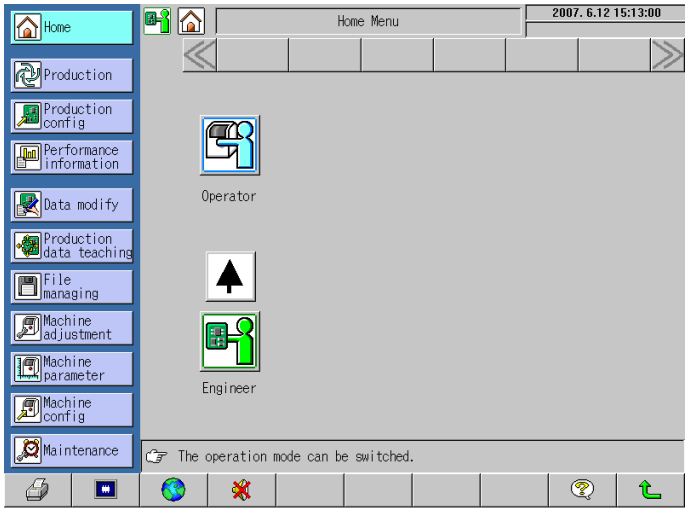
Main Procedure

1	Turn the power ON. (“Home Menu” screen)
2	Change to the Operator level.
3	Call up the Data Modify Menu screen.
4	Select the PCB data.
5	Create the PCB data.
6	Input the PCB data name.
7	Call up the PCB Data Input screen.
8	Input in each parameters.
9	Complete the PCB Data input.

■ Procedure for Creating the PCB Data

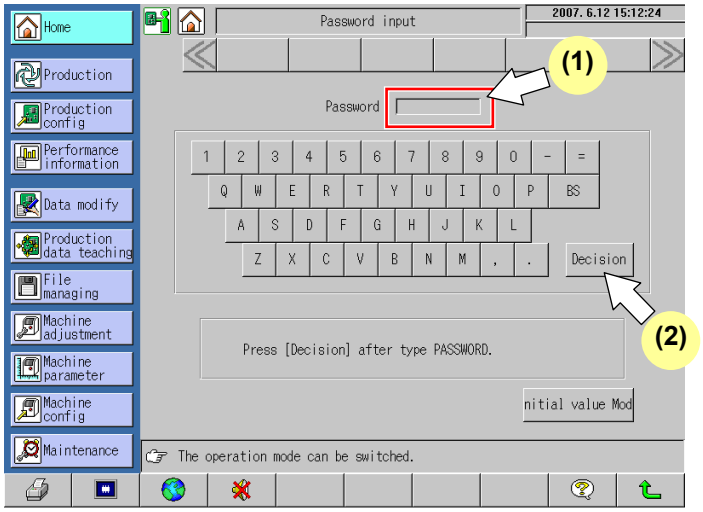
26. Turn the power ON.

- "Home Menu" screen appears on the display.



27. Change to the Operator level.

- Change to the operator level at which the data can be modified.
  (1) Enter the password.
  (2) Press [Decision] .

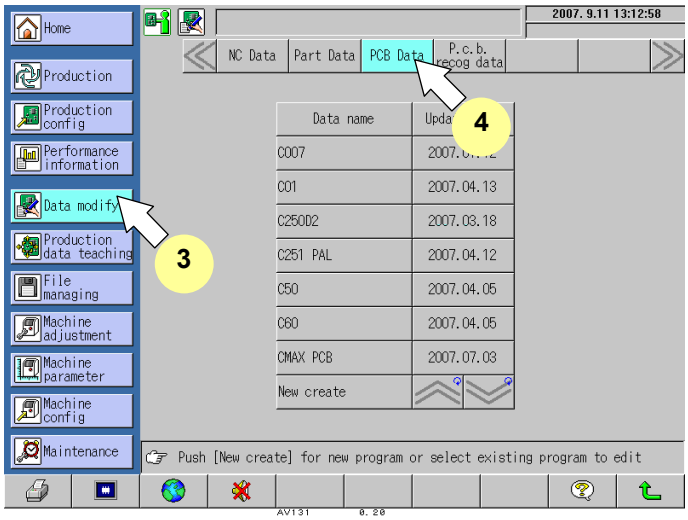


28. Call up the Data Modify screen.

- Press [Data Modify] on the "Home Menu" screen.

29. Select part data.

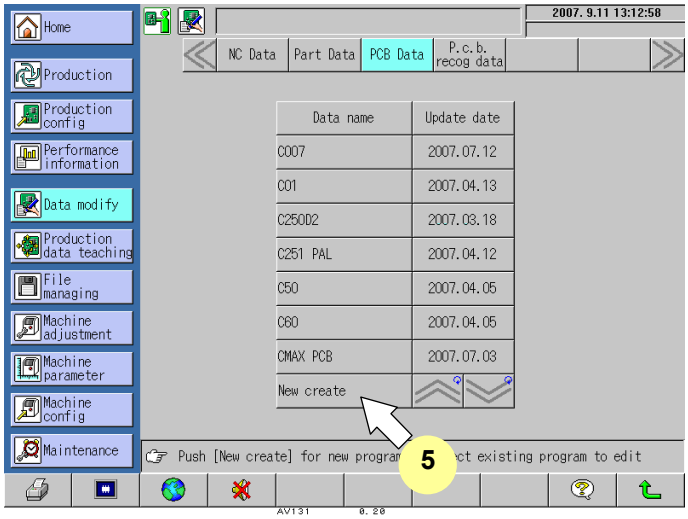
- Press [Part Data] .



30. Create a PCB data.

Ex.) Create a new PCB data named as ‘D007’.

- Press [New Create] .

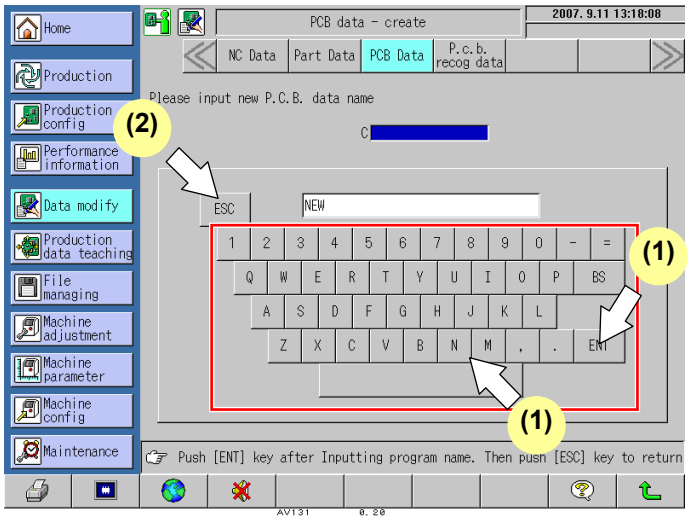


31. Input the PCB data name.

(1) Input a PCB data name.

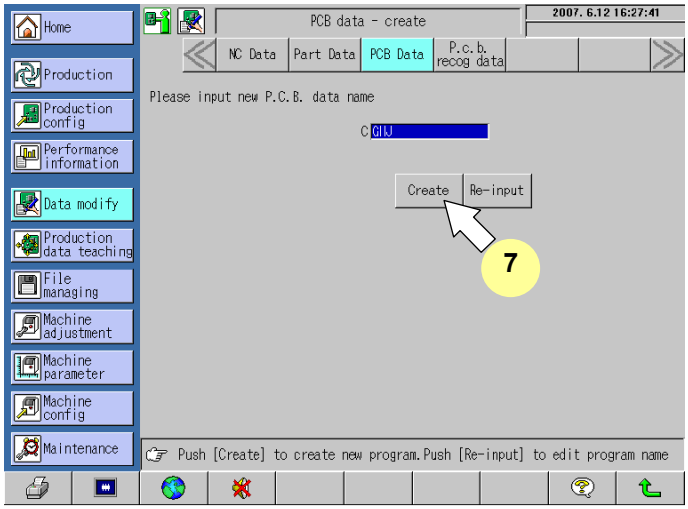
Ex.) Input ‘007’, and press [ENT] to set the data.

(2) Press [ESC] to close the input window.



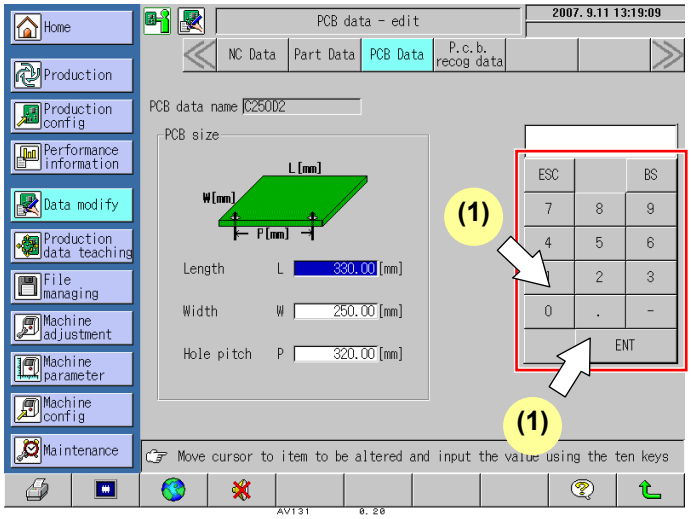
32. Call up the PCB Data Input screen.

- Press [Create] to switch to the PCB Data Input screen.



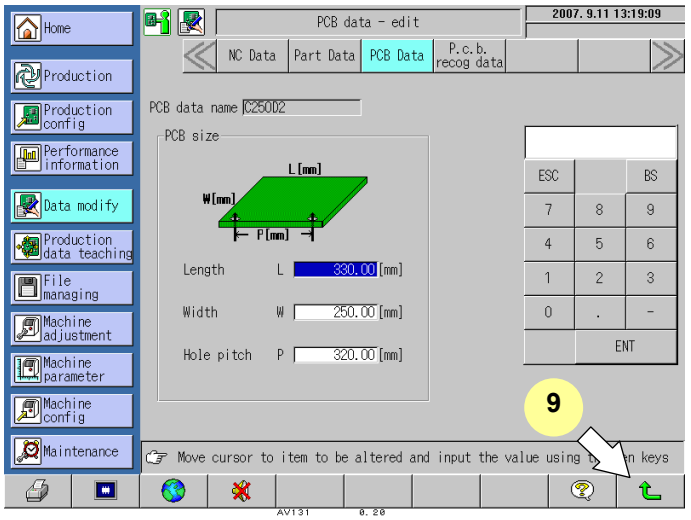
33. Input parameters.

- Input the PCB size.
  Ex.) Length: 330.00 mm
  Width: 250.00 mm
  Hole pitch: 320.00 mm
  (1) Input the PCB size.
  Ex.) Designate the Length field, input '330.00', then press [ENT] to set the data.
  Designate the Width field, input '250.00', then press [ENT] to set the data.
  Designate the Hole pitch field, input '320.00', then press [ENT] to set the data.



34. Complete the PCB data input.

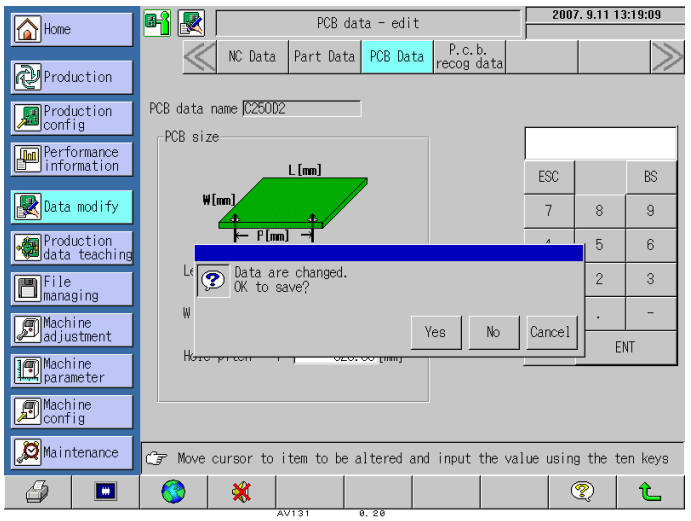
- After all the parameters are input, press .
  The PCB Data Selection screen is displayed again.



35. Confirm the save.

The confirmation dialog appears. Press the desired button.

- [Yes]
  Pressing this button saves the data and closes the dialog.
- [No]
  Pressing this button discards the changes you made and closes the dialog.
- [Cancel]
  Pressing this button continues the editing.



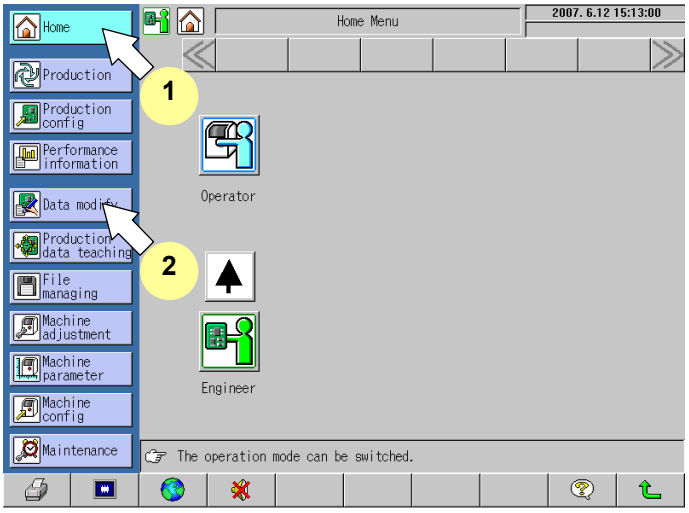
5.2.2 Editing the PCB Data

Access the PCB data - Edit screen via the Data modify menu.

■ Procedure for Editing the PCB Data

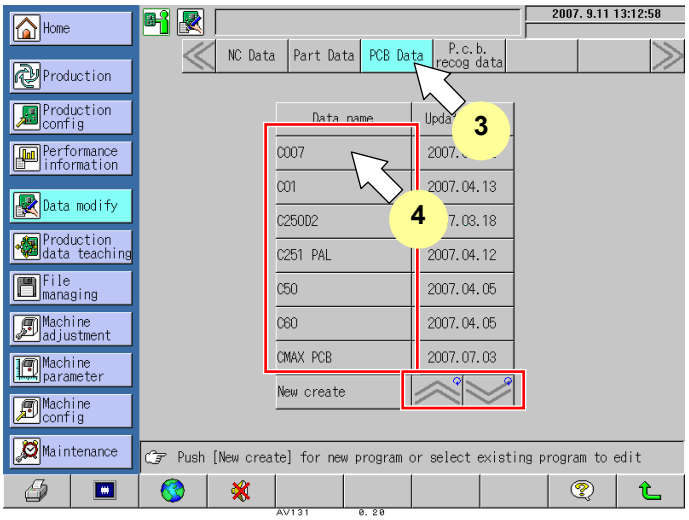
1. Call up the “Home Menu” screen.
- Press [Home] .

2. Call up the Data Modify Menu screen.
- Press [Data modify] .

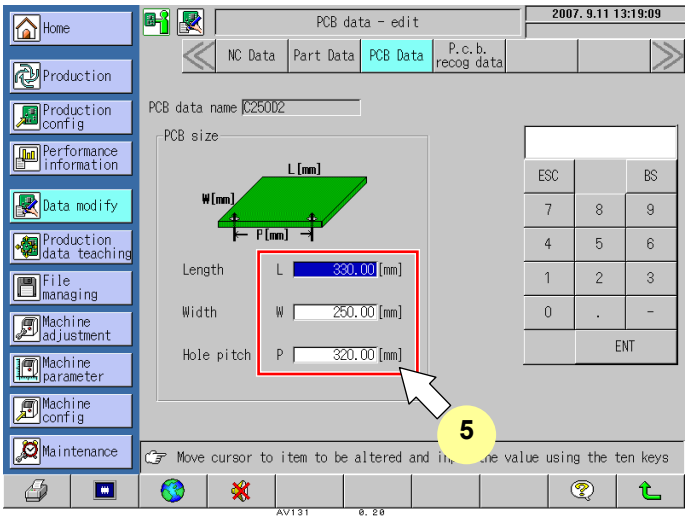


3. Call up the PCB Data Modify Menu screen.
- Press [PCB Data] .

4. Designate the PCB data.
- Select the data to modify from the list of data names displayed on the screen.
  =HINT=
  Use to scroll through the list of data displayed on the screen.

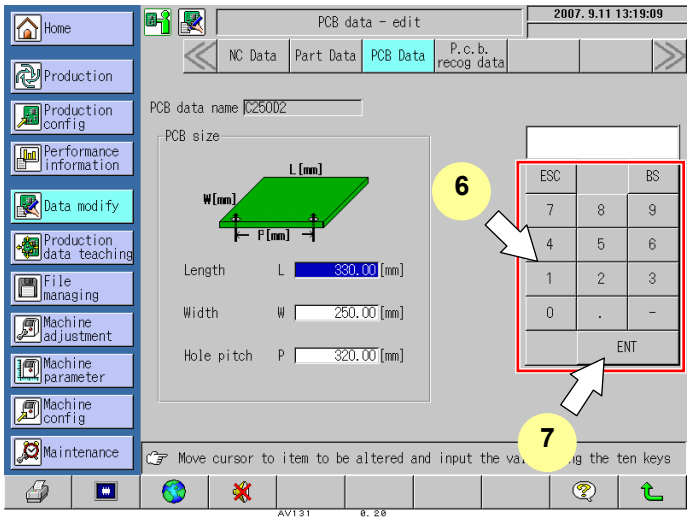


5. Press the input field of the data to be modified or corrected.



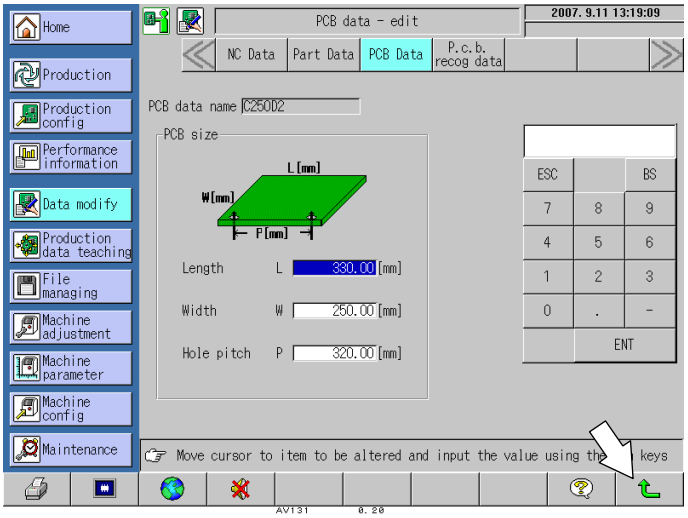
6. Input the new data.

7. Register the data.
- Press [ENT] to register the data.

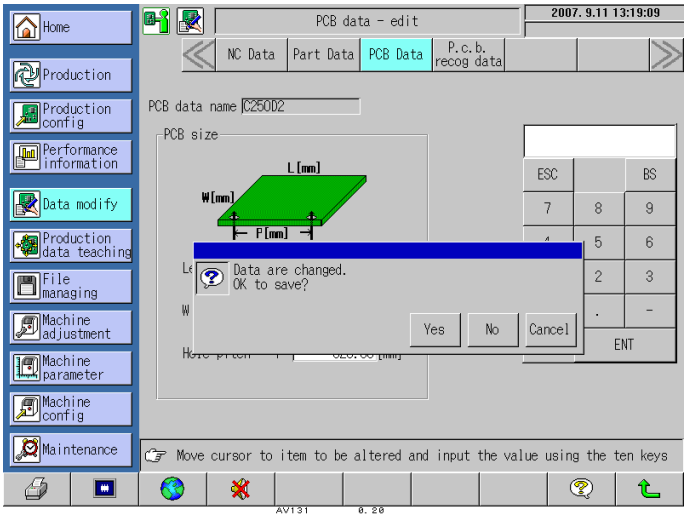


◆ Exiting the Editing Mode

1. Press to exit the editing mode.



2. Confirm the save.
The confirmation dialog appears. Press the desired button.
- [Yes]
  Pressing this button saves the data and closes the dialog.
- [No]
  Pressing this button discards the changes you made and closes the dialog.
- [Cancel]
  Pressing this button continues the editing.



6. P.C.B. RECOGNITION DATA CREATION

This chapter describes the procedure for creating the P.C.B. recognition data.

6.1. Setting Items of the PCB Recognition Data

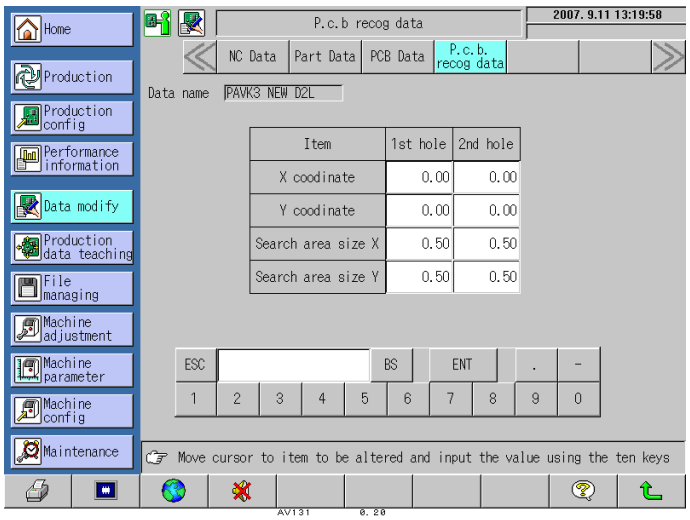
D79PEC-4X-010-A0

6.1.1 Parameters

The PCB recognition data consists of the positional data of PCB holes to be recognized during PCB recognition. PCB recognition is performed to correct the angle and position of the PC board based on this data.

The PCB recognition data includes the following parameters (data items).

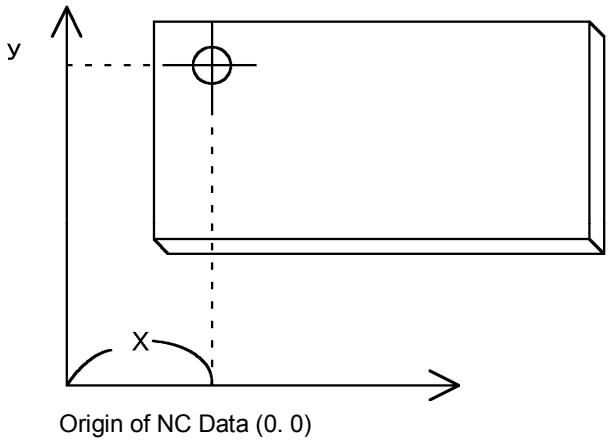
Parameter	Input range / Standard Setting	Details
X	0.00 to 999.99 (mm)	X coordinate of the recognition hole position
Y	0.00 to 999.99 (mm)	Y coordinate of the recognition hole position
Search area size X	0 to 511 (pixel)	Size of the recognition search area in X direction
Search area size Y	0 to 479 (pixel)	Size of the recognition search area in Y direction



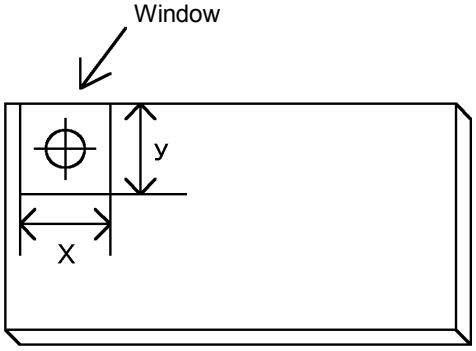
6.1.2 Creating the Recognition Data

■ Procedure for Creating the Recognition Data

1. Determine the recognition coordinates (0.00 to 999.99 mm for both X and Y).



2. Determine the window sizes. (X: 0 to 511, Y: 0 to 479)



=HINT=
For the recognition coordinate, designate an optional hole that functions as the basis of the PCB. Any holes on the PC board can be designated as the recognition coordinates.

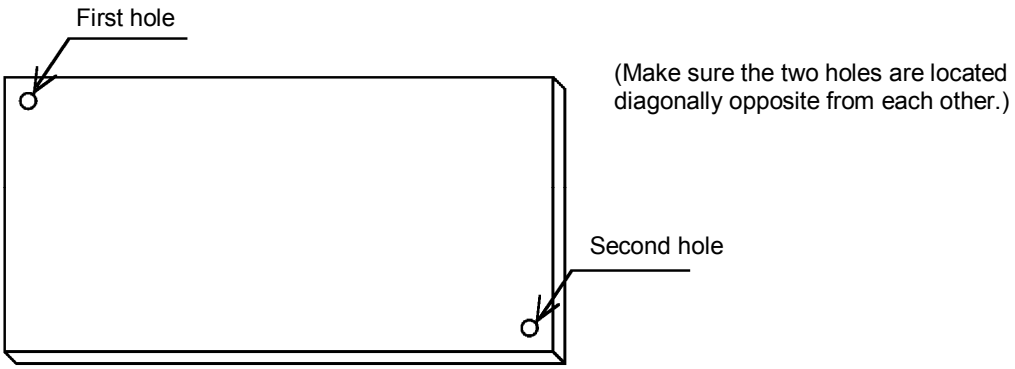
3. To offset slanting PCBs, determine the second hole in the same way.

=REMARKS=

The recognition data name is the same as the NC data name (cannot be changed).

=HINT=

Although any holes on the PC board can be designated as the two reference holes, setting them as in the positions shown below will provide better effects in recognition offset.



6.2. Editing the PCB Recognition Data

D79PEC-4X-020-A0

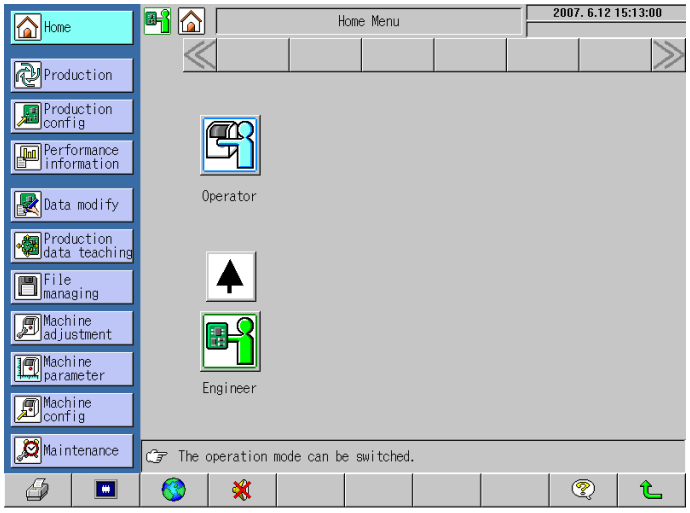
6.2.1 Creating and Editing the PCB Recognition Data

Take the following steps to input PCB recognition data.

■ Procedure for Creating the PCB Recognition Data

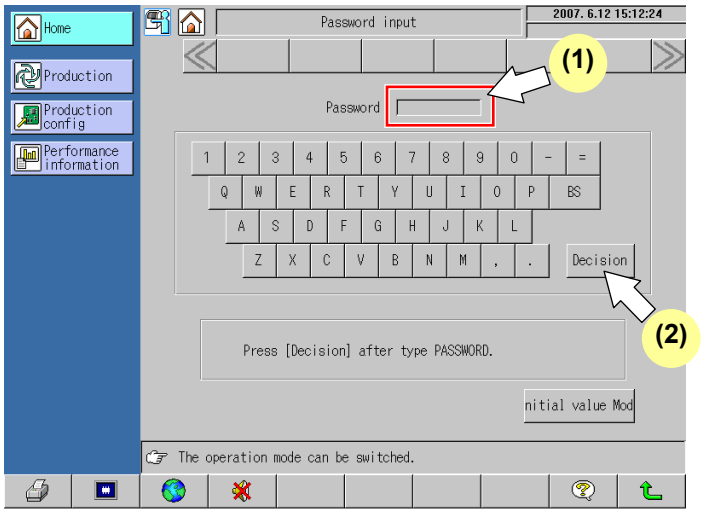
1. Turn the power ON.

- "Home Menu" screen appears on the display.



2. Change to the Operator level.

- Change to the operator level at which the data can be modified.
  (1) Enter the password.
  (2) Press [Decision] .

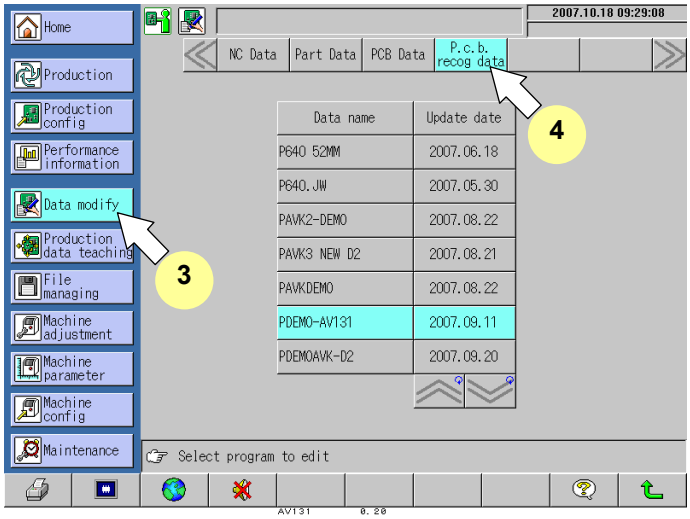


3. Call up the Data Modify Menu screen.

- Press [Data modify] on the "Home Menu" screen.

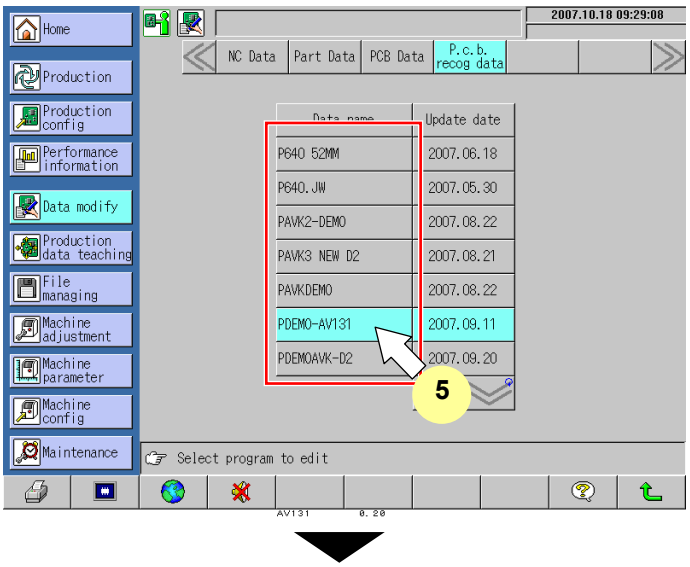
4. Select PCB recognition data.

- Press [P.c.b. recog data] .



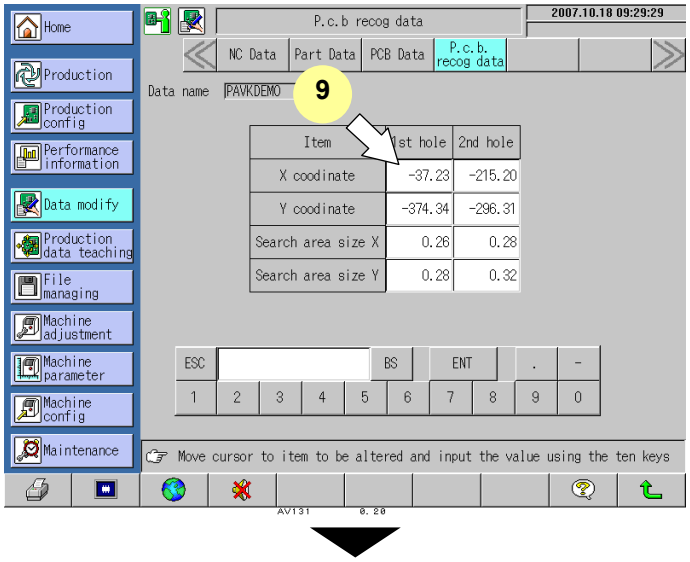
5. Select PCB recognition data.

Specify the PCB recognition data to be edited. (The data name is linked with the NC data name.)



6. Select the data item to be edited.

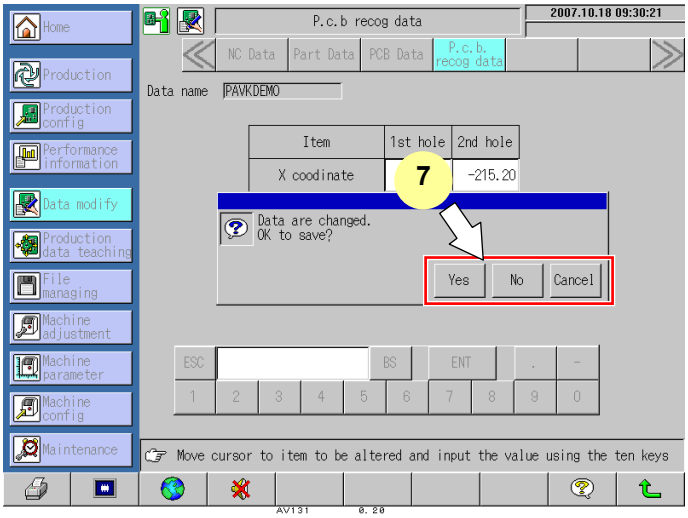
Touch the desired item directly and enter a value using the numeric keypad.



7. Complete editing.

Pressing displays a confirmation dialog asking you if you want to save the data.

- To save the data, press [Yes] .
- To discard the changes you made, press [No]. To continue the editing, press [Cancel] .



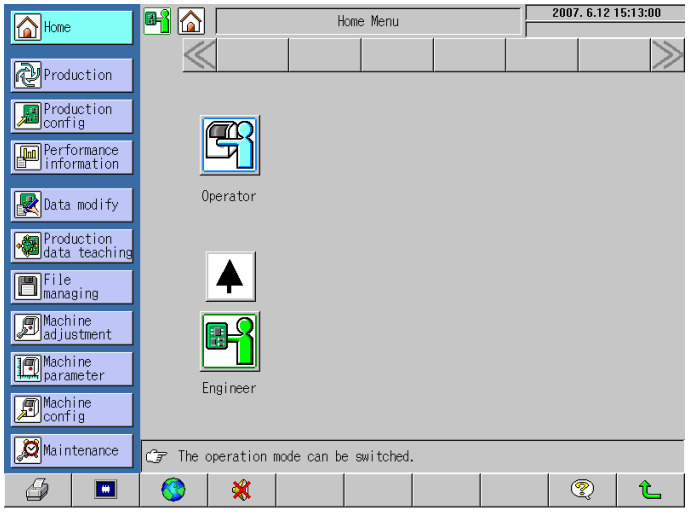
6.2.2 Teaching PCB Recognition Data

Use the following steps to teach PCB recognition data.

■ Procedure for Teaching the PCB Recognition Data

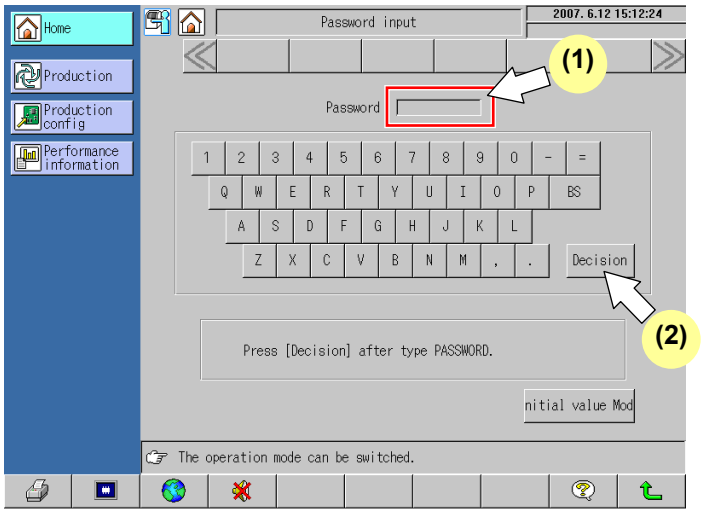
1. Turn the power ON.

- "Home Menu" screen appears on the display.



2. Change to the Operator level.

- Change to the operator level at which the data can be modified.
  (1) Enter the password.
  (2) Press [Decision] .

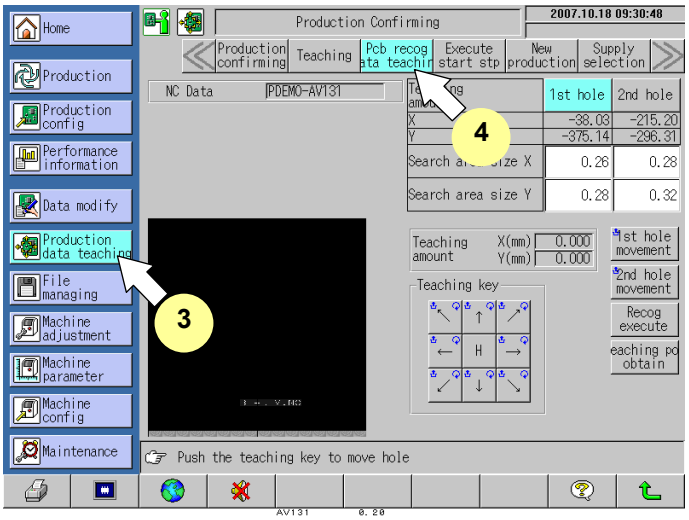


3. Call up the Production Data Teaching Menu screen.

- Press [Production data teaching] on the "Home Menu" screen.

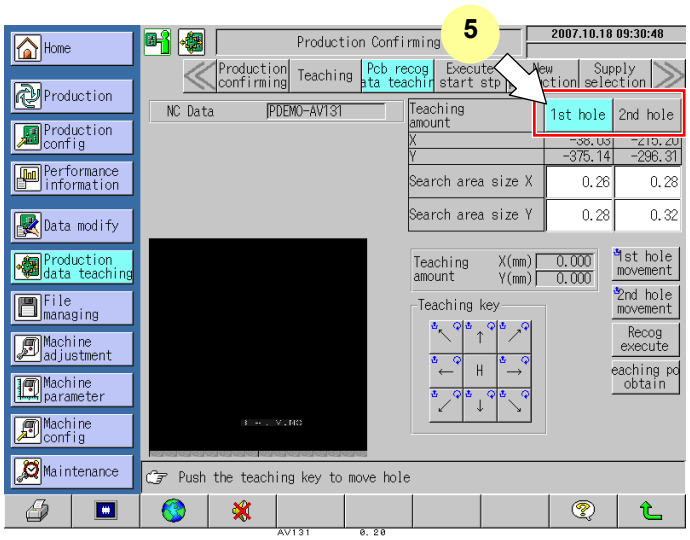
4. Select PCB recognition data teaching.

- Press [Pcb recog data teaching] .



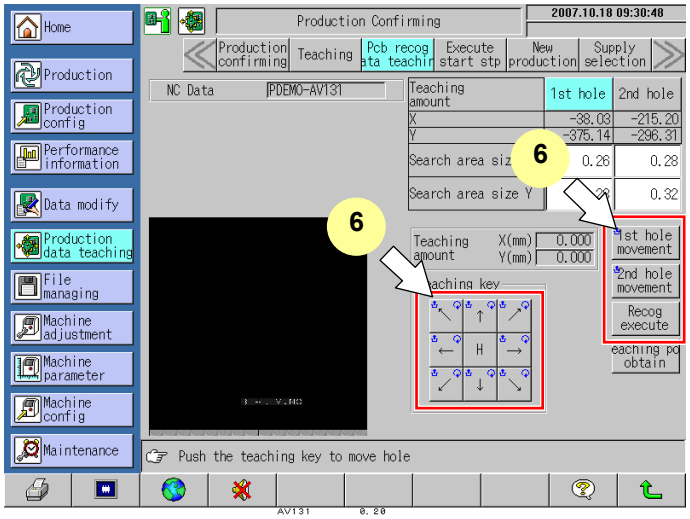
5. Select the target hole.

Press either [1st hole] or [2nd hole] on the screen to select the hole to be taught.



6. Teaching

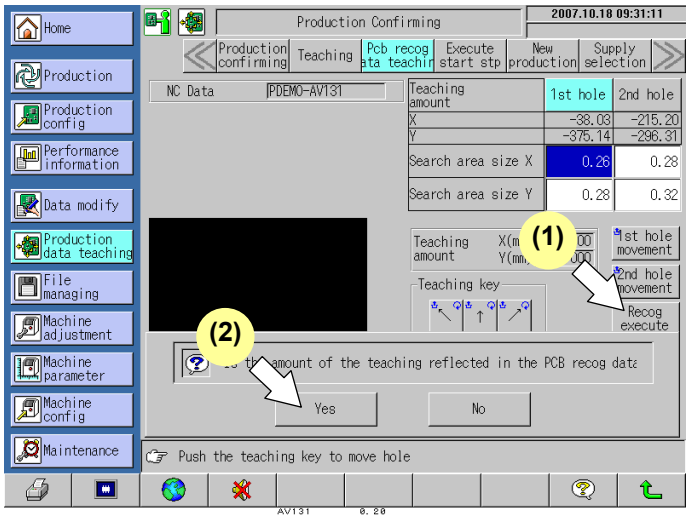
- Using the teaching keys, move the X-Y table and teach the hole.
- [1st hole movement] moves the X-Y table to the 1st hole coordinates specified in the PCB recognition data. [2nd hole movement] also moves the X-Y table to the 2nd hole coordinates in the same way as with [1st hole movement] .
- [Recog. execute] executes hole recognition and moves the X-Y table based on the recognition result.



7. Apply the teaching result.

(1) To apply the teaching data in step 6 to the PCB recognition data, press [Teaching pos obtain].

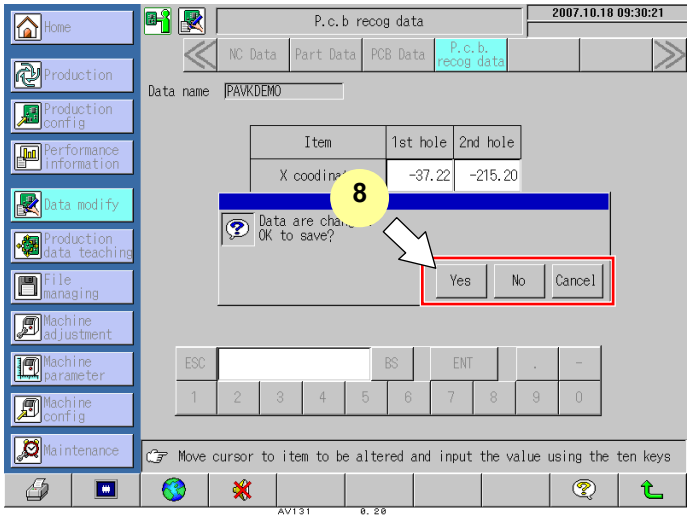
(2) A dialog appears asking you whether to apply the data. Press [Yes] to apply the teaching amount.



8. Complete editing.

Pressing displays a confirmation dialog asking you if you want to save the data.

- To save the data, press [Yes] .
- To discard the changes you made, press [No]. Press [Cancel] to continue the editing.



7. DATA CHECK PROCEDURE

This chapter describes the procedure for checking data.

7.1. Production Confirming (Check Operation)

D79PEC-4W-020-A0

7.1.1 Production Confirming Operation

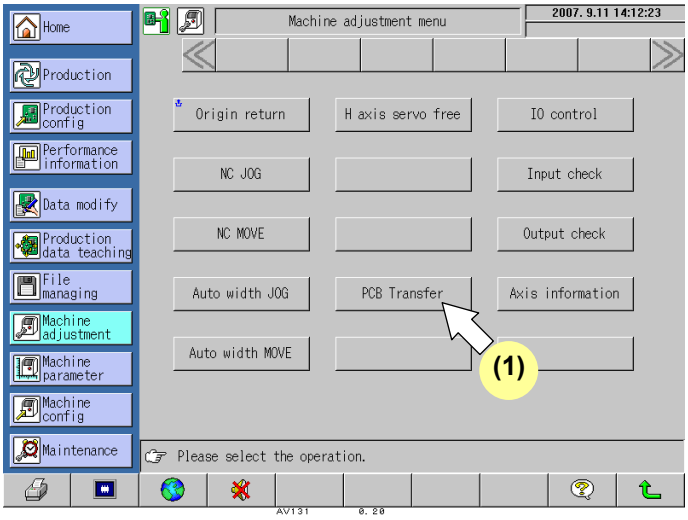
Before starting actual production with new NC data and part data, it is necessary to check the insertion position, insertion direction and insertion pitch. When NC data or part data have been modified, it is also necessary to check that there is no error in the modified data.

Production confirming is to check for proper insertion position, insertion direction and insertion pitch by operating the machine without mounting components.

◆ Procedure for Production Confirming Operation

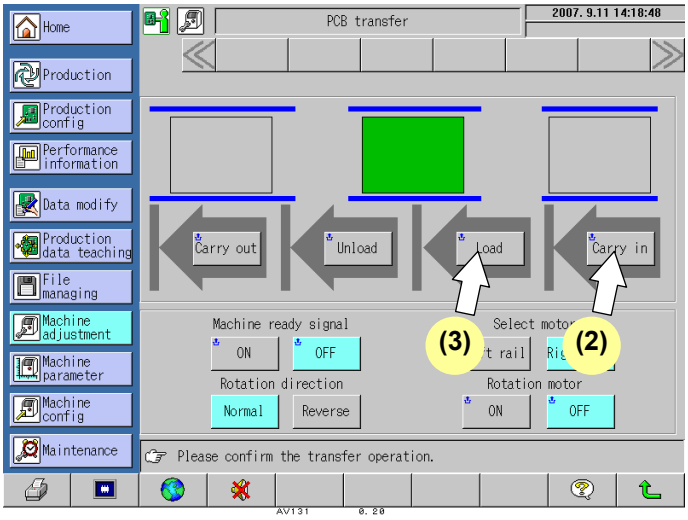
1. Place a PCB.

(1) Press [Machine adjustment] on the main menu, followed by [PCB Transfer] .



(2) When the “PCB transfer” screen appears, put a PCB on the right rail and press [Carry in] to load the PCB.

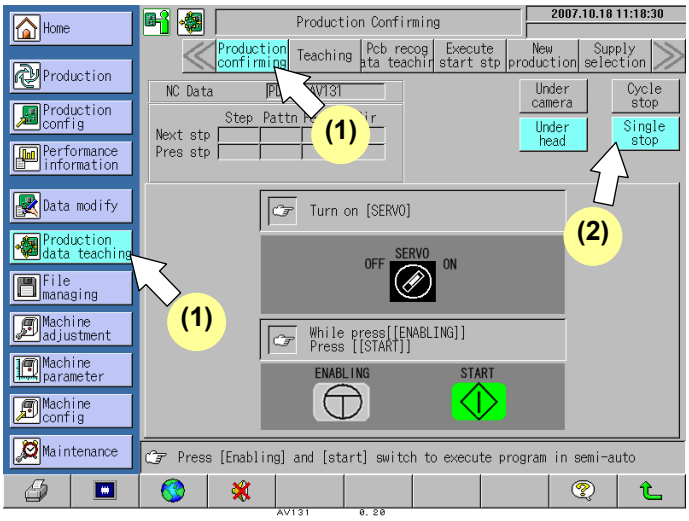
(3) Pressing [Load] sets the PCB in the X-Y table.



2. Confirm production.

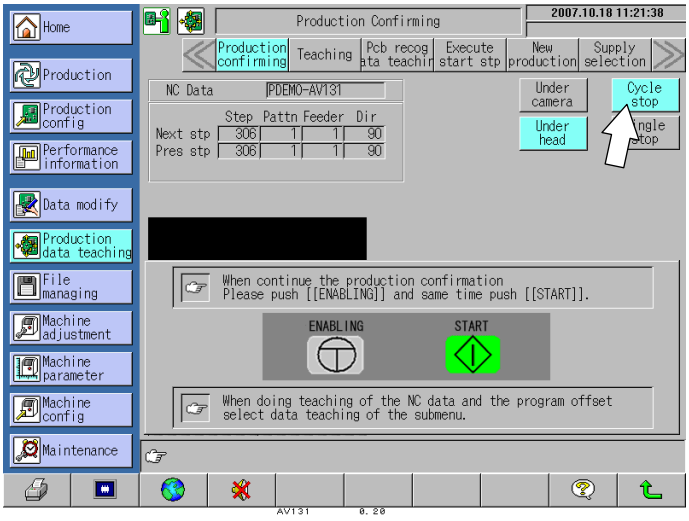
(1) Press [Production data teaching] on the main menu and [Production confirming] on the submenu.
(2) Check the NC data name and make sure that the target data is selected.

- Press [Single stop] , then hold down <ENABLING> and press <START> to enable the insertion positions to be checked one by one.



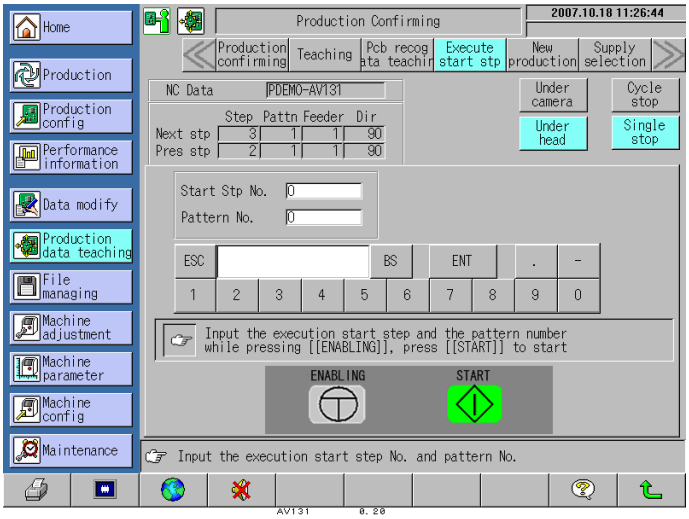
- Press [Single stop] , then hold down <ENABLING> and press <START> to allow all insertion positions to be checked.

- If any error is found in the NC data after production confirming operation, correct it. If no error is found, you can proceed to production via the [Production] menu.



3. Execution start step function

Operation can be started from any step in the NC data during production confirming process.
While the machine is stopped, pressing [Execute start stp] on the submenu displays a dialog prompting you to input the start step.
Input the start step and pattern No., then press [ENT] .
Hold down <ENABLING> and press <START> to start the operation from the specified step.



7.2. Data Teaching

D79PEC-4W-030-A0

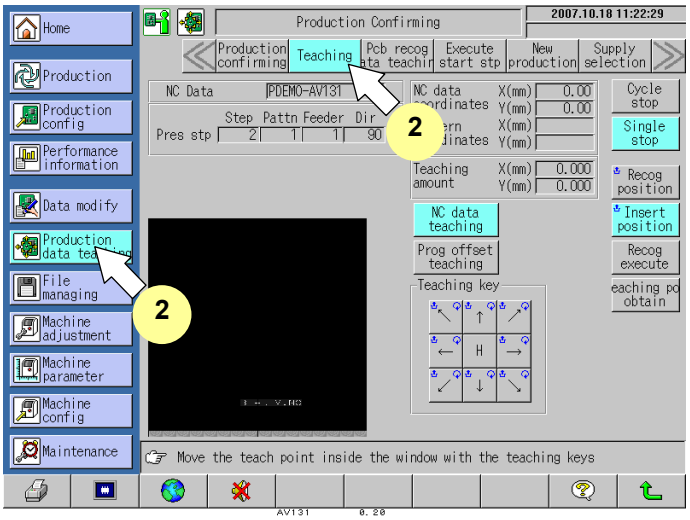
7.2.1 NC Data Teaching

The NC data teaching function permits you to teach the XY coordinates while checking the holes on the PCB.
The insertion point shown at 'Pres stp' is the target for teaching.

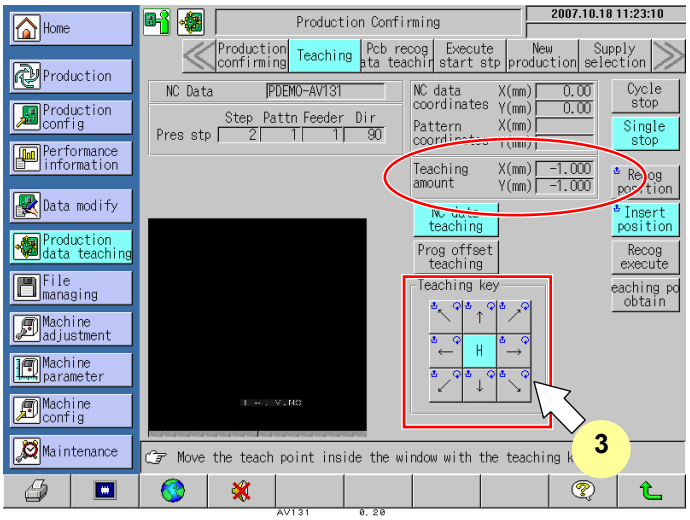
◆ Procedure for NC Data Teaching

1. Stop the machine at the insertion point to teach.

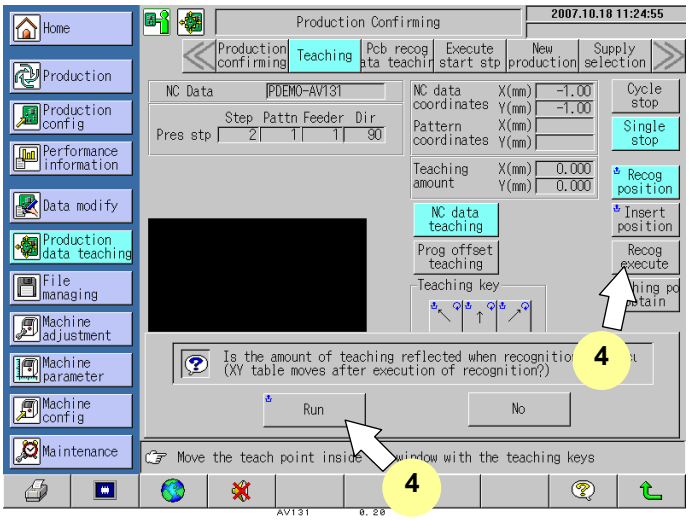
2. Press [Production data teaching] on the main menu and [Teaching] on the submenu.



3. Using the teaching keys, move the insertion point to the optimum position.
   The distance moved appears in the 'Teaching amount' field.
   - Pressing [Recog position] (while holding down <ENABLING>) moves the insertion point to the position directly under the camera. (The teaching position can be checked on the recognition screen.)
   - Pressing [Insert position] (while holding down <ENABLING>) moves the insertion point to the position directly under the insertion head.

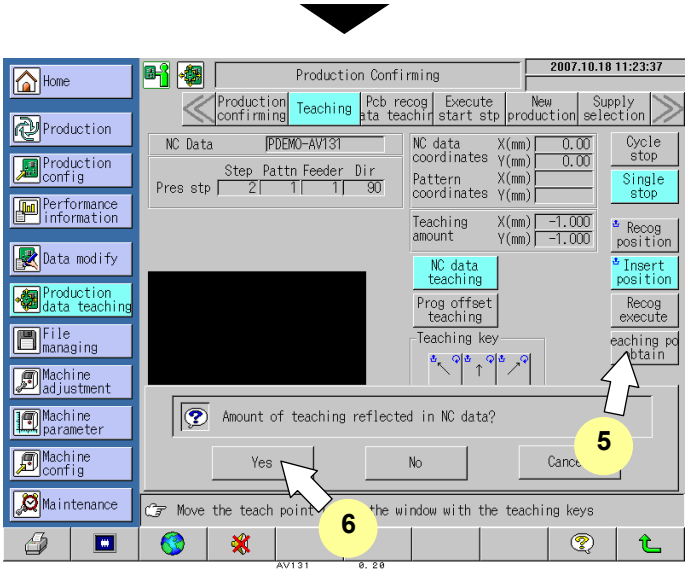


4. Pressing [Recog execute] recognizes the insertion hole to enable automatic teaching. With the dialog shown, press [Run] (while holding down <ENABLING>).



5. Pressing [Teaching pos obtain] displays a dialog asking you if you want to apply the teaching amount to the data.

6. Press [Yes] to apply to the data.

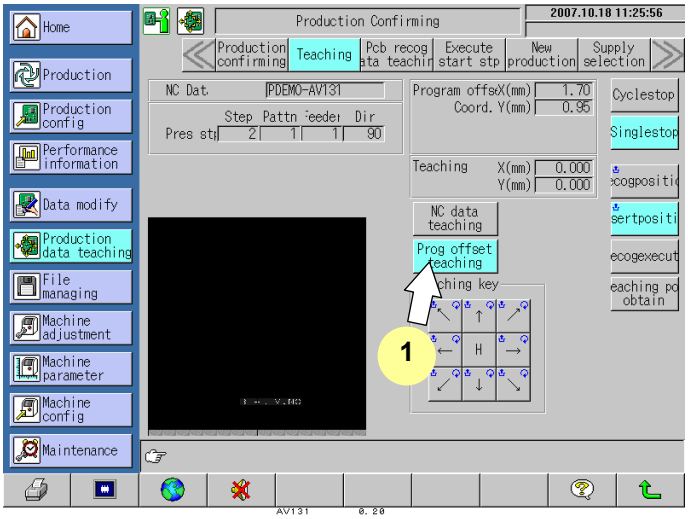


7.2.2 Program Offset Teaching

The program offset teaching function allows you to teach the XY coordinates while checking the holes on the PCB.

◆ Procedure for Program Offset Teaching

1. Press [Program offset teaching] to call up the program offset teaching screen.



Teaching procedures are the same as for NC data teaching.

8. FILE MANAGING

This chapter describes how to manage the data files.

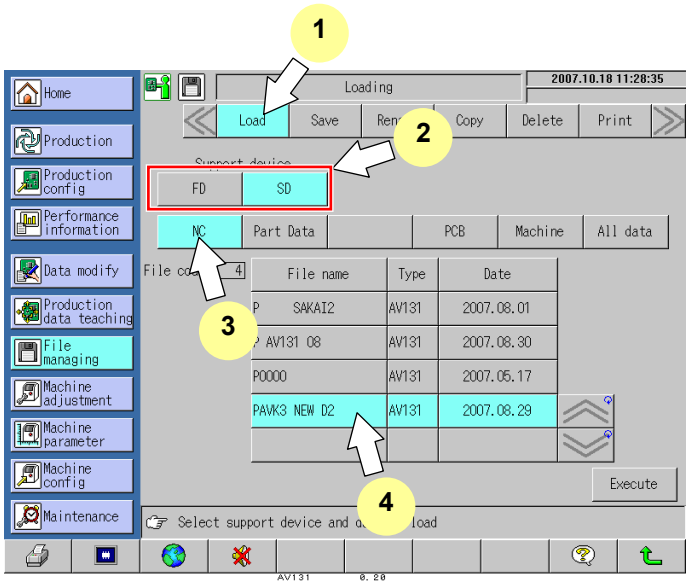
8.1. File Managing

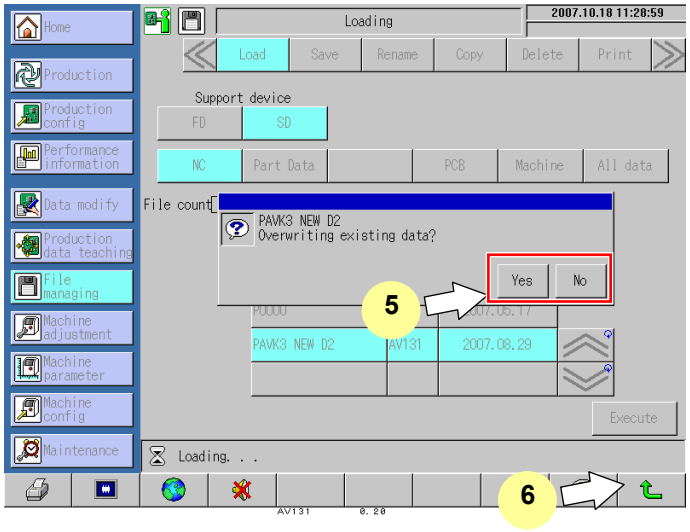
D79PEC-5F-010-A0

8.1.1 Load

This is used to load data saved on a floppy disk or SD card into the machine memory.

◆ Loading Procedure ('NC' Data)





1. Select the loading function.

- Press [Load] on the submenu.

2. Select the target device.

- Press [FD] or [SD] .

3. Select the data.

- Press [NC] .
  The list of data files is displayed.

4. Designate the file name.

Ex.) To load 'PAVK3 NEW D2'.

=HINT=

Use the to scroll through the list of the files displayed on the screen.

=HINT=

More than one file can be selected.
If an unnecessary file is selected, press the file name again to cancel it.

5. Load the data.

- Press [Execute] .

=REMARKS=

When an existing data in the memory has the same name as the data to be loaded from the device, the message at right appears.
Press [Yes] to overwrite the existing data with the data to be loaded.
Press [No] to cancel loading the data.

6. Complete loading.

- After the loading is completed, press .
  The screen returns to the Home Menu.

To load data of 'Part Data', 'PCB', 'Machine' or 'All data', specify one in step 3. (Steps thereafter are the same as those for NC data.)
When 'Machine' or 'All data' is selected, no file name needs to be specified.

8.1.2 Save

This function is used to save data of the memory into a floppy disk or SD card.

◆ Saving Procedure ('NC' Data)

1. Select the saving function.

- Press [Save] on the submenu.

2. Select the device.

- Press [FD] or [SD] .

3. Select the data.

- Press [NC] .
  The list of data files is displayed.

4. Designate the file name.

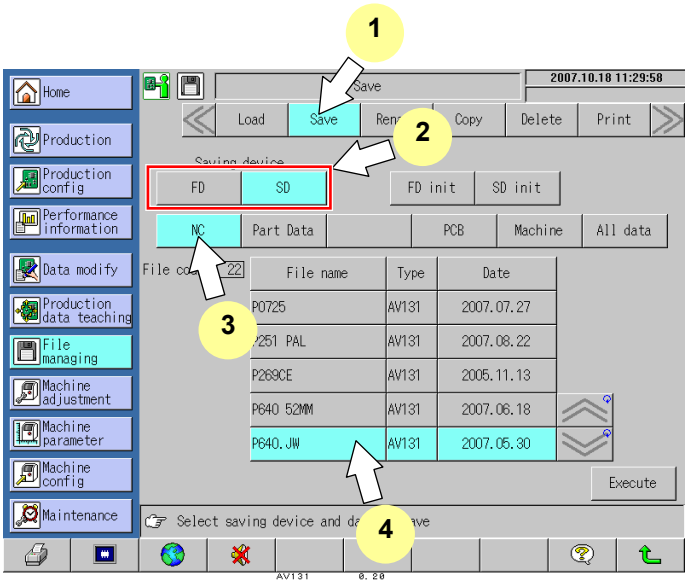
Ex.) To save 'P640. JW'.

=HINT=

Use the to scroll through the list of the files displayed on the screen.

=HINT=

More than one file can be selected.
If an unnecessary file is selected, press the file name again to cancel it.

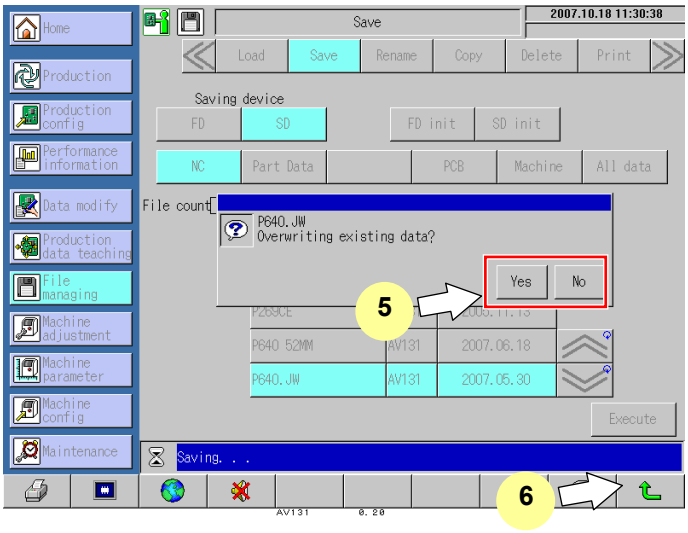


5. Save the data.

- Press [Execute] .

=REMARKS=

When the selected device has a data file with the same name as the data to be saved, the message at right appears.
Press [Yes] to overwrite the data on the selected device with the data in memory.
Press [No] to cancel saving the data.



6. Complete saving.

- After saving data is completed, press .
  The screen returns to the Home Menu.

To save data of 'Part Data', 'PCB', 'Machine' or 'All data', specify one in step 3. (Steps thereafter are the same as those for NC data.)
When 'Machine' or 'All data' is selected, no file name needs to be specified.

8.1.3 Rename

This function is used to change the data name in the devices.

◆ Procedure for Renaming Data ('NC' Data)

1. Select the renaming function.

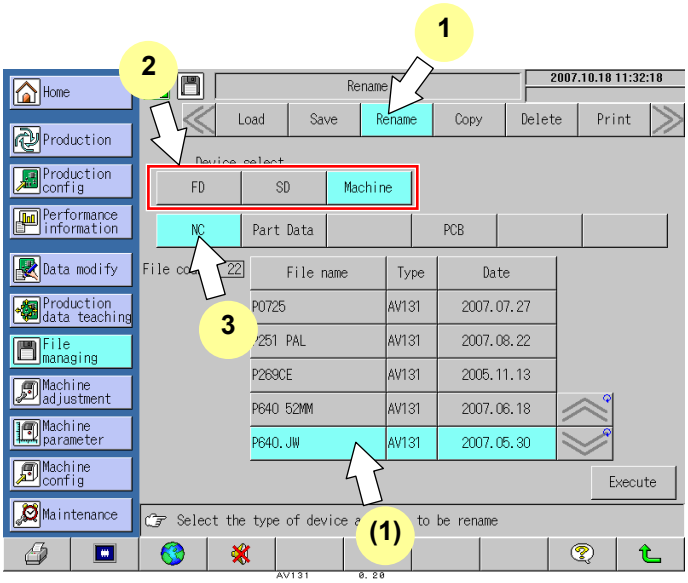
- Press [Rename] on the submenu.

2. Select the device.

- Press [FD], [SD], or [Machine] .

3. Select the data.

- Press [NC] .
  The list of data files is displayed.



4. Designate the file name.

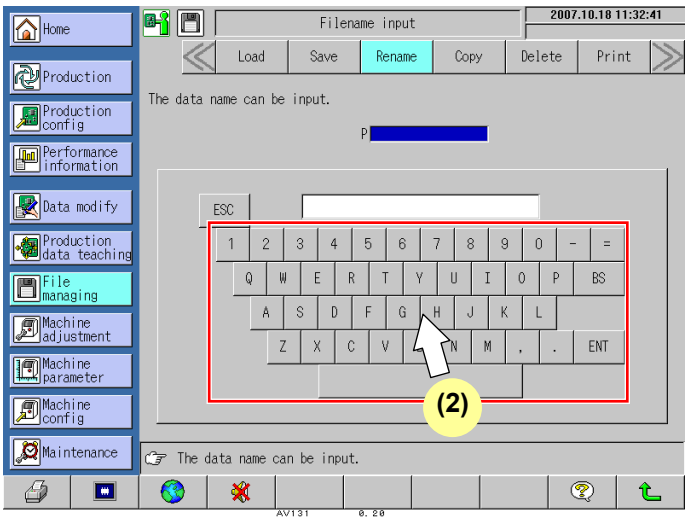
Ex.) To rename 'P640. JW' in the machine memory to 'PTEST'.

=HINT=

Use the to scroll through the list of the files displayed on the screen.

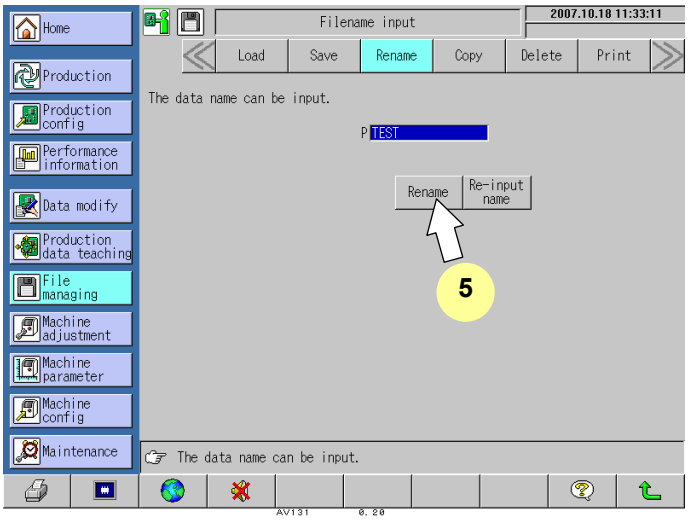
(1) Pressing a file name and then [Execute] button displays the Input window.

(2) Input a new file name.
Ex.) Input 'TEST', and press [ENT] to set the data.



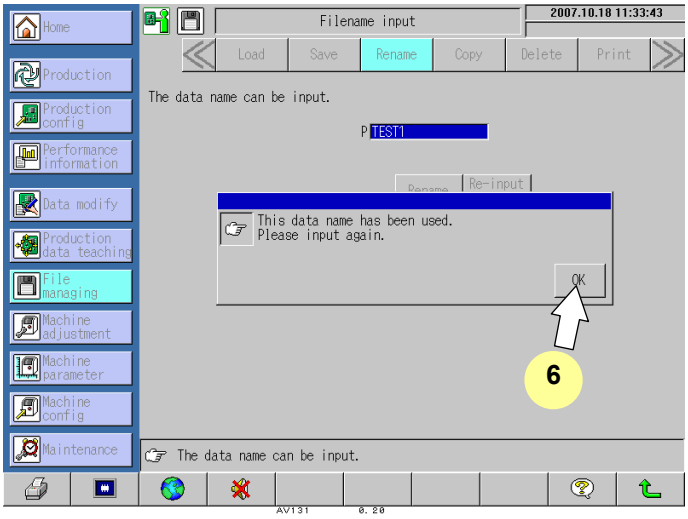
5. Rename the file.

- Press [Rename] to rename the file.



6. Re-input.

- If the specified data name already exists, a dialog to that effect is displayed. If such a dialog appears, press [OK] and enter a name again.



8.1.4 Copy

This function is used to copy data within the device.

◆ Data Copy Procedure ('NC' Data)

1. Select the copying function.

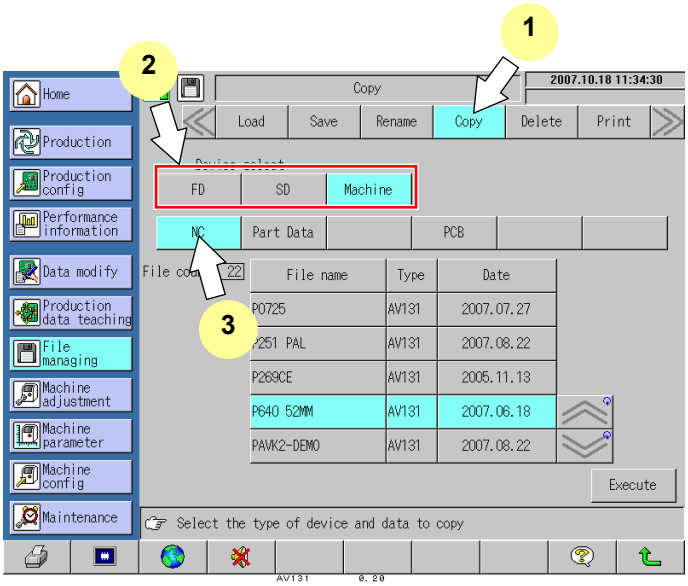
- Press [Copy] on the submenu.

2. Select the device.

- Press [FD], [SD], or [Machine] .

3. Select the data.

- Press [NC] .
  The list of data files is displayed.



4. Designate the file name.

Ex.) To copy 'P640 52MM' in the machine memory and to save it as 'PTEST'.

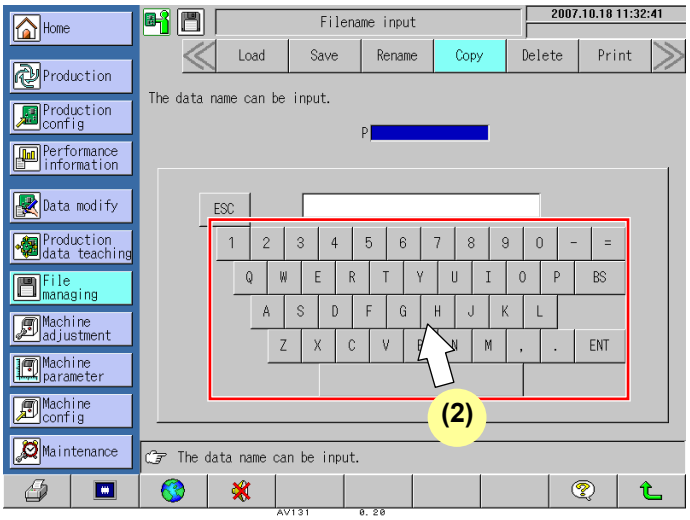
=HINT=

Use the to scroll through the list of the files displayed on the screen.

(1) Pressing a file name and then [Execute] displays the Input window.

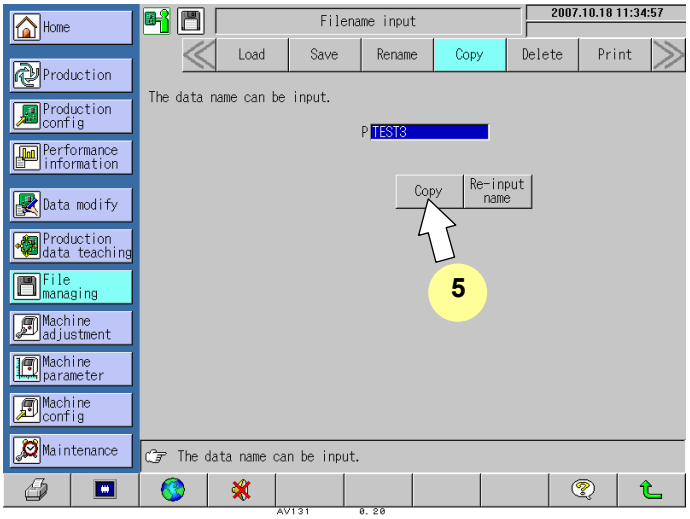
(2) Input the file name.

Ex.) Input 'TEST', and press [ENT] to set the data.



5. Copy the file.

- Press [Rename] to copy the file.



8.1.5 Delete

This function is used to delete the data in the device.

◆ Procedure for Deleting Data ('NC' Data)

1. Select the deleting function.

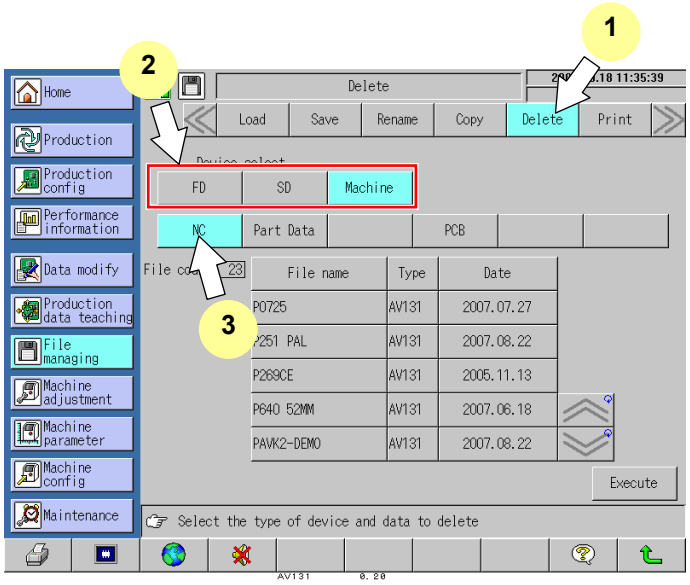
- Press [Delete] on the submenu.

2. Select the device.

- Press [FD], [SD], or [Machine] .

3. Select the data.

- Press [NC] .
  The list of data files is displayed.



4. Designate the file name.

Ex.) To delete 'P269CE' in the machine memory.

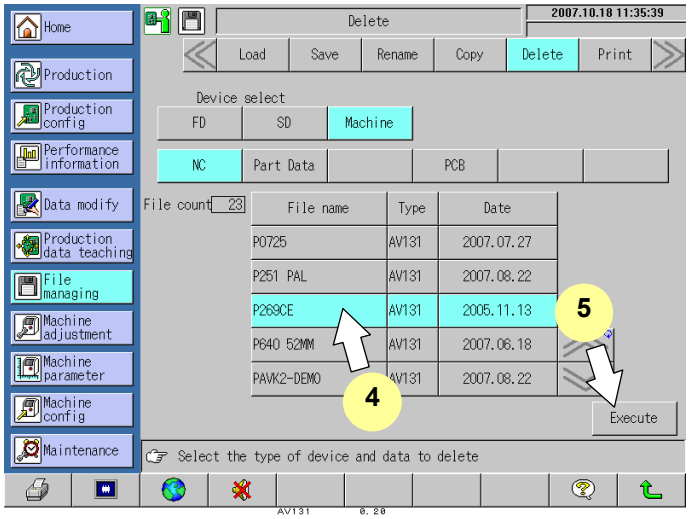
=HINT=

Use the to scroll through the list of the files displayed on the screen.

- Press the file name.

5. Delete.

- Press [Execute] .



8.1.6 Print

This function is used to print the data in machine memory. (Printer is optional.)

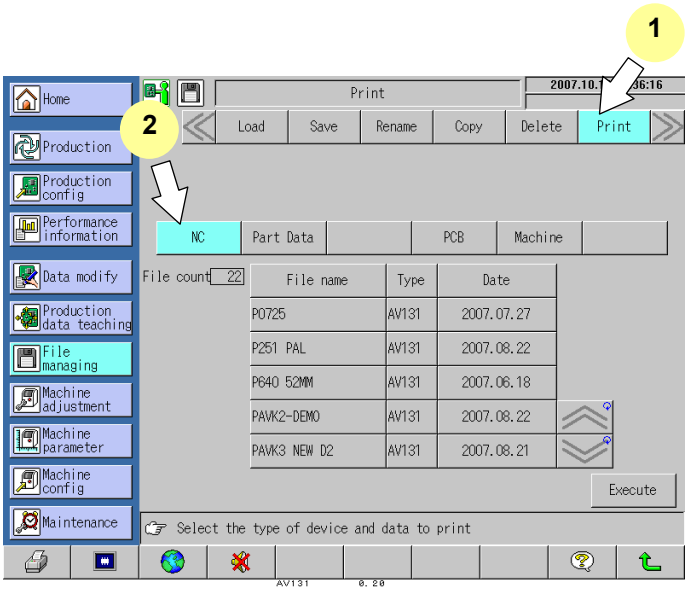
◆ Procedure for Printing Data ('NC' Data)

1. Select the printing function.

- Press [Print] on the submenu.

2. Select the data.

- Press [NC] .
  The list of data files is displayed.

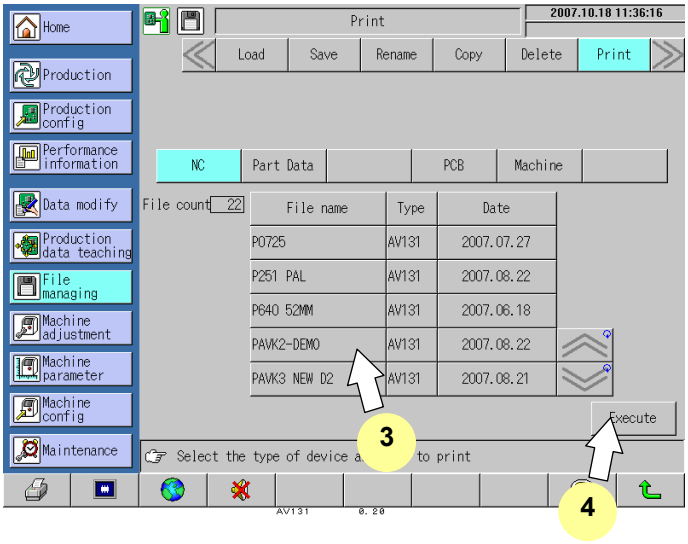


3. Designate the file name.

- Specify the data name directly on the screen.
  =HINT=
  Use the to scroll through the list of the files displayed on the screen.
- Press the file name.

4. Print the file.

- Press [Execute] to print the data.
  =HINT=
  It is also possible to specify multiple data files for printing.



8.1.7 Disk Initialization

This function is used to initialize a floppy disk or SD card.

◆ Procedure for Initialization

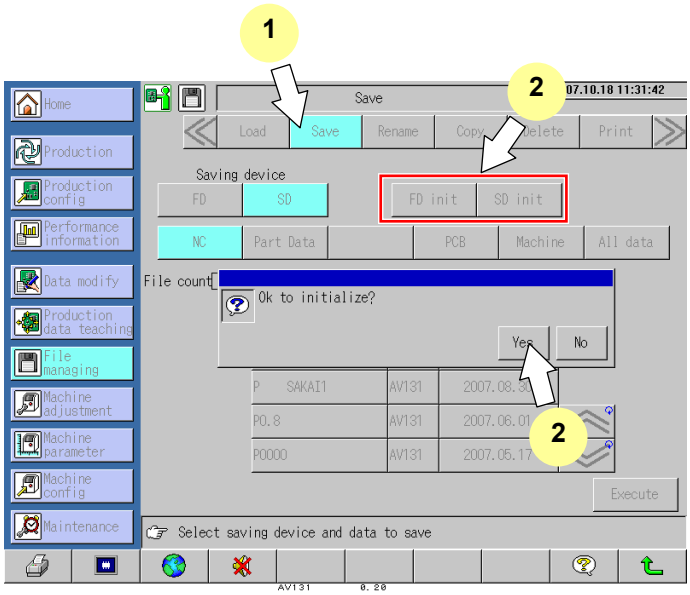
1. Select the initialization function.

- Press [Save] on the submenu.

2. Initialize the device.

- Pressing [FD init] or [SD init] displays a dialog asking you if you want to initialize a floppy disk or SD card.

  Press [Yes] to initialize the specified device.
  Press [No] to cancel initializing the device.



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